SETS, Relations and Functions

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Chapter 1

Preliminaries

1.1 Relations

Q: Show that the relation R in the set of real numbers defined as $R = \{(a, b) : \sqrt{a} + 3 \leqslant b\}$ is neither reflexive nor symmetric nor transitive.

1.1.1 Problems for Practice

1.1.1.1 Matrix match type problems

Matrix 1: Under column-I properties of relations are mentioned. Match the stated property with the relations listed under column-II , that satisfies it

Column I	Column II
(A) Equivalence	(P) The relation R in the set of natural numbers N, defined as $R = \left\{(a, b) : a > b^2\right\}$
(B) Reflexive	(Q) Relation R defined in the set $\{1, 2, 3, 4, 5\}$ defined by $R = \left\{(a, b) : a^2 + ab - 2b^2 = 0\right\}$
(C) Symmetric	(R) Relation R defined on the set of real numbers $\mathbb{R}$, defined by $R = \{(a, b) : \sqrt{a} + 5 < b\}$
(D) Transitive	(S) Relation R in the set $A = \{x \epsilon \mathbb{Z}; 0 \leq x \leq 9\}$,given by R=$\{(a, b) : (a - b)^4$ is divisible by 3$\}$

1.2 Real Numbers

When we talk of functions, we talk of only real valued functions. So, it is important that we brush up our theory of real numbers done in junior classes.

Definition

Any decimal fraction, terminating or nonterminating, is called a real number. The set of real numbers is represented by the symbol $\mathbb{R}$.

1.2.1 Rational Numbers

Every rational number can be written in the form $\frac{p}{q}$ where p and q are integers. The set of rational numbers is reqresented by the symbol $\mathbb{Q}$. i.e. $\mathbb{Q} = \boldsymbol{\{}\frac{p}{q} \mid q \neq 0,\ p\ \boldsymbol{\&}\ q\ \epsilon\ \mathbb{Z}\boldsymbol{\}}$ where $\mathbb{Z}$ is the set of Integers.

The decimal representation of a rational number can be either terminating or non-terminating reccuring. e.g.

3.24 is a terminating decimal represenation.

$5.\overline{3}$ is a non-terminating recurring decimal representation.

Properties

- The set of rational numbers is **closed** with respect to the operations of addition, subtraction and product. i.e. If p and q are two rational numbers, then $p+q$, $p-q$ and pq are rational. Also $\frac{p}{q}$ is rational provided $q \neq 0$.**[Closure Property]**
- Between any two rational numbers p & q , there exist infinite rational numbers.[**Property of Denseness**]

Q. Find all the rational values of x for which $y = \sqrt{x^2+x+5}$ is a rational number.

Sol. Suppose x and y are rational numbers . Then the sum $x+y = q$ is a rational number.We now express x in terms of q .

$$y + x = \sqrt{x^2+x+5} + x = q$$

$$\sqrt{x^2+x+5} = q - x$$

$$x^2 + x + 5 = q^2 - 2qx + x^2$$

$$x = \frac{q^2-5}{1+2q}, q \neq -\frac{1}{2}$$

To prove the reverse i.e. $y = \sqrt{x^2+x+5}$ is a rational number if $x = \frac{q^2-5}{1+2q}, q \neq -\frac{1}{2}$

Its easy to check

$$y = \sqrt{x^2+x+5} = \sqrt{\left(\frac{q^2-5}{1+2q}\right)^2 + \left(\frac{q^2-5}{1+2q}\right) + 5}$$

$$= \sqrt{\frac{q^4+2q^3+11q^2+10q+25}{(1+2q)^2}}$$

$$= \sqrt{\left(\frac{q^2+q+5}{1+2q}\right)^2}$$

It may be observed that q^2+q+5 is positive for all values of q.Hence

$$y = \frac{q^2+q+5}{|1+2q|}$$

which is rational for $q \neq -\frac{1}{2}$.

Note : The sole purpose of adding x to y was to eliminate the square term of x . The same goal can be achieved by subtracting x from y.

1.2.2 Irrational Numbers

Non-terminating non-repeating decimal fractions are called Irrational numbers . In fact, all real numbers which are not rational are Irrational numbers. The set of Irrational numbers can be represented by the symbol T or $\mathbb{R} - \mathbb{Q}$.

Properties

- The sum,difference,product and quotient of two Irrational numbers may be rational or irrational.ie. The set **T** is not closed with respect to addition,subtraction,multiplication and division.**[Closure Property]**
- The sum, difference,product and quotient of a rational number and an Irrational number is Irrational.
- There exist infinite rational and irrational numbers between two real numbers.**[Property of Denseness]**

Q. Prove that for every rational number r satisfying $r^2 < 3$ one can always find a larger rational number $r + k(k > 0)$for which $(r + k)^2 < 3$.

Sol. We know, there exist infinite rational numbers between a rational number and an irrational number. So, there will be infinite rational numbers between r and $\sqrt{3}$. So, there exist infinite values of k satisfying the condition.Let us take the case when r is sufficiently close to $\sqrt{3}$.In that case, we can chose a k such that $k \to 0$. This implies $k^2 < k$. i.e. $(r + k)^2 < r^2 + 2rk + k$. It is sufficient to put $r^2 + 2rk + k = 3$.i.e.

$$k = \frac{3 - r^2}{1 + 2r}$$

It may be observed that any rational number smaller than $\dfrac{3 - r^2}{1 + 2r}$ in magnitude will also satisfy the condition.

1.2.3 Problems for practice

1.2.3.1 Subjective Problems

Q1. Find all the rational values of x for which $y = \sqrt{49x^2 + 5x + 3}$ is a rational number.

Q2. Prove that for every rational number s satisfying $s^2 > 5$ one can always find a smaller rational number $s - k(k > 0)$for which $(s - k)^2 > 5$.

Q3. Does 219.151551555515555555551......... represent a rational number?. Give reason.

Q4. Represent $\sqrt{31} - 3$ and $\sqrt{52}$ on the number line.

1.2.3.2 Multiple Answer MCQ's

Q1. Which of the following are rational numbers.

a) $\dfrac{\sqrt{3}}{2 + \sqrt{3}}$

b) $\dfrac{2}{6}$

c) $\left(\sqrt{5} - 2\right)\left(\sqrt{5} + 2\right)$

d) $\dfrac{11\pi}{7\pi}$

1.2.3.3 Write the Answer Type Questions

Q1: The solution set of $\dfrac{7}{x-2} \geq 5$ is

1.2.3.4 Assertion-Reason Type Questions

Each question in this section has four choices (a),(b),(c) and (d) out of which only one is correct. Mark your choices as follows:

(a) STATEMENT-1 is True, STATEMENT-2 is True, STATEMENT-2 is the correct explaination of STATEMENT-1.

(b) STATEMENT-1 is True, STATEMENT-2 is True, STATEMENT-2 is **NOT** the correct explaination of STATEMENT-1.

(c) STATEMENT-1 is True, STATEMENT-2 is False.

(d) STATEMENT-1 is False, STATEMENT-2 is True

Q1: STATEMENT-1: For every rational number r satisfying $r^2 < 2$ one can always find a larger rational number $r+k(k>0)$for which $(r+k)^2 < 2$.

STATEMENT-2: There exist infinite rational numbers between any two rational numbers.

1.2.3.5 Hints and Solutions

Subjective Problems

Q1 $x = \dfrac{3-q^2}{14q-5}$, $q \epsilon \mathbb{Q}$ & $q \neq \dfrac{5}{14}$ {Hint: Put $y-7x=q$ to get the answer. Also, equivalently if we had put $y+7x=q$ we would have got , $x = \frac{q^2-3}{14q+5}$, $q \neq -\frac{5}{14}$}

Q2 {Hint: Let us take $s^2-2sk=5$ which would imply $s^2-2sk+k^2>5$. ie $k=\frac{s^2-5}{2s}$}

Q3 No , it is not a rational number. {Hint. 2^{n-1}fives between nth and $(n+1)$th unity. The pattern is definite but non periodic.}

Q4 {Hint. $\sqrt{n}$ can be drawn by the following method of construction. Draw an arc from the origin of length $\frac{n-1}{2}$. From the point where it cuts the y axis, draw an arc of length $\frac{n+1}{2}$ intersecting the x-axis at P. The distance OP is $\sqrt{n}$.}

Multiple Answer MCQ's

Q1 B,C,D

Write the Answer Type Questions

Q1 the half open interval (2,17/5][1]

Assertion Reason Type Questions

Q1 B

1.3 Absolute value of a real number

The magnitude of a number is called the Absolute value of the number.Also,it is the distance of the point representing the number from origin on the number line . e.g. $|2-\sqrt{5}| = \sqrt{5}-2$.

[1]Intervals can be of the following kinds

1. Open Interval i.e of the kind (a,b)
2. Closed Interval i.e. of the kind [a,b]
3. Semi-Closed Interval(or Semi-open/Half-open) ie. of the kind (a,b] or [a,b)

Mathematical Definition

$$|x| = \begin{cases} -x & , x < 0 \\ x & , x \geqslant 0 \end{cases}$$

Some alternate definitions of $|x|$ are

$$|x| = \begin{cases} \sqrt{x^2} \\ max\{x, -x\} \\ x.sgn(x) \end{cases}$$

1.3.1 Properties of Absolute value

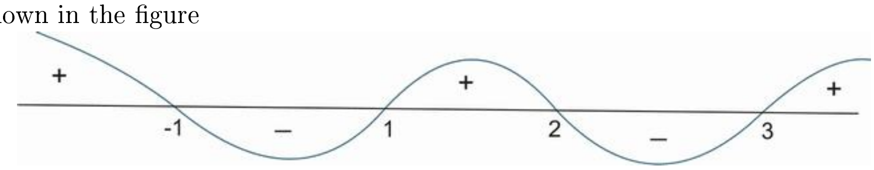
1. $|x| = \alpha(\alpha > 0) \Rightarrow x = \pm\alpha$
2. $|x| \leqslant \alpha(\alpha > 0) \Rightarrow -\alpha \leqslant x \leqslant \alpha$
3. $|x| \geqslant \alpha\,(\alpha > 0) \Rightarrow x \geqslant \alpha$or $x \leqslant -\alpha$
4. $|x+y| \leqslant |x| + |y|$ {Equality occurs when x and y are of same sign. i.e. $xy \geqslant 0$ e.g.$|\,(-5)+(-2)\,| = |-5| + |-2|$ as (-5).(-2) is positive }[**The Triangle Inequality**][2]
   - $|x-y| \leqslant |x| + |y|$ {Equality occurs when x and y are of opposite sign. i.e. $xy \leqslant 0$}
   - $|x+y| \geqslant ||x| - |y||$ {Equality occurs when x and y are of opposite sign. i.e. $xy \leqslant 0$}
   - $|x+y| \geqslant |x| - |y|$ {Equality occurs when x and y are of opposite sign i.e. $xy \leqslant 0$ and $|x| \geqslant |y|$}
   - $|x-y| \geqslant ||x| - |y||$ {Equality occurs when x and y are of same sign. i.e. $xy \geqslant 0$}
   - $|x-y| \geqslant |x| - |y|$ {Equality occurs when x and y are of same sign . i.e. $xy \geqslant 0$ and $|x| \geqslant |y|$}
5. $|xy| = |x||y|$
6. $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}, y \neq 0$

1.3.2 The Wavey-Curve Technique

Let us consider, that we have an expression of the form

$$e = \frac{(x+1)\,(x-1)\,(x-3)}{x-2}$$

and we have to find the values of x for which this expression is > 0 , < 0 or $= 0$. To find the sign of the expression, we first of all mark these points on the number line . Then we draw a curve like the one shown in the figure



We start from the right hand side , draw a positive section of curve between ∞ and the right most point. Then, at the rightmost point(3 in this case) , we switch the sign of the curve to negative and

[2]A proof of the Triangle Inequality
$|x+y|^2 = (x+y)^2$
$= x^2 + 2xy + y^2$
$\leq x^2 + 2\,|x|\,|y| + y^2$
$= |x|^2 + 2\,|x|\,|y| + |y|^2$
$= (|x| + |y|)^2$
$\Longrightarrow |x+y| \leq$ |x|+|y|

take it below the axis. Like this , we change sign of the curve at every Critical point(ie. at 3,2,1,-1 in this case). Now from this curve, we try to find out the sign of the expression

$$e = \frac{(x+1)(x-1)(x-3)}{x-2}$$

$e > 0$ whenever + sign appears in the wavey-curve(i.e. when wavey-curve is above the axis)
$\Rightarrow e > 0$ for $x \epsilon (-\infty, -1) \cup (1,2) \cup (3, \infty)$[3]
$e < 0$ whenever - sign appears in the wavey-curve(i.e. when wavey-curve is below the axis)
$\Rightarrow e < 0$ for $x \epsilon (-1, 1) \cup (2, 3)$
$e = 0$ for all the critical points in the numerator(if and only if they don't appear in the denominator too)
$\Rightarrow e = 0$ for $x \epsilon \{-1, 1, 3\}$
$e =$Not defined for all the critical points in the denominator.
$e =$Not defined for $x \epsilon \{2\}$
{Note: It may be noted, that if we have to find x such that $e \geq 0$, then we will take the union of the values of x found out for $e > 0$ and $e = 0$
i.e. $e \geq 0$ for $x \epsilon ((-\infty, -1) \cup (1, 2) \cup (3, \infty)) \cup \{-1, 1, 3\}$. The same expression can be written by using a closed interval at the points where $e = 0$ i.e. $e \geq 0$ for $x \epsilon (-\infty, -1] \cup [1, 2) \cup [3, \infty)$
In a similar way, we can find the values of x for which $e \leq 0$. The solution is $x \epsilon [-1, 1] \cup (2, 3]$. }

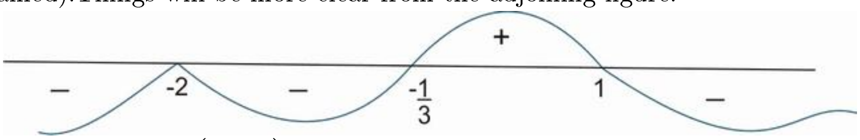
Example

We have to find the sign of $E = \dfrac{(x+2)^6 (3x+1)^3}{(1-x)}$

To find the sign of such an expression, we first of all make the coefficients of x as 1 in all terms. This gives

$$E = -\frac{27 (x+2)^6 \left(x + \frac{1}{3}\right)^3}{x-1}$$

The critical points of this expression are $x = \left\{-2, -\frac{1}{3}, 1\right\}$. The wavey-curve will start with a negative sign at ∞ as there is a - sign in front of E. The curve will then change sign at the first critical point -1. Thereafter , the curve will change sign 3 times at the point $x = -\frac{1}{3}$. (It may be noted that if a curve changes sign an odd number of times, it is equivalent to change of sign just one time) . Then the curve changes sign 6 times at $x = -2$.(change of sign an even number of times means the sign is retained).Things will be more clear from the adjoining figure.



$\Rightarrow E > 0$ for $x \epsilon \left(-\frac{1}{3}, 1\right)$

Similarly, $E < 0$ for $x \epsilon (-\infty, -2) \cup \left(-2, -\frac{1}{3}\right) \cup (1, \infty)$.

Also, $E = 0$ for $x \epsilon \left\{-2, -\frac{1}{3}\right\}$ and E is not defined for $x \epsilon \{1\}$

Q Solve for x

a) $|2x + 5| = 6$

[3]The actual reason for the positive sign of e in the interval $(3, \infty)$ is that in this interval the expressions $x - 3 > 0$, $x - 2>0$, $x - 1 > 0$ and also $x + 1 > 0$. Due to this reason, the expression e which is obtained by multiplying 3 positive quatities and dividing the resultant with a positive quantity is positive.
Similarly, in the interval $(2, 3)$ the quantities $x - 2>0$, $x - 1 > 0$ & $x + 1 > 0$ while the quantity $x - 3 < 0$. Hence the expression e is negative in this interval.
In this way we can investigate for other intervals also.

b) $|5x+7| = |7x-9\ |$

c) $|3x-5| < 1$

d) $|x^2-5x+6| > x^2-5x+6$

e) $|x| = x+7$

f) $|x| = x-5$

g) $\left|\dfrac{(x-5)^3}{(x+2)(x-3)}\right| > \dfrac{(x-5)^3}{(x+2)(x-3)}$

h) $|\cos x| = \cos x + 1$

i) $|(x^2+2x+5)+(3-5x)| = |x^2+2x+5| + |3-5x|$

j) $\left|\left(x^8-4\right)-\left(x^4+2\right)\right| = \left|x^8-4\right| - \left|x^4+2\right|$

k) $x^2 - |x| - 2 = 0$

l) $|x+2| = |x-2| + 2|x|$

m) $\left|\dfrac{3x+5}{5x+7}\right| - \left|\dfrac{7x+9}{5x+7}\right| = 1$

n)$2^{|x+1|} - |2^x - 1| = 2^x + 1$

Sol. a) $2x+5 = \pm 6$

$\Rightarrow 2x = 1$ or $2x = -11$

$\Rightarrow x = \dfrac{1}{2}$ or $-\dfrac{11}{2}$

b) $|5x+7| = |7x-9\ |$

$\Rightarrow (5x+7) = \pm(7x-9)$

$\Rightarrow 5x+7 = 7x-9$ or $5x+7 = -(7x-9)$

$\Rightarrow 2x = 16$ or $12x = 2$

$\Rightarrow x = 8$ or $x = \dfrac{1}{6}$ i.e. $x\epsilon\left\{8, \dfrac{1}{6}\right\}$

c) **Method 1**: $|3x-5| < 1$

$\Rightarrow -1 < 3x-5 < 1$ (Property 2)

$\Rightarrow 4 < 3x < 6$

$\Rightarrow \dfrac{4}{3} < x < 2$ i.e $x\epsilon\left(\dfrac{4}{3}, 2\right)$

Method 2: $|3x-5| < 1$

Squaring both sides, we get

$(3x-5)^2 < 1$

$\Rightarrow 9x^2 - 30x + 25 < 1$

$\Rightarrow 9x^2 - 30x + 24 < 0$

$\Rightarrow 3x^2 - 10x + 8 < 0$

$\Rightarrow (3x-4)(x-2) < 0$

$\Rightarrow x\epsilon\left(\dfrac{4}{3}, 2\right)$

d) $|x^2-5x+6| > x^2-5x+6$

{Note: It can be observed that $|y|$ is strictly greater than y if $y < 0$}

$\Rightarrow x^2 - 5x + 6 < 0$

$\Rightarrow (x-2)(x-3) < 0$

$\Rightarrow 2 < x < 3$ i.e. $x \epsilon (2,3)$

e) **Case I**: $x \geq 0$

This assumption would imply that $|x| = x$

$\Rightarrow x = x + 7$ which is not true for any x. Hence , no solution exists.

Case II: $x < 0$

$\Rightarrow -x = x + 7$

$\Rightarrow 2x = -7$ or $x = -\dfrac{7}{2}$ (which satisfies the condition $x < 0$)

f)**Case I**: $x \geq 0$

$\Rightarrow x = x - 5$ which is not true for any x. Hence , no solution exists for $x \geq 0$.

Case II: $x < 0$

$\Rightarrow -x = x - 5$

$\Rightarrow 2x = 5$ or $x = \dfrac{5}{2}$

This value of x does not satisfy $x < 0$. Hence , no solution exists for $x < 0$ also.

g)$\left|\dfrac{(x-5)^3}{(x+2)^2(x-3)}\right| > \dfrac{(x-5)^3}{(x+2)^2(x-3)}$

It may be observed that $|y| > y$ is only possible when $y < 0$

$\Rightarrow \dfrac{(x-5)^3}{(x+2)^2(x-3)} < 0$

$\Rightarrow x \epsilon (3,5)$

[Note:The solution to such equations can be found out with the help of **Wavey-Curve Technique** as discussed in the Lectures]

h) $|\cos x| = \cos x + 1$

Case I: $\cos x \geq 0$

$\Rightarrow \cos x = \cos x + 1$

which is not possible for any x. Hence , no solution exists for $\cos x \geq 0$.

Case II:$\cos x < 0$

$\Rightarrow -\cos x = \cos x + 1$

$\Rightarrow 2\cos x = -1$

$\Rightarrow \cos x = -\dfrac{1}{2}$ (which satisies our assumption that $\cos x < 0$)

$\Rightarrow x = 2n\pi \pm \dfrac{2\pi}{3}$, $n \epsilon Z$.

i) $|(x^2 + 2x + 5) + (3 - 5x)| = |x^2 + 2x + 5| + |3 - 5x|$

$|x + y| = |x| + |y| \Rightarrow xy \geq 0$(equality condition in Property 4)

i.e x & y are of the same sign

Now $x^2 + 2x + 5 = (x+1)^2 + 4 > 0$

$\Rightarrow 3 - 5x \geq 0$

i.e $x \leq \dfrac{3}{5}$ or $x \epsilon (-\infty, \dfrac{3}{5})$

j)$\left|(x^8 - 4) - (x^4 + 2)\right| = \left|x^8 - 4\right| - \left|x^4 + 2\right|$

We know, $|x-y| = |x| - |y|$ if $xy \geq 0$ & $|x| \geqslant |y|$

Also , it may be noted that $x^4 + 4 > 0 \forall x$

$\Rightarrow$Our conditions reduce to $x^8 - 4 \geq x^4 + 2$

$\Rightarrow x^4 - 2 \geq 1$

$\Rightarrow x^4 \geq 3$

$\Rightarrow x \geq \sqrt[4]{3}$ or $x \leq -\sqrt[4]{3}$ i.e. $x\epsilon \left(-\infty, -\sqrt[4]{3}\right) \cup \left(\sqrt[4]{3}, \infty\right)$

k)$x^2 - |x| - 2 = 0$

$\Rightarrow |x|^2 - |x| - 2 = 0$

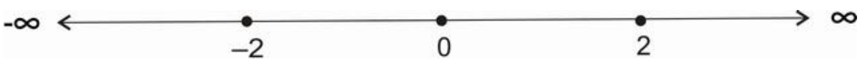
$\Rightarrow (|x| - 2)(|x| + 1) = 0$

$\Rightarrow |x| = 2$ or $|x| = -1$(Rejected)

$\Rightarrow x = \pm 2$ i.e $x\epsilon \{-2, 2\}$

l)$|x+2| = |x-2| + 2|x|$

We first of all find the critical points. $|x+2| = 0$ gives the point -2. Similarly, the points 2,0 are obtained by euating the other modulus terms to zero. So, there are 3 critical points ie at $x = \{-2, 0, 2\}$. We plot these points on the number line and divide the whole number line at these critical points.



Case I : $x < -2$

The equation becomes

$-(x+2) = -(x-2) - 2x$

$\Rightarrow 2x = 4$

$\Rightarrow x = 2$ (which is rejected as it does not belong to the interval $(-\infty, -2)$)

Case II: $-2 \leq x < 0$

The equation becomes

$x + 2 = -(x-2) - 2x$

$\Rightarrow x = 0$ (Which is rejected as it does not belong to the interval $[-2, 0)$)

Case III : $0 \leq x < 2$

The equation becomes

$x + 2 = -(x-2) + 2x$

which is true for all x , but our interval is $[0, 2)$

$\Rightarrow x\epsilon [0, 2)$ is a solution

Case IV : $x \geq 2$

The equation in this case becomes

$x + 2 = (x-2) + 2x$

$\Rightarrow x = 2$ (which is accepted as it lies in the interval chosen i.e. $[2, \infty)$)

Hence $x = 2$ is a solution

Now we combine all the solutions obtained from different intervals. So, the final solution is the union of all the cases i.e. $x\epsilon [0, 2) \cup \{2\}$

$\Rightarrow$The complete solution set is $x\epsilon [0, 2]$

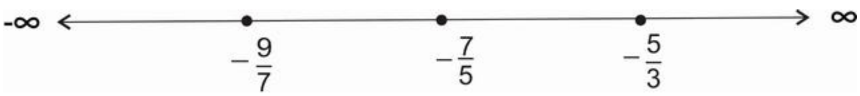
m)$\left|\dfrac{3x+5}{5x+7}\right| - \left|\dfrac{7x+9}{5x+7}\right| = 1$

First thing to be noted is that $5x + 7 \neq 0$

i.e. $x \neq -\frac{7}{5}$

$\Rightarrow |3x+5| - |7x+9| = |5x+7|$

Now in this case, the critical points are $-\frac{9}{7}$, $-\frac{7}{5}$ and $-\frac{5}{3}$. Plotting them on the number line, we get the intervals as shown below



Case I: $x < -\frac{9}{7}$

$\Rightarrow -(3x+5) + (7x+9) = -(5x+7)$

$\Rightarrow 9x = -11$

$\Rightarrow x = -\frac{11}{9}$ (which lies in the interval defined by $x < -\frac{9}{7}$)

Case II: $-\frac{9}{7} \leq x < -\frac{7}{5}$

$\Rightarrow -(3x+5) - (7x+9) = -(5x+7)$

$\Rightarrow -5x = 7$

$\Rightarrow x = -\frac{7}{5}$ (which is rejected because in the beginning we fixed $x \neq -\frac{7}{5}$. Also the point $-\frac{7}{5}$ does not lie in the interval $-\frac{9}{7} \leq x < -\frac{7}{5}$)

Case III : $-\frac{7}{5} \leq x < -\frac{5}{3}$

$\Rightarrow -(3x+5) - (7x+9) = +(5x+7)$

$\Rightarrow -15x = 21$

$\Rightarrow x = -\frac{7}{5}$ (which is rejected because in the beginning we fixed $x \neq -\frac{7}{5}$ due to domain restrictions.)

Case IV: $x \geq -\frac{5}{3}$

$\Rightarrow (3x+5) - (7x+9) = (5x+7)$

$\Rightarrow -9x = 11$

$\Rightarrow x = -\frac{11}{9}$ (Rejected because $-\frac{11}{9}$ does not belong to the interval $x \geq -\frac{5}{3}$)

Hence, the only solution is $x = -\frac{11}{9}$.

n)$2^{|x+1|} - |2^x - 1| = 2^x + 1$

The critical points lie at $|x+1| = 0$ and $|2^x - 1| = 0$. i.e at $x = -1$ and $x = 0$ respectively. So , dividing the number line as done previously

Case I: $x < -1$

$\Rightarrow 2^{-(x+1)} + (2^x - 1) = 2^x + 1$

$\Rightarrow 2^{-(x+1)} = 2$

$\Rightarrow -(x+1) = 1$

$\Rightarrow x = -2$ (Accepted)

Case II: $-1 \leq x < 0$

$\Rightarrow 2^{x+1} + 2^x - 1 = 2^x + 1$

$\Rightarrow 2^{x+1} = 2$

$\Rightarrow x + 1 = 1$

$\Rightarrow x = 0$ (Rejected as it does not lie in the interval $-1 \leq x < 0$)

Case III: $x \geq 0$

$\Rightarrow 2^{x+1} - (2^x - 1) = 2^x + 1$

$\Rightarrow 2^{x+1} = 2.2^x$ (which is true for all x belonging to the interval $x \geq 0$)

$\Rightarrow$The solution to Case III is $x \epsilon [0, \infty)$

Hence, the general solution to this problem is $x \epsilon \{-2\} \cup [0, \infty)$.

1.3.3 Problems for Practice

1.3.3.1 Subjective Problems

Q1: Solve for x

a) $|\sin x| = -\sin x + 2$

b) $\left|\dfrac{(x+3)^4(x-3)^3}{(x+1)(x-1)}\right| < -\dfrac{(x+3)^4(x-3)^3}{(x+1)(x-1)}$

c) $\left|x^2+3x+2\right| = -\left(x^2+3x+2\right)$

1.3.3.2 Single Answer MCQ's

Q1: The complete solution set of $\dfrac{1}{\left|\dfrac{4}{x}-10\right|} < 3$ is

a) $x\epsilon\mathbb{R} - \left(\dfrac{12}{31}, \dfrac{12}{29}\right)$

b) $x\epsilon \left(\dfrac{12}{31}, \dfrac{12}{29}\right) - \left\{\dfrac{2}{5}\right\}$

c) $x\epsilon\mathbb{R} - \{0\} - \left[\dfrac{12}{31}, \dfrac{12}{29}\right]$

d) None of these

Q2: The point in the co-ordinate plane satisfying $|7x+5| = -|3-3y|$ is

a)$(-2, 3)$

b)$\left(-\dfrac{5}{7}, 1\right)$

c) $(-5, -3)$

d) None of these

Q3: The complete solution set of $|\cot x| = \cot x + 2\sqrt{3}$ is

a)$\{x : x = n\pi + \dfrac{\pi}{3}\ , n\epsilon\mathbb{Z}\}$

b)$\{x : x = n\pi - \dfrac{\pi}{3}\ , n\epsilon\mathbb{Z}\}$

c)$\{x : x = n\pi + \dfrac{\pi}{6}\ , n\epsilon\mathbb{Z}\}$

d)$\{x : x = n\pi - \dfrac{\pi}{6}\ , n\epsilon\mathbb{Z}\}$

Q4: The area bounded by the curves given by $|x| + |2y| = 2$ is

a) 2 units

b) 4 units

c) 6 units

d) 8 units

1.3.3.3 Multiple answer MCQ's

Q1: The solution set of $2|\tan x| = \tan x + \sqrt{3}$ contains

a)$\{x : x = n\pi + \frac{\pi}{3}\ ,\ n\epsilon Z\}$

b)$\{x : x = n\pi - \frac{\pi}{3}\ ,\ n\epsilon Z\}$

c)$\{x : x = n\pi + \frac{\pi}{6}\ ,\ n\epsilon Z\}$

d)$\{x : x = n\pi - \frac{\pi}{6}\ ,\ n\epsilon Z\}$

1.3.3.4 Matrix Match Type Problems

Matrix 1: Under Column I , some equations are given . Under Column II, some solutions satisfying some of the equations are given. Match the entry in Column I with the solution satisfying it in Column II.{Note: [ ] is the Greatest Integer Function}

Column I	**Column II**
(P)$\left\|\sin^{-1} x\right\| = \sin^{-1} x + \frac{\pi}{6}$	**(A)**-$\frac{1}{2}$
(Q)$\ln\|x\| = \|\ln x\|$	**(B)**1
(R)$\|[x]\| = [\|x\|]$	**(C)**$\frac{3}{2}$
(S)$\left\|x^2 - 3x + 2\right\| > x^2 - 3x + 2$	**(D)**2

1.3.3.5 Linked Comprehension Type Problems

Comprehension 1: Consider the equation $y = \left|\left(x^4 - 9\right) - \left(x^2 - 3\right)\right| - \left|x^4 - 9\right| + \left|x^2 - 3\right|$. The sign of y depends on the value of x. Read the following questions based on this information and answer them carefully.

Q1: The value(s) of x for which $y > 0$ is/are :

a)$x\epsilon R$

b)$x\epsilon\left(-\sqrt{3}, \sqrt{3}\right)$

c)$x\epsilon\left(-\sqrt[4]{3}, \sqrt{3}\right)$

d)None of these

Q2: The value(s) of x for which $y \le 0$ is/are :

a)$x\epsilon R$

b)$x\epsilon\left(-\sqrt{3}, \sqrt{3}\right)$

c)$x\epsilon\left(-\sqrt[4]{3}, \sqrt{3}\right)$

d)None of these

Comprehension 2: The first 3 terms of an A.P. are $|3x|$, $|3x - 1|$and $|3x + 1|$. Based on this information, give the answers to the questions below

Q1: The value of x is

a) 1

b) 2

c) -1

d) None of these

Q2: The 10th term of the A.P. is

a) $\frac{1}{4}$

b) $\frac{5}{4}$

c) $\frac{19}{4}$

d) None of these

Q3: The sum of first 15 terms of the series is

a) $\frac{225}{4}$

b) $\frac{359}{4}$

c) $\frac{433}{4}$

d) None of these

Comprehension 3: A function $f_n(x)$ is defined for all $n\epsilon N$ and $f_{n+m}(x)$ is defined as $f_{n+m}(x) = f_n(f_m(x))$ where $f_1(x) = \frac{2x-1}{x+1}$ for $x\epsilon R - \{-1\}$. Using this definition , $f_2(x) = f_1(f_1(x)) = \frac{x-1}{x}$ for $x\epsilon R - \{-1, 0\}$. Similarly $f_3(x) = f_1(f_2(x)) = \frac{2-x}{1-2x}$ for $x\epsilon R - \left\{-1, 0, \frac{1}{2}\right\}$and so on. Based on the information, answer the questions below.

Q1: The domain of definition of $g(x) = |\ln(f_{34}(x))|$ is

a) $(-\infty, 1) - \left\{-1, 0, \frac{1}{2}\right\}$

b) $(-\infty, 1] - \left\{-1, 0, \frac{1}{2}\right\}$

c) $(1, \infty) - \{2\}$

d) None of these

Q2: The complete solution set of $|f_{71}(x)| > \left|\frac{1}{f_{73}(x)}\right|$ is

a) $(-1, 1)$

b) $\mathbb{R} - \{2\} - [-1, 1]$

c) $\mathbb{R} - \{-1, 1, 2\}$

d) None of these

Q3: The values of x for which $|f_{56}(x)| > f_{56}(x)$ belong to the interval

a) $\left(\frac{1}{2}, 2\right)$

b) $(2, \infty)$

c) $(0, 1)$

d) None of these

1.3.3.6 Hints and Solutions

Subjective Questions

Q1 a)$x = 2n\pi + \dfrac{\pi}{2}$, $n \epsilon Z$

{Hint: The solution of the equation is all values of x which satisfy the equation , $\sin x = 1$}

b) No Solution i.e $x \epsilon \phi$

{Hint: It may be observed that $|y| < -y$ is never true . Even $|y| < y$ also gives no solution.}

c) $x \epsilon [-3, -2]$

{Hint: $|y| = -y \Rightarrow y \leq 0$ }

Single Answer MCQ's

Q1) C

{Hint: $\dfrac{1}{\left|\dfrac{4}{x} - 10\right|} < 3$

$\Rightarrow \left|\dfrac{4}{x} - 10\right| > \dfrac{1}{3}$

Case I : $\dfrac{4}{x} - 10 > \dfrac{1}{3}$

$\Rightarrow \dfrac{4}{x} > \dfrac{31}{3}$

$\Rightarrow \dfrac{1}{x} > \dfrac{31}{12}$

$\Rightarrow 0 < x < \dfrac{12}{31}$

Case II: $\dfrac{4}{x} - 10 < -\dfrac{1}{3}$

$\Rightarrow \dfrac{4}{x} < \dfrac{29}{3}$

$\Rightarrow \dfrac{1}{x} < \dfrac{29}{12}$

$\Rightarrow \dfrac{1}{x} < 0$ or $0 < \dfrac{1}{x} < \dfrac{29}{12}$

$\Rightarrow x < 0$ or $x > \dfrac{12}{29}$

Combining all cases, we get $x \epsilon (-\infty, 0) \cup \left(0, \dfrac{12}{31}\right) \cup \left(\dfrac{12}{29}, \infty\right)$ i.e. $x \epsilon \mathbb{R} - \{0\} - \left[\dfrac{12}{31}, \dfrac{12}{29}\right]$}

Q2) B

Q3) D

Q4) B

Multiple Answer MCQ's

Q1) A,D

Matrix Match Type Problems

Matrix 1:

$P \rightarrow A$

$Q \rightarrow B, C, D$

$R \rightarrow B, C, D$

$S \rightarrow C$

Linked Comprehension Type Problems

Comprehension 1: Q1) D Q2)A

{Hint:a) $y > 0$

$\Rightarrow |(x^4 - 9) - (x^2 - 3)| - |x^4 - 9| + |x^2 - 3| > 0$

$\Rightarrow \left|\left(x^4-9\right)-\left(x^2-3\right)\right| > \left|x^4-9\right|-\left|x^2-3\right|$

Now we know that $|x-y| \geqslant |x|-|y|$ {Equality occurs when x and y are of same sign . i.e. $xy \geqslant 0$ and $|x| \geqslant |y|$}

$\Rightarrow \left|\left(x^4-9\right)-\left(x^2-3\right)\right| = \left|x^4-9\right|-\left|x^2-3\right|$ when $\left(x^4-9\right)\left(x^2-3\right) \geqslant 0$ and $\left|x^4-9\right| \geqslant \left|x^2-3\right|$

i.e when $\left(x^2-3\right)^2\left(x^2+3\right) \geqslant 0$ and $\left|x^2-3\right|\left(x^2+3\right) \geqslant \left|x^2-3\right|$

i.e when $\left(x^2-3\right)^2\left(x^2+3\right) \geqslant 0$ and $\left|x^2-3\right|\left(x^2+2\right) \geqslant 0$ which is true for all x.

i.e $\forall x \epsilon \mathbb{R}$, y is identically equal to zero.

Hence $y > 0$ has no solution

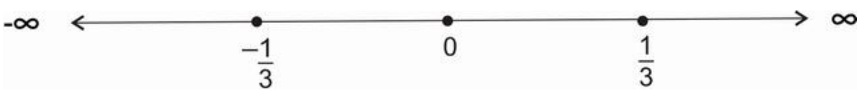
and $y \leqslant 0$ is true for all $x \epsilon \mathbb{R}$. i.e (b)$\rightarrow$A}

Comprehension 2: Q1) D Q2)C Q3)A

{Hint: a)$|3x|$, $|3x-1|$and $|3x+1|$are in A.P.

$\Rightarrow 2 \times |3x-1| = |3x| + |3x+1|$

The critical points in this case are $\left\{-\frac{1}{3}, 0, \frac{1}{3}\right\}$



Case I : $x \epsilon \left(-\infty, -\frac{1}{3}\right)$

In this case equation becomes

$2 \times -(3x-1) = -3x-(3x+1)$

$\Rightarrow -6x+2 = -6x-1$ which gives no solution

Case II: $x \epsilon \left[-\frac{1}{3}, 0\right)$

In this case , the equation becomes

$2 \times (3x-1) = -3x-(3x+1)$

$\Rightarrow 6x-2 = -6x-1$

$\Rightarrow x = \frac{1}{12}$ (which is accepted as $\frac{1}{12} \epsilon \left[-\frac{1}{3}, 0\right)$)

Case III : $x \epsilon \left[0, \frac{1}{3}\right)$

In this case , the equation becomes

$2 \times (3x-1) = 3x-(3x+1)$

$\Rightarrow 6x-2 = -1$

$\Rightarrow x = \frac{1}{6}$ (which is rejected as $\frac{1}{6} \notin \left[0, \frac{1}{3}\right)$

Case IV :$x \epsilon \left[\frac{1}{3}, \infty\right)$

In this case, the equation becomes

$2 \times (3x-1) = 3x+(3x+1)$

i.e $6x-2 = 6x+1$. Hence no solution in this case also.

Combining all the cases, we get $x = \frac{1}{12}$

b) Substituting the value of x , the series becomes $\left\{\frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \cdots\cdots\cdots\right\}$

Hence, the series has first term $a = \frac{1}{4}$ and common difference $d = \frac{1}{2}$.

$T_{10} = a + (10-1)\,d = \frac{1}{4} + 9 \times \frac{1}{2} = \frac{19}{4}$

c) The sum of first 15 terms of the series is

$S_{15} = \frac{n}{2}\left(2a+(n-1)\,d\right) = \frac{15}{2}\left(2 \times \frac{1}{4} + 9 \times \frac{1}{2}\right) = \frac{15}{2} \times \frac{15}{2} = \frac{225}{4}$}

Comprehension 3: Q1) A Q2)B Q3)C

{Hint: $f_1(x) = \dfrac{2x-1}{x+1}$ for $x\epsilon R - \{-1\}$

$f_2(x) = f_1(f_1(x)) = \dfrac{x-1}{x}$ for $x\epsilon R - \{-1,0\}$

$f_3(x) = f_1(f_2(x)) = \dfrac{2-x}{1-2x}$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2}\right\}$

On simlar lines, we can find $f_4(x) = f_1(f_3(x)) = \dfrac{1}{1-x}$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1\right\}$

$f_5(x) = f_1(f_4(x)) = \dfrac{1+x}{2-x}$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

$f_6(x) = f_1(f_5(x)) = x$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

It may be noted that this function repeats itself after a gap of 6 .i.e $f_{6n+r}(x) = f_6(f_6...ntimes(f_r(x)))) = f_r(x)$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

Q1: $f_{34}(x) = f_4(x)$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

$\Rightarrow f_{34}(x) = \dfrac{1}{1-x}$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

$\Rightarrow g(x) = |\ln(f_{34}(x))|$ has domain given by

$\dfrac{1}{1-x} > 0$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

i.e $x < 1$ and $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

i.e $x\epsilon \left((-\infty,1) - \left\{-1,0,\dfrac{1}{2}\right\}\right)$

Q2: $|f_{71}(x)| > \left|\dfrac{1}{f_{73}(x)}\right|$

i.e we have to solve $|f_5(x)| > \left|\dfrac{1}{f_1(x)}\right|$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

i.e $\left|\dfrac{1+x}{2-x}\right| > \left|\dfrac{1}{\dfrac{2x-1}{x+1}}\right|$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

i.e $|2x-1| > |2-x|$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

Squaring both sides we get

$4x^2 - 4x + 1 > 4 + x^2 - 4x$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

i.e. $3x^2 > 3$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

$\Rightarrow x\epsilon\left(\mathbb{R} - \{2\} - [-1,1]\right)$

Q3: $|f_{56}(x)| > f_{56}(x)$

$\Rightarrow |f_2(x)| > f_2(x)$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

$\Rightarrow f_2(x) < 0$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

$\Rightarrow \dfrac{x-1}{x} < 0$ for $x\epsilon R - \left\{-1,0,\dfrac{1}{2},1,2\right\}$

$\Rightarrow x\epsilon\left((0,1) - \left\{\dfrac{1}{2}\right\}\right)$

These values are a substet of $(0,1)$.}

Chapter 2

Functions (Preliminaries II)

Functions are one of the most important areas of mathematics because they lie at the heart of much of mathematical analysis.

2.1 Domain of Definition and Range

Q: For the function $f(x) = \dfrac{x}{x+2}$, find $f(x+3)$, $f(3x)$,$3f(x)$, $3f(3x+3)$, $f\left(x^3\right)$, $(f(x))^3$ with their respective domains of definition.

Sol: $f(x+3) = \dfrac{x+3}{x+5}$ where $x+5 \neq 0$ i.e $D_{f(x+3)} = \mathbb{R} - \{-5\}$

$f(3x) = \dfrac{3x}{3x+2}$ where $3x+2 \neq 0$ i.e. $D_{f(3x)} = \mathbb{R} - \left\{-\dfrac{2}{3}\right\}$

$3f(x) = \dfrac{3x}{x+2}$ where $x+2 \neq 0$ i.e. $D_{3f(x)} = \mathbb{R} - \{-2\}$

$3f(3x+3) = \dfrac{3x}{3x+5}$ where $3x+5 \neq 0$ i.e. $D_{3f(3x+3)} = \mathbb{R} - \left\{-\dfrac{5}{3}\right\}$

$f\left(x^3\right) = \dfrac{x^3}{x^3+2}$ where $x^3+2 \neq 0$ i.e. $D_{f\left(x^3\right)} = \mathbb{R} - \{-\sqrt[3]{2}\}$

$(f(x))^3 = \left(\dfrac{x}{x+2}\right)^3$ where $x+2 \neq 0$ i.e. $D_{(f(x))^3} = \mathbb{R} - \{-2\}$

Q: Find the domains of definition of the following functions:

a) $\sqrt{2-x} - \sqrt[3]{x} + \sqrt[4]{2+x}$
b) $\sqrt{x^2+x-2}$
c) $\dfrac{\sqrt{x^2-3x+2}}{\sqrt{4-x^2}}$
d) $\sqrt{\tan x}$
e) $\sqrt{-\sin x - \dfrac{1}{2}}$
f) $\sqrt{1 - cosecx}$
g) $\log\left(\dfrac{x^3\left(x^2-4\right)}{\left(x^2-1\right)(x+3)}\right)$
h) $\sqrt{\log x}$
i) $\sqrt{\log_x (2-x)}$

j) $\cos^{-1}\left(\frac{x}{2}-2\right)-\log\left(\frac{x}{2}-2\right)$

k) $\sqrt{[x]-x}$ where [ ] represents the greatest integer function

l) $\sqrt{\sin\left(\frac{[x]}{2}\pi\right)}$

m) $\frac{1}{\sqrt{|x|-x}}$

2.1.1 Problems for Practice

2.1.1.1 Subjective Problems

Q1: (i) Prove that for the logarithmic function $y=\ln|x|$, if the argument takes values in a G.P. , then the corresponding values of the function y are in A.P.

(ii)Prove that for the exponential function $y=e^x$, if the argument takes values in a A.P.. , then the corresponding values of the function y are in G.P.

2.1.1.2 Hints and Solutions

Subjective Problems

Q1 : {Hint: (i) Let the arguments be defined by the general term $x_n = ar^{n-1}$. Then the value of the function corresponding to it will be $y_n=\ln\left|ar^{n-1}\right|=\ln|a|+(n-1)\ln|r|$. This means that two consecutive terms y_n and y_{n-1} differ by a constant .i.e. $y_n-y_{n-1}=\ln|r|$. Hence , these terms are in A.P.

(ii)Proceeding in a similar manner , if we take the argument as a general term of an A.P. then the corresponding values of the function will have a constant common difference.}

2.2 Periodicity

2.2.1 Problems for practice

2.2.1.1 Subjective Problems

Q1: Find the range of the function defined on **R** given by

$$f(x)=\begin{cases} e^{x-n} & ;n\leqslant x\leqslant n+\frac{2}{3} \\ \sin\left(\frac{3\pi}{2}x\right) & ;n+\frac{2}{3}<x<n+1 \end{cases}$$

where $n\epsilon Z$.

2.3 Injectivity, Surjectivity and Bijectivity

2.3.1 Problems for Practice

2.3.1.1 Subjective Problems

Q1: Check whether the function $f:(-\infty,-3)\to\left[-\frac{1}{4},0\right]$ given by $f(x)=\frac{x+1}{x^2+2x+5}$ is bijective in its domain or not. Given that $f(x)$ is strictly decreasing in the domain.

2.3.1.2 Single Answer MCQ's

Q1: A function $f:R\to[1,2]$ given by $f(x)=\log_3(9-6|\cos x|)$ is

a) One-one only
b) Onto only
c) One-one and onto
d) None of these

Q2: Let $f : [0, \infty) \to S$ defined as $f(x) = 2^{x(1-x)}$ be onto, then the set S is

a) $[0, \infty)$
b) $[1, \infty)$
c) $\left[2^{-\frac{1}{4}}, 1\right]$
d) $\left[2^{-\frac{1}{4}}, \infty\right)$

2.3.1.3 Hints and Solutions

Single Answer MCQ's

Q1: B
Q2: D

2.4 Inverse of a function

2.4.1 Problems for practice

2.4.1.1 Subjective Problems

Q1: Check whether the function $f : (-\infty, 0) \to (0, \infty)$ given by $f(x) = \dfrac{1}{\sqrt{|x| - x}}$ is one-one and onto. Find it's inverse if possible.

Q2: A function $f : R - \{-2, 2\} \to \left(-\infty, \dfrac{5}{2}\right] \cup (3, \infty)$ given by $f(x) = 3 + \dfrac{1}{|x| - 2}$. Divide the domain into minimum number of suitable subsets so that the function becomes invertible on each individual domain. Also find the inverse on each of these domains.

2.4.1.2 Single Answer MCQ's

Q1: A function on $f : [2, 4) \to (-\infty, 0]$ is given by $f(x) = -2 + \log_2\left(-x^2 + 4x\right)$. Then $f^{-1}(x)$ is given by

a) $2 \pm 2\sqrt{1 - 2^x}$
b) $2 - 2\sqrt{1 - 2^x}$
c)$2 + 2\sqrt{4 - 2^x}$
d) None of these

2.4.1.3 Hints and Solutions

Single Answer MCQ's

Q1: D

Chapter 3

Logarithms Worksheet

3.1 Logarithmic Differentiation(Revisiting Logarithms)

Let us first learn the basic definition of logarithms. First of all , we have a exponential equation of the form $a^\alpha = b$. This equation can be written in the logarithmic form as$\alpha = log_a b$. So, we understand that logarithms is just another way of writing an equation which has exponents involved in it.

Logarithms have few basic properties :

- $\log_a p = \dfrac{\log_b p}{\log_b a}$ (Base Change Formula)
- $\log_a pq = \log_a p + \log_a q$
  $\log_a p^n = n \log_a p$

Now suppose, we have a function of the form,

$y = f(x) = [u(x)]^{v(x)}$.

By taking logarithm (to base e) the above may be rewritten as

$\ln y = v(x) \ln[u(x)]$

Using chain rule we may dierentiate this to get

$\dfrac{1}{y}.\dfrac{dy}{dx} = v(x).\dfrac{1}{u(x)}.u'(x) + v'(x)\ln[u(x)]$

The main point to be noted in this method is that f (x) and u(x) must always be positive as otherwise their logarithms are not defined.

Q: Differentiate the following functions w.r.t. x

a) $f(x) = \sqrt{\dfrac{(x-3)(x^2+8)}{x^2+3x+4}}$

b) $f(x) = x^{\sin x}, x > 0$

c) $f(x) = cosx.cos2x.cos3x$

d) $f(x) = (\ln x)^{\cos x}$

e) $f(x) = (\ln x)^x + x^{\ln x}$

f) $f(x) = (\sin x)^x + \sin^{-1}\sqrt{x}$

Q1: Find the domains of definition of the following functions:

a)$\log\left(\dfrac{x^3\left(x^2-4\right)}{\left(x^2-1\right)(x+3)}\right)$

b)$\sqrt{\log x}$

c) $\sqrt{\log_x(2-x)}$

Q2 Matrix 1: Under Column I , some equations are given . Under Column II, some solutions satisfying some of the equations are given. Match the entry in Column I with the solution satisfying it in Column II.{Note: [ ] is the Greatest Integer Function}

Column I	**Column II**
(P) $\left\|\sin^{-1} x\right\| = \sin^{-1} x + \frac{\pi}{6}$	**(A)**-$\frac{1}{2}$
(Q) $\ln\|x\| = \|\ln x\|$	**(B)** 1
(R) $\|[x]\| = [\|x\|]$	**(C)** $\frac{3}{2}$
(S) $\left\|x^2 - 3x + 2\right\| > x^2 - 3x + 2$	**(D)** 2

Q3 Comprehension 1: A function $f_n(x)$ is defined for all $n \epsilon N$ and $f_{n+m}(x)$ is defined as $f_{n+m}(x) = f_n(f_m(x))$ where $f_1(x) = \frac{2x-1}{x+1}$ for $x \epsilon R - \{-1\}$. Using this definition , $f_2(x) = f_1(f_1(x)) = \frac{x-1}{x}$ for $x \epsilon R - \{-1, 0\}$. Similarly $f_3(x) = f_1(f_2(x)) = \frac{2-x}{1-2x}$ for $x \epsilon R - \left\{-1, 0, \frac{1}{2}\right\}$and so on. Based on the information, answer the questions below.

1: The domain of definition of $g(x) = |\ln(f_{34}(x))|$ is

a) $(-\infty, 1) - \left\{-1, 0, \frac{1}{2}\right\}$

b) $(-\infty, 1] - \left\{-1, 0, \frac{1}{2}\right\}$

c) $(1, \infty) - \{2\}$

d) None of these

2: The complete solution set of $|f_{71}(x)| > \left|\frac{1}{f_{73}(x)}\right|$ is

a) $(-1, 1)$

b) $\mathbb{R} - \{2\} - [-1, 1]$

c) $\mathbb{R} - \{-1, 1, 2\}$

d) None of these

3: The values of x for which $|f_{56}(x)| > f_{56}(x)$ belong to the interval

a) $\left(\frac{1}{2}, 2\right)$

b) $(2, \infty)$

c) $(0, 1)$

d) None of these

Q4: (i) Prove that for the logarithmic function $y = \ln|x|$, if the argument takes values in a G.P. , then the corresponding values of the function y are in A.P.

(ii)Prove that for the exponential function $y = e^x$, if the argument takes values in a A.P.. , then the corresponding values of the function y are in G.P.

Previous Years Questions

Advanced (Logarithms)

Numerical

2022

Q1: The product of all positive real values of x satisfying the equation $x^{\left(16(\log_5 x)^3 - 68\log_5 x\right)} = 5^{-16}$ is __________.

2018

Q2: The value of $\left((\log_2 9)^2\right)^{\frac{1}{\log_2(\log_2 9)}} \times \left(\sqrt{7}\right)^{\frac{1}{\log_4 7}}$

Previous Years Questions

Mains (Logarithms)

MCQ (Single Correct Answer)

2026

Q1: The sum of all the real solutions of the equation $\log_{(x+3)}\left(6x^2+28x+30\right)=5-2\log_{(6x+10)}\left(x^2+6x+9\right)$ is equal to :

(a) 1
(b) 4
(c) 0
(d) 2

2025

Q1: The product of all solutions of the equation $e^{5(\log_e x)^2+3}=x^8, x>0$, is :

(a) e^2
(b) e
(c) $e^{6/5}$
(d) $e^{8/5}$

2023

Q1: The number of integral solutions x of $\log_{\left(x+\frac{7}{2}\right)}\left(\frac{x-7}{2x-3}\right)^2 \geq 0$ is :

(a) 8
(b) 7
(c) 5
(d) 6

Q2: If the solution of the equation $\log_{\cos x} \cot x + 4\log_{\sin x} \tan x = 1, x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha + \sqrt{\beta}}{2}\right)$, where α, β are integers, then $\alpha + \beta$ is equal to:

(a) 3 (b) 6

(c) 4 (d) 5

Numerical

2023

Q1: Let a, b, c be three distinct positive real numbers such that $(2a)^{\log_e a} = (bc)^{\log_e b}$ and $b^{\log_e 2} = a^{\log_e c}$. Then $6a + 5bc$ is equal to __________.

Q2: Let $S = \left\{\alpha : \log_2\left(9^{2\alpha-4} + 13\right) - \log_2\left(\frac{5}{2}.3^{2\alpha-4} + 1\right) = 2\right\}$. Then the maximum value of β for which the equation $x^2 - 2\left(\sum_{\alpha \in S} \alpha\right)^2 x + \sum_{\alpha \in S} (\alpha + 1)^2 \beta = 0$ has real roots, is __________.

2021

Q1: The number of solutions of the equation:

$\log_{(x+1)}\left(2x^2 + 7x + 5\right) + \log_{(2x+5)}(x+1)^2 - 4 = 0, x > 0$, is __________.

Q2: The number of solutions of the equation $\log_4(x-1) = \log_2(x-3)$ is: __________.

2020

Q1: The number of distinct solutions of the equation

$\log_{\frac{1}{2}} |\sin x| = 2 - \log_{\frac{1}{2}} |\cos x|$ in the interval $[0, 2\pi]$, is __________.

Assignment

Logarithms

Level I

Part I

Q1: Prove that $\log_{34} 5$ lies in the open interval $\left(\frac{1}{3}, \frac{1}{2}\right)$.

Q2: Show that $\sum_{r=2}^{43} \frac{1}{\log_r n} = \log_n (43)!$.

Q3: Given $n^4 < 10^n$ for a fixed positive integer $n \geq 2$, prove that $(n+1)^4 < 10^{n+1}$.

Q4: If $\frac{\log x}{y-z} = \frac{\log y}{z-x} = \frac{\log z}{x-y}$, prove that $x^{y+z}.y^{z+x}.z^{x+y} = 1$. Also prove that $x^{y+z} + y^{z+x} + z^{x+y} \geq 3$.

Q5: Prove that $(yz)^{\log \frac{y}{z}}.(zx)^{\log \frac{z}{x}}.(xy)^{\log \frac{x}{y}} = 1$.

Q6: Solve $\frac{6}{5} a^{\log_a x . \log_{10} a . \log_a 5} - 3^{\log_{10} \frac{x}{10}} = 9^{\log_{100} x + \log_4 2}$.

Q7: Solve for x : $\log_{0.1} \sin 2x + \log_{10} \cos x = \log_{10} 7$.

Q8: Solve $\log_{\cos x} \sin x + \log_{\sin x} \cos x = 2$.

Q9: If $1 + \log_2 (1 + \cos 2x) > \log_4 2$ then find x.

Part II

Exercise

Q1: Show that $\log_{300} 4$ lies in the interval $\left(\frac{1}{5}, \frac{1}{4}\right)$.

Q2: If $x = \log_{2a} a, y = \log_{3a} 2a$ and $z = \log_{4a} 3a$ then prove that $xyz + 1 = 2yz$.

Q3: If $x = \log_a (bc)$, $y = \log_b (ca)$ and $z = \log_c (ab)$, prove that $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} = 1$.

Q4: Prove that $\log_a (bc).\log_b (ca).\log_c (ab) = 2 + l\, og_a (bc) + \log_b (ca) + \log_c (ab)$.

Q5: If $\frac{x(y+z-x)}{\log x} = \frac{y(z+x-y)}{\log y} = \frac{z(x+y-z)}{\log z}$, prove that $y^z.z^y = z^x.x^z = x^y.y^x$.

Q6: If $\frac{\log x}{y-z} = \frac{\log y}{z-x} = \frac{\log z}{x-y}$, prove that $x^x + y^y + z^z \geq 3$.

Q7: Prove that $x^{\log y - \log z}.y^{\log z - \log x}.z^{\log x - \log y} = 1$.

Q8: Solve: $\log_{10} \sin x + \log_{100} (8\cos x) = \frac{1}{4} \log_{\sqrt{10}} 3$.

Q9: Solve: $\log_2 \tan x + \log_2 \tan 2x = 0, 0 \leq x \leq 2\pi$.

Q10: Solve $1 + \log_4 \sin x + 2\log_{16} \cos x > 0$.

Fill in the blanks

11. $\log_3 \log_2 \log_{\sqrt{3}} 81 =$ __________.

12. If $\log x + \log y = \log(x + y)$ then y as a function of x is given by y __________.

13. The sum of the series $\sum_{r=0}^{\infty} \left| \log_3 \left(e^2\right)^r \right|^{-1}$ = __________.

14. If $a^2 + b^2 = 23ab$ then $\log \frac{a+b}{5}$ is the AM of __________ and __________.

15. If $0 < x < 1$ then $\sum_{r=0}^{\infty} \log\left(1 + x^2\right)^r = \log($__________$)$.

16. If $\log 2 = 0.30103$, $\log 3 = 0.4771213$ then the number of digits in the number equal to $3^{12} \times 2^{8}$ is __________.

17. If $\log 3 = 0.4771213$ then the number of zeros between the decimal point and the first significant digit in the value of 9^{-16} is __________.

18.If $\log_2\left(\sin\frac{x}{2}\right) < -1$ then the set of general values of x is __________.

Objective Questions

19. $2\log a + 2\log a^2 + 2\log a^3 + ... + 2\log a^n = 55.\log a$ then the value of n is

(a) 10 (b) 11

(c) 20 (d) 5

20. If $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b}$ then the value of xyz is

(a) 1 (b) 3

(c) 2 (d) none of these

21. $a^{\log_a b \times \log_b c \times \log_c d}$ is equal to

(a) b (c) c

(c) d (d) none of these

22. If a, b, c are positive numbers $(\neq 1)$ in GP and $x > 0$ but $\neq 1$, then the value of $\log_b x\left(\log_x a + \log_x c\right)$ is equal to

(a) 1 (b) 2

(c) 3 (d) none of these

Part III

Chapter Test

1. In each of the following, fill in the blank so that the resulting sentence becomes true.

(a) If for a ΔABC, $\log_{0.1}\cos A > 1$ then $A \in$

(b) $\log\cos\frac{x}{2}+\log\cos\frac{x}{2^2}+\log\cos\frac{x}{2^3}+.....++\log\cos\frac{x}{2^n}+\log\sin\frac{x}{2^n}=\log(\underline{\qquad\qquad\qquad})$.

2. Prove that in an acute-angled ΔABC,

$\log\left(\tan A+\tan B+\tan C\right)=\log\tan A+\log\tan B+\log\tan C.$

Can this hold for any triangle ABC?

Q3: Prove that $4+\log_2\sin\frac{\pi}{5}+\log_2\sin\frac{2\pi}{5}+\log_2\sin\frac{3\pi}{5}+\log_2\sin\frac{4\pi}{5}=\log_2 5.$

4. Solve $1+\log_2\sin x+\log_2\sin 3x>0, 0\le x\le 2\pi.$

Level II

Objective Questions Type I [Only one correct answer]

Q1: If $\log_2 3 = a, \log_3 5 = b$ and $\log_7 2 = c$, then the logarithm of the number 63 to base 140 is

(a) $\dfrac{1+2ac}{2c+abc+1}$ (b) $\dfrac{1-2ac}{2c-abc-1}$

(c) $\dfrac{1-2ac}{2c+abc+1}$ (d) $\dfrac{1+2ac}{2c-abc-1}$

Q2: If $\log_5 120 + (x-3) - 2\log_5\left(1-5^{x-3}\right) = -\log_5\left(0.2-5^{x-4}\right)$, the x is

(a) 1 (b) 2

(c) 3 (d) 4

Q3: If $x_n > x_{n-1} > \ldots > x_2 > x_1 > 1$, then the value of $\log_{x_1}\log_{x_2}\log_{x_3}\ldots.\log_{x_n} x_n^{x_{n-1}^{\cdot^{\cdot^{x_1}}}}$, is

(a) 0 (b) 1

(c) 2 (d) undefined

Q4: The number $\log_{20} 3$ lies in

(a) $\left(\dfrac{1}{4},\dfrac{1}{3}\right)$ (b) $\left(\dfrac{1}{3},\dfrac{1}{2}\right)$

(c) $\left(\dfrac{1}{2},\dfrac{3}{4}\right)$ (d) $\left(\dfrac{3}{4},\dfrac{4}{5}\right)$

Q5: The number of real values of the parameter λ for which $(\log_{16} x)^2 - \log_{16} x + \log_{16}\lambda = 0$ with real coefficients will have exactly one solution is

(a) 1 (b) 2
(b) 3 (d) 4

Q6: If $y = 2^{\frac{1}{\log_x 4}}$, then x is equal to :

(a) $\sqrt{y}$ (b) y

(c) y^2

(d) y^3

Q7: If $\log_2 x + \log_2 y \geq 6$, then the least value of $x + y$ is

(a) 4

(b) 8

(c) 16

(d) 32

Q8: A rational number which is 50 times its own logarithm to the base 10 is

(a) 1

(b) 10

(c) 100

(d) 1000

Q9: If $x = \log_5(1000)$ and $y = \log_7(2058)$, then

(a) $x > y$

(b) $x < y$

(c) $x = y$

(d) none of these

Q10: If $f(x) = \ln\left(\frac{1+x}{1-x}\right)$, then

(a) $f(x_1).f(x_2) = f(x_1 + x_2)$

(b) $f(x+2) - 2f(x+1) + f(x) = 0$

(c) $f(x) + f(x+1) = f(x^2 + x)$

(d) $f(x_1) + f(x_2) = f\left(\frac{x_1 + x_2}{1 + x_1 x_2}\right)$

Q11: $\log_{10} 2, \log_{10}(2^x + 1), \log(2^x + 3)$ are in AP, then

(a) $x = 0$

(b) $x = 1$

(c) $x = \log_{10} 2$

(d) $x = \frac{1}{2}\log_2 5$

Q12: If $\log_2 \log_2 \log_4 256 + 2\log_{\sqrt{2}} 2$, then A is equal to

(a) 2

(b) 3

(c) 5

(d) 7

Q13. $7\log\left(\frac{16}{15}\right) + 5\log\left(\frac{25}{24}\right) + 3\log\left(\frac{81}{80}\right)$ is equal to

(a) 0

(b) 1

(c) $\log 2$ (d) $\log 3$

14. For $y=\log_a x$ to be defined $'a'$ must be

(a) any positive real numbers (b) any number

(c) $\geq e$ (d) any positive real number $\neq 1$

15. If $\log_{10} 3 = 0.477$, the number of digit in 3^{40} is

(a) 18 (b) 19

(c) 20 (d) 21

16. If $x=\log_3 5, y=\log_{17} 25$, which one of the following is correct?

(a) $x<y$ (b) $x=y$

(c) $x>y$ (d) None of these

17. The domain of the function $\sqrt{(\log_{0.5} x)}$ is

(a) $(1,\infty)$ (b) $(0,\infty)$

(c) $(0,1]$ (d) $(0.5,1)$

18. The number $\log_2 7$ is

(a) an integer (b) a rational number

(c) an irrational number (d) a prime number

19. The value of $\frac{1}{\log_2 n}+\frac{1}{\log_3 n}+....+\frac{1}{\log_{43} n}$ is

(a) $\frac{1}{\log_{43!} n}$ (b) $\frac{1}{\log_{43} n}$

(c) $\frac{1}{\log_{42} n}$ (d) $\frac{1}{\log_{43} n!}$

20. $\log_{10} \tan 1^o + \log_{10} 2^o + ... + \log_{10} \tan 89^o$ is equal to

(a) 0 (b) 1

(c) 27

(d) 81

21. If $\log_{12} 27 = a$, then $\log_6 16$ is equal to

(a) $2\left(\frac{3-a}{3+a}\right)$

(b) $3\left(\frac{3-a}{3+a}\right)$

(c) $4\left(\frac{3-a}{3+a}\right)$

(d) $5\left(\frac{3-a}{3+a}\right)$

22. $\log_7 \log_7 \sqrt{7\sqrt{\left(7\sqrt{7}\right)}}$ is equal to

(a) $3\log_2 7$

(b) $3\log_7 2$

(c) $1-3\log_7 2$

(d) $1-3\log_2 7$

23. If $\frac{\log x}{b-c} = \frac{\log y}{c-a} = \frac{\log z}{a-b}$, then $x^a y^b z^c$ is equal to

(a) xyz

(b) abc

(c) 0

(d) 1

24. $\frac{1}{1+\log_a bc} + \frac{1}{1+\log_b ca} + \frac{1}{1+\log_c ab}$ is equal to

(a) 0

(b) 1

(c) 2

(d) 3

25. If $(4)^{\log_9 3} + (9)^{\log_2 4} = (10)^{\log_x 83}$, then x is equal to

(a) 2

(b) 3

(c) 10

(d) 30

26. If x, y, z are in GP and $a^x = b^y = c^z$, then

(a) $\log_b a = \log_c b$

(b) $\log_c b = \log_a c$

(c) $\log_a c = \log_b a$

(d) $\log_a b = 2\log_a c$

27. If $\frac{x\left(y+z-x\right)}{\log x}=\frac{y(z+x-y)}{\log y}=\frac{z(x+y-z)}{\log z}$,then $x^y y^x = z^y y^z$ is equal to

(a) $z^x x^z$

(b) $x^z y^x$

(c) $x^y y^z$

(d) $x^x y^y$

28. If $\log_3 2, \log_3\left(2^x-5\right), \log_3\left(2^x-7/2\right)$ are in AP, then x is equal to

(a) 1

(b) 2

(c) 3

(d) 4

29. If $y=a^{\frac{1}{1-\log_a x}}$ and $z=a^{\frac{1}{1-\log_a y}}$, then x is equal to

(a) $a^{\frac{1}{1+\log_a z}}$

(b) $a^{\frac{1}{2+\log_a z}}$

(c) $a^{\frac{1}{1-\log_a z}}$

(d) $a^{\frac{1}{2-\log_a z}}$

30. If $\log_{\sqrt{8}} b = 3\,{}^1\!/_3$, then b is equal to

(a) 2

(b) 8

(c) 32

(d) 64

31. If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, then x lies in the interval

(a) $\left(-\infty,1\right)$

(b) $\left(1,2\right)$

(c) $\left(2,\infty\right)$

(d) none of these

32. The value of $3^{\log_4 5} - 5^{\log_4 3}$ is

(a) 0

(b) 1

(c) 2

(d) none of these

33. If $x^{18} = y^{21} = z^{28}$, then $3, 3\log_y x, 3\log_z y, 7\log_x z$ are in

(a) AP

(b) GP

(c) HP

(d) AGP

34.If $\dfrac{1}{\log_3 \pi}+\dfrac{1}{\log_4 \pi}>x,$ then x be

(a) 2 (b) 3

(d) π (d) none of these

35. If $\ln\left(\dfrac{a+b}{3}\right)=\left(\dfrac{\ln a+\ln b}{2}\right),$ then $\dfrac{a}{b}+\dfrac{b}{a}$ is equal to

(a) 1 (b) 3

(c) 5 (d) 7

36. If $\log_3\{5+4\log_3(x-1)\}=2,$ then x is equal to

(a) 2 (b) 4

(c) 8 (d) $\log_2 16$

37. If $2x^{\log_4 3}+3^{\log_4 x}=27,$ then x is equal to

(a) 2 (b) 4

(c) 8 (d) 16

38. The interval of x in which the inequality $5^{\frac{1}{4}\left(\log^2{}_5 x\right)}\geq 5x^{\frac{1}{5}(\log_5 x)}$

(a) $\left(0,5^{-2\sqrt{5}}\right]$ (b) $\left[5^{2\sqrt{5}},\infty\right)$

(c) both (a) and (b) (d) none of these

39. The solution set of the equation $\log_x 2\log_{2x} 2=\log_{4x} 2$ is

(a) $\left\{2^{-\sqrt{2}},2^{\sqrt{2}}\right\}$ (b) $\{1/2,2\}$

(c) $\left\{1/4,2^2\right\}$ (d) none of these

40. The least value of the expression $2\log_{10} x-\log_x 0.01$ is

(a) 2 (b) 4

(c) 6 (d) 8

41. The solution of the equation $\log_7 \log_5(\sqrt{x+5}+\sqrt{x})=0$ is

(a) 1 (b) 3

(c) 4 (d) 5

42. The number of solutions of $\log_4(x-1)=\log_2(x-3)$ is

(a) 3 (b) 1

(c) 2 (d) 0

Objective Questions Type II {One or more than one correct answer(s)}

Q1: It is given that $x=9$ is a solution of the equation $\ln\left(x^2+15a^2\right)-\ln(a-2)=\ln\left(\frac{8ax}{a-2}\right)$, then

(a) $a=3$ (b) $a=\frac{3}{5}$

(c) other solution is $x=3/5$ (d) other solution is $x=15$

Q2: The value of $\log\{\log_b a.\log_c b.\log_d c.\log_a d\}$ is

(a) 0 (b) $\log abcd$

(b) $\log 1$ (d) 1

Q3: If $\frac{\log_2 x}{4}=\frac{\log_2 y}{6}=\frac{\log_2 z}{3k}$ and $x^3y^2z=1$, then k is equal to

(a) -8 (b) -4

(c) 0 (d) $\log_2\left(\frac{1}{256}\right)$

Q4: If $\frac{\log a}{(b-c)}=\frac{\log b}{(c-a)}=\frac{\log c}{(a-b)}$, then $a^{b+c}\cdot b^{c+a}\cdot c^{a+b}$ is equal to

(a) 0 (b) 1

(c) $a+b+c$ (d) $\log_b a\cdot\log_c b\cdot\log_a c$

Q5: The expression $5^{\log_{1/5}(1/2)}+\log_{\sqrt{2}}\left(\frac{4}{\sqrt{7}+\sqrt{3}}\right)+\log_{1/2}\left(\frac{1}{10+2\sqrt{21}}\right)$ simplifies to

(a) 6

(b) 4

(c) $\sqrt{6\sqrt{6\sqrt{6\sqrt{6....\infty}}}}$

(d) $3^{\log_{1/3}\left(\frac{1}{6}\right)}$

Q6. If $\log_a x = \alpha, \log_b x = \beta, \log_c x = \gamma$ and $\log_d x = \delta, x \neq 1$ and $a,b,c,d \neq 0, >1$, then $\log_{abcd} x$ equals

(a) $\leq \dfrac{\alpha+\beta+\gamma+\delta}{16}$

(b) $\geq \dfrac{\alpha+\beta+\gamma+\delta}{16}$

(c) $\dfrac{1}{\alpha^{-1}+\beta^{-1}+\gamma^{-1}+\delta^{-1}}$

(d) $\dfrac{1}{\alpha\beta\gamma\delta}$

Q7: $\log_p \log_p \sqrt[p]{\sqrt[p]{\sqrt[p]{....\sqrt[p]{p}}}}$, $p > 0$ and $p \neq 1$, is equal to

(a) n

(b) –n

(c) $\dfrac{1}{n}$

(d) $\log_{1/p}\left(p^n\right)$

Q8: Sum of the roots of the equation $x+1 = 2\log_2\left(2^x+3\right) - 2\log_4(1980-2^{-x})$ is

(a) $\log_{11} 2$

(b) $\log_2 11$

(c) $\log_{11}(0.5)$

(d) $\log_{0.5}\left(\dfrac{1}{11}\right)$

Q9: If $a^x = b,\ b^y = c,\ c^z = a,\ x = \log_b a^{k_1},\ y = \log_c b^{k_2},\ z = \log_a c^{k_3}$, then $k_1k_2k_3$ is equal to

(a) 1

(b) abc

(c) (xyz)

(d) 0

Q10: If $\dfrac{\ln a}{b-c} = \dfrac{\ln b}{c-a} = \dfrac{\ln c}{a-b},\ a,b,c > 0$, then

(a) $a^{b+c}\cdot b^{c+a}\cdot c^{a+b} = 1$

(b) $a^{b+c}+b^{c+a}+c^{a+b} \geq 3$

(c) $a^{b+c}\cdot b^{c+a}\cdot c^{a+b} = 3$

(d) $a^{b+c}+b^{c+a}+c^{a+b} \geq 3(3)^{1/3}$

Q11: If $\dfrac{\ln a}{y-z}=\dfrac{\ln b}{z-x}=\dfrac{\ln c}{x-y}$, then

(a) $a^x \cdot b^y \cdot c^z = 1$

(b) $a^{y^2+yz+z^2} \cdot b^{z^2+zx+x^2} \cdot c^{x^2+xy+y^2} = 1$

(c) $a^{y+z} \cdot b^{z+x} \cdot c^{x+y} = 1$

(d) $abc = 1$

Q12: The solution of the equation $3^{\log_a x} + 3x^{\log_a 3} = 2$ is given by

(a) $a^{\log_3 a}$

(b) $(2/a)^{\log_3 2}$

(c) $a^{-\log_3 2}$

(d) $2^{-\log_3 a}$

Linked-Comprehension Type

Passage I

An equation of the form $2m\log_a f(x) = \log_a g(x), a > 0 < a \neq 1, m \in N$

$\begin{cases} \dfrac{f(x)>0}{f^{2m}(x)=g(x)} \end{cases}$ *is equivalent to the system*

1. The number of solutions of $2\log_e 2x = \log_e(7x-2-2x^2)$ is

(a) 1

(b) 2

(c) 3

(d) infinite

2. The number of solutions of $\ln 2x = 2\ln(4x-15)$ is

(a) 0

(b) 1

(c) 2

(d) infinite

3. The number of solutions of $\log(3x^2+x-2) = 3\log(3x-2)$ is

(a) 1

(b) 2

(c) 3

(d) 0

4. Solution set of the equation $\log_{(x^3+6)}(x^2-1) = \log_{(2x^2+5x)}(x^2-1)$ is

(a) {-2}

(b) {1}

(c) {3}

(d) {-2,1,3}

5. Solution set of the equation $\log(x-9)+2\log\sqrt{(2x-1)}=2$ is

(a) $\{\phi\}$ (b) {1}

(c) {2} (d) {13}

Passage II

Equations of the form (i) $f(\log_a x)=0, a>0, a\neq 1$ and (ii) $g(\log_x A)=0$, $A>0$, then Eq. (i) is equivalent to $f(t)=0$, where $t=\log_a x$.

If $t_1, t_2, t_3, \ldots, t_k$ are the roots of $f(t)=0$, then $\log_a x=t_1$, $\log_a x=t_2$, ... $\log_a x=t_k$ and Eq. (ii) is equivalent to $f(y)=0$, where $y=\log_x A$. If $y_1, y_2, y_3, \ldots, y_k$ are the roots of $f(y)=0$, then $\log_x A=y_1$, $\log_x A=y_2$, ..., $\log_x A=y_k$.

On the basis of above information, answer the following question

1. The number of solutions of the equation $\dfrac{1-2(\log x^2)^2}{\log x-2(\log x)^2}=1$ is

(a) 0 (b) 1

(c) 2 (d) infinite

2. The number of solutions of the equation $\log^3{}_x 10-6\log^2{}_x 10+11\log_x 10-6=0$ is

(a) 0 (b) 1

(c) 2 (d) 3

3. The solution set of $(\log_5 x)^2+\log_5 x+1=\dfrac{7}{\log_5 x-1}$ contains

(a) (1,3) (b) {1}

(c) {25} (d) {1,25}

4. The set of all x satisfying the equation $x^{\log_3 x^2+(\log_3 x)^2-10}=\dfrac{1}{x^2}$ is

(a) {1,9} (b) $\{9, \dfrac{1}{81}\}$

(c) $\{1,4,\frac{1}{81}\}$

(d) $\{1,9,\frac{1}{81}\}$

5. If $\frac{(\ln x)^2-3\ln x+3}{\ln x-1}<1$, then x belongs to

(a) (0,e)

(b) (1,e)

(c) (1,2e)

(d) (0,3e)

Matrix-Match Type

Given below are Matching Questions, with two columns (each having some items) each. Each item of Column I has to be matched with the items of Column II, by encircling the correct match(es).

1. Observe the following columns :

Column I	Column II
(A)If $\log_{34}5$ lies in the interval (a,b), then	(P) [10a+10b]=8, where [.] denotes the greatest integer function
(B) If $\log_{300}4$ lies in the interval (a,b), then	(Q) (10a+10b)=5, where (.) denotes the greatest integer function
(C) If $\log_{400}3$ lies in the interval (a,b), then	(R) [6b-3a]=2, where [.] denotes the greatest integer function
	(S) [10a+10b]=3, where [.] denotes the greatest integer function
	(T) (6b-3a)=1, where (.) denotes the greatest integer function

(A) P Q R S T

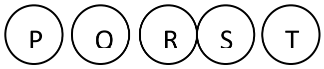
(B) P Q R S T

(C) P Q R S T

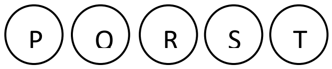
2. Observe the following columns :

Column I	Column II
(A) The solution set of $\log_{100}\lvert x+y\rvert=\frac{1}{2}$, $\log_{10}y-\log_{10}\lvert x\rvert=\log_{100}4$ is	(P) $\{\sqrt{2},2\}$
(B) The solution set of $4\log^2{}_2 x+1=2\log_2 y$ and $\log_2 x^2\geq\log_2 y$	(Q) {1,1}
(C) The solution set of $\log_4 x-\log_2 y=0$ and $x^2-5y^2+4=0$	(R) {-10,20}
	(S){4,2}
	(T) $\left\{\frac{10}{3},\frac{20}{3}\right\}$

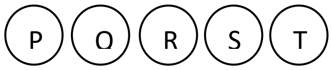
(A)



(B)



(C)



Level III

Q1: Prove that $\dfrac{\log_a x}{\log_{ab} x} = 1 + \log_a b$, for permissible values of letters involved in the result.

Q2: Find the value of x for which $\log(x+3)+\log(x-3)=\log 27$.

Problem 1: If $\ln 2 . \log_b 625 = \log_{10} 16 . \ln 10$, then find the value of b.

Problem 2: Find the value of $(0.16)^{\log_{2.5}\left(\frac{1}{4}\right)}$.

Problem 3: If $n = 1000!$, evaluate $\dfrac{1}{\log_2 n} + \dfrac{1}{\log_3 n} + ... + \dfrac{1}{\log_{1000} n}$.

Problem 4: Solve for x, $\log_7 \log_5\left(\sqrt{x+5}+\sqrt{x}\right)=0$

Objective

Problem 1: The sum of the series $\dfrac{1}{\log_2 a} + \dfrac{1}{\log_4 a} + \dfrac{1}{\log_8 a} + ...$ upto n terms is

(a) $\dfrac{n(n+1)}{2}\log_a 2$ (b) $\dfrac{n}{2}\log_a 2$

(c) $\dfrac{(n+1)}{2}\log_a 2$ (d) $\dfrac{n(n+1)(2n+1)}{6}\log_a 2$

Problem 2: The value of 'b' satisfying $\log_{\sqrt{8}} b = 3\, {}^{1}\!/_{3}$ is

(a) 30 (b) 31

(c) 32 (d) 33

Problem 3: The value of x, satisfying $\log_a\left(1+\log_b\left\{1+\log_c\left(1+\log_p x\right)\right\}\right)=0$, is

(a) 4 (b) 3

(c) 2 (d) 1

Problem 4: For $0 < x < 1$, the value of $\log(1+x)+\log\left(1+x^2\right)+\log\left(1+x^4\right)+.....$ to ∞ is

(a) $\log(1-x)$ (b) $-\log(1-x)$

(c) $\log x$ (d) $-\log x$

Problem 5: The value of x satisfying $18^{4x-3} = \left(54\sqrt{2}\right)^{3x-4}$ is

(a) 2 (b) 6
(b) 3 (d) 4

Problem 6: If $a^2 + b^2 = 14ab$, then $\log_2\left(\sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}}\right)$ is equal to

(a) 2 (b) 6

(c) 3 (d) 4

Problem 7: If $\log 2 = a, \log 3 = b, \log 7 = c$ and $6^x = 7^{x+4}$, then x is equal to

(a) $\frac{4c}{a+b+c}$ (b) $\frac{4c}{a+b-c}$

(c) $\frac{4c}{a-b-c}$ (d) none of these

Problem 8: If $\log_{0.3}(x-1) < \log_{0.09}(x-1)$, then x lies in the interval

(a) $(2,\infty)$ (b) $(-2,-1)$

(c) $(1,2)$ (d) none of these

Exercise 2

Q1: Prove that $\frac{1}{\log_a abc} + \frac{1}{\log_b abc} + \frac{1}{\log_c abc} = 1$.

Q2: Solve $\log_e(x-a) + \log_e x = 1.(a > 0)$

Q3: If for $x, y \in R^+$, $\ln\frac{x+y}{2} = \frac{1}{2}(\ln x + \ln y)$, show that $x = y$.

Q4: If $\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b}$, then prove that $a^a b^b c^c = 1$.

Q5: Show that $3\log 4 - 2\log 6 + \log(18)^{3/2} = \log 96\sqrt{2}$

Q6: Solve for x: $\log 2 + \log(x+2) - \log(3x-5) = \log 3.$

Q7: Find x if $\log_a x = 3\log_a y - \frac{1}{2}\log_a z.$

Q8: Solve for x : $x^{\log_x (x+3)^2} = 16.$

Q9: Find x if it is given by $\log_8\left(\frac{8}{x^2}\right) = 3.(\log_8 x)^2$

Q10: Find the least value of the expression $2\log_{10} x - \log_x(0.01)$ for $x > 1.$

Q11: Find the solution set of the inequality $\log_{|\sin x|}\left(x^2 - 8x + 23\right) > \frac{3}{\log_2 |\sin x|}.$

Q12: Solve for x the following equation:

$$\log_{(2x+3)}\left(6x^2 + 23x + 21\right) = 4 - \log_{(3x+7)}\left(4x^2 + 12x + 9\right)$$

Level IV

Integer Type/Fill in the blanks

1. The value of $6+\log_{3/2}\left(\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\cdots}}}\right)$ is __________.

2. The solution of the equation $\log_7 \log_5\left(\sqrt{x+5}+\sqrt{x}\right)=0$ is__________.

MCQ-Single Correct

Q1: Let (x_0, y_0) be solution of the following equations

$$(2x)^{\ln 2}=(3y)^{\ln 3}$$

$$3^{\ln x}=2^{\ln y}$$

Then x_0 is

(a) $\frac{1}{6}$

(b) $\frac{1}{3}$

(b) $\frac{1}{2}$

(d) 6

Q2: The number of solutions of $\log_4(x-1)=\log_2(x-3)$

(a) 3

(b) 1

(c) 2

(d) 0

Q3: The number $\log_2 7$ is

(a) an integer

(b) a rational number

(c) irrational number

(d) a prime number

MCQ – Multi Correct

Q1: If $3^x = 4^{x-1}$, then $x =$

(a) $\dfrac{2\log_3 2}{2\log_3 2-1}$

(b) $\dfrac{2}{2-\log_2 3}$

(c) $\dfrac{1}{1-\log_4 3}$

(d) $\dfrac{2\log_2 3}{2\log_2 3-1}$

Q2: The equation $x^{3/4(\log_2 x)^2+\log_2 x-(5/4)} = \sqrt{2}$ has

(a) at least one real solution

(b) exactly three real solutions

(c) exactly one irrational solution

(d) complex roots.

Subjective

1. If n is a natural number such that $n = p_1^{\alpha_1}.p_2^{\alpha_2}.p_3^{\alpha_3}....p_k^{\alpha_k}$ and $p_1, p_2, ..., p_k$ are distinct primes, then show that $\ln k \geq \ln 2$.

SETS

Section 1

- **Definition:** A set is a well-defined collection of objects.

There are two methods of representing a set

(i) **Roaster or tabular form**

e.g. natural numbers less than 5 = {1, 2, 3, 4}

(ii) **Set builder form**

e.g. : Vowels in English alphabet
= {x : x is a vowel in the English alphabet }

Section 2

- **Types of sets**

(i) **Empty set or Null set or void set:** A set which does not contain any element is called null set.

(ii) **Finite set:** A set which consists of a definite number of elements is called Finite set.

(iii) **Infinite set:** A set which consists of a indefinite number of elements is called infinite set.

(iv) **Singleton set:** A set having exactly one element is called singleton set.

- Subset :– A set A is said to be a subset of set B if $a \in A \Rightarrow a \in B, \forall a \in A$

Note: If $A \subset B$ and $A \neq B$, then A is called a *proper subset of* B and B is called *superset* of A.

- Equal sets :– Two sets A and B are equal if they have exactly the same elements i.e A = B if $A \subset B$ and $B \subset A$

- Power set : The collection of all subsets of a set A is called power set of A, denoted by P(A) i.e. $P(A) = \{ B : B \subset A \}$

- Equivalent sets : Two finite sets A and B are equivalent, if their cardinal numbers are same i.e., n(A) = n(B).

- **Types of Intervals**
Open Interval $(a, b) = \{ x \in R : a < x < b \}$
Closed Interval $[a, b] = \{ x \in R : a \le x \le b \}$
Semi open or Semi closed Interval,

$(a,b] = \{ x \in R : a < x \le b\}$
$[a,b) = \{ x \in R : a \le x < b\}$

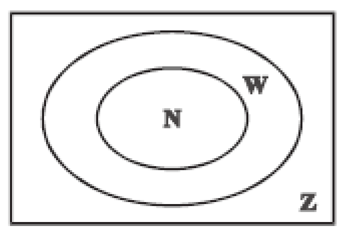
Section 3

- **Universal set**

This is a basic set; in a particular context whose elements and subsets are relevant to that particular context. For example, for the set of vowels in English alphabet, the universal set can be the set of all alphabets in English. Universal set is denoted by **U**

- **Venn diagrams**

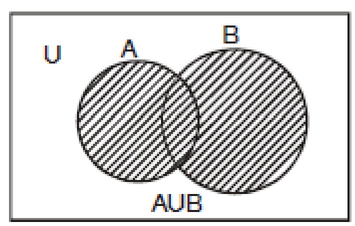
Venn Diagrams are the diagrams which represent the relationship between sets. For example, the set of natural numbers is a subset of set of whole numbers which is a subset of integers. We can represent this relationship through Venn diagram in the following way.



- **Operations on sets**

  - **Union** of two sets A and B is,

$$A \cup B = \{ x : x \in A \text{ or } x \in B \}$$

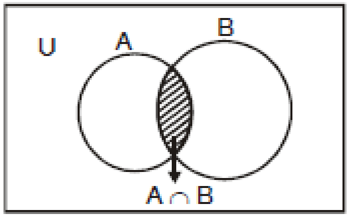


Properties of the operation of union.

(i) $A \cup B = B \cup A$ (ii) $(A \cup B) \cup C = A \cup (B \cup C)$
(iii) $A \cup \varphi = A$ (iv) $A \cup A = A$ (v) $U \cup A = U$

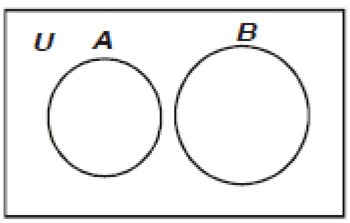
- **Intersection** of two sets A and B is,

$A \cap B = \{ x : x \in A \text{ and } x \in B\}$



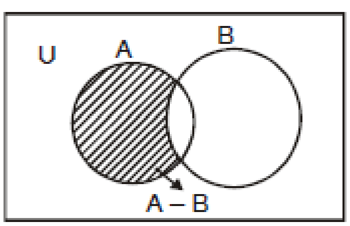
Properties of the operation of intersection
(i) $A \cap B = B \cap A$ (ii) $(A \cap B) \cap C = A \cap (B \cap C)$
(iii) $\varphi \cap A = \varphi$; $U \cap A = A$ (iv) $A \cap A = A$
(v) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
(vi) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

- Disjoint sets : Two sets A and B are said to be disjoint if $A \cap B = \phi$



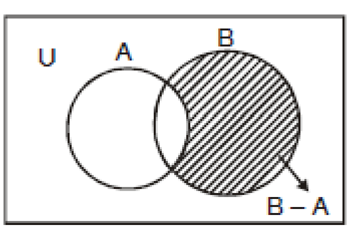
- Difference of sets A and B is,

$A - B = \{ x : x \in A \text{ and } x \notin B\}$



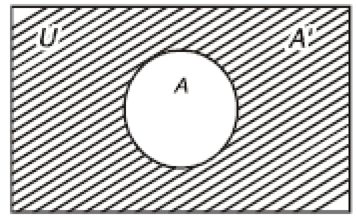
- Difference of sets B and A is,

$B - A = \{ x : x \in B \text{ and } x \notin A \}$



- Complement of a set A, denoted by A' or A^c is

$A' = A^c = U - A = \{ x : x \in U \text{ and } x \notin A\}$



Properties of complement sets :

1. Complement laws
(i) $A \cup A' = U$ (ii) $A \cap A' = \phi$ (iii) $(A')' = A$

2. De Morgan's Laws
(i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$

3. $\phi' = U$ and $U' = \phi$

<u>Section 4</u>

Laws of Algebra of sets.
(i) $A \cup \phi = A$
(ii) $A \cap \phi = \phi$

- $A - B = A \cap B'$

- Commutative Laws :–

(i) $A \cup B = B \cup A$ (ii) $A \cap B = B \cap A$

- Associative Laws :–

(i) $(A \cup B) \cup C = A \cup (B \cup C)$ (ii) $(A \cap B) \cap C = A \cap (B \cap C)$

- Distributive Laws :–

(i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
(ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

- If $A \subset B$, then $A \cap B = A$ and $A \cup B = B$

- When A and B are disjoint $n(A \cup B) = n(A) + n(B)$

- When A and B are not disjoint

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

- $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

Some important symbols

N : the set of all natural numbers
Z : the set of all integers
Q : the set of all rational numbers
R : the set of real numbers
Z^+ : the set of positive integers
Q^+ : the set of positive rational numbers, and
R^+ : the set of positive real numbers.

Worksheet CSBE

Q1 Write the set $\left\{\frac{1}{2},\frac{2}{3},\frac{3}{4},\frac{4}{5},\frac{5}{6},\frac{6}{7}\right\}$,in the set builder form

Ans: $\left\{x : x = \frac{n}{n+1}, n \in N, n \le 6\right\}$

Q2 Write the following sets in the roaster form.
(i) A = {x : x is a positive integer less than 10 and $2^x - 1$ is an odd number}
(ii) C = {x : $x^2 + 7x - 8 = 0$, $x \in \mathbf{R}$}

Ans: i) {1, 2, 3, 4, 5, 6, 7, 8, 9}.

ii) {– 8, 1}

Q3 From 50 students taking examinations in Mathematics, Physics and Chemistry, each of the student has passed in at least one of the subject, 37 passed Mathematics, 24 Physics and 43 Chemistry. At most 19 passed Mathematics and Physics, at most 29 Mathematics and Chemistry and at most 20 Physics and Chemistry. What is the largest possible number that could have passed all three examination?

Ans: 14

Q4 Each set X_r contains 5 elements and each set Y_r contains 2 elements and $\bigcup_{r=1}^{20} X_r = S = \bigcup_{r=1}^{n} Y_r$. If each element of S belong to exactly 10 of the X_r'*s* and to exactly 4 of the Y_r'*s*, then *n* is equal to.

Ans: 20

Q5 In a town of 10,000 families, it was found that 40% families buy newspaper A, 20% families buy newspaper B and 10% families buy newspaper C. 5% families buy A and B, 3% buy B and C and 4% buy A and C. If 2% families buy all the three papers. Find the no. of families which buy
(i) A only (ii) B only (iii) none of A, B, and C.

Ans: i)3300
ii)1400
iii)4000

Q6 Two finite sets have m and n elements. The total no. of subsets of the first set is 56 more than the total no. of subsets of second set. Find the value of m and n.

Ans: 6

Q7 A college awarded 38 medals in football, 15 in basketball and 20 in cricket. If these medals went to a total of 58 men and only three men got medals in all the three sports, how many received medals in exactly two of the three sports ?

Ans: 9

Q8 Let A and B be sets. If A ∩ X = B ∩ X = φ and A ∪ X = B ∪ X for some set X, show that A = B.

Q9 List all the subsets of the set {-1,0,1}

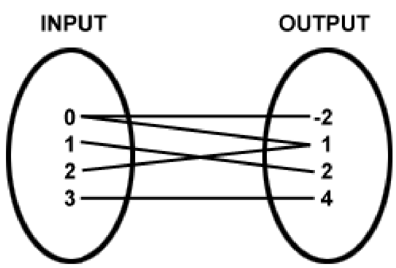
Ans: $\phi, \{-1\}, \{0\}, \{1\}, \{-1,0\}, \{0,1\}\{1,-1\}, \{-1,0,1\}$

Q10 Let U = {1, 2, 3, 4, 5, 6} A = {2, 3} and B = {3, 4, 5} Find $A' \cap B'$, $A \cup B$ and hence show that $(A \cup B)' = A' \cap B'$.

Ans: $(A \cup B)' = A' \cap B'$ ={1, 6}

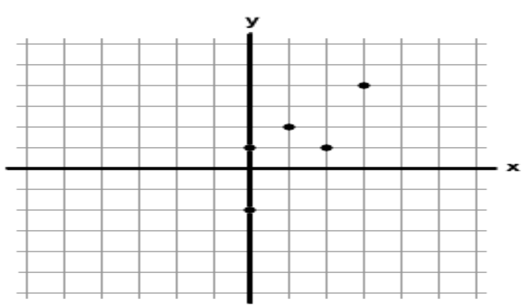
MATHEMATICS LECTURES BY
MANISH KALIA
(B.TECH Delhi College of Engineering)
TOPIC-RELATIONS

A relation is a set of inputs and outputs, often written as ordered pairs (input, output). We can also represent a relation as a mapping diagram or a graph. For example, the relation can be represented as:



Mapping Diagram of Relation

Lines connect the inputs with their outputs. The relation can also be represented as:



Thus the relation in a set A is a subset of A × A. Thus, the empty set φ and A × A are twoextreme relations.

Types of Relations

1]Empty relation: A relation R in a set A is called *empty relation*, if no element of A is related to any element of A, i.e., R = φ ∈ A × A.

2] **Universal relation:** A relation R in a set A is called *universal relation*, if each element of A is related to every element of A, i.e., R = A × A.

3] **Identity relation:** A relation R in a set A is called *identity relation*, if R={(a,b):a∈A,b∈A and a=b}

4] **Reflexive relation:** A relation *R* is said to be reflexive over a set *A* if (a,a) € *R* for every *a* € *R*.

Examples:If A is the set of all males in a family, then the relation "is brother of" is not reflexive over A. Because any person from the set A cannot be brother of himself.

The relation R = {(1,1)(2,2)(3,3)} is reflexive over the set A = {1,2,3}.

5] **Symmetric relation:** A relation *R* is said to be symmetric if (a,b) ∈ R => (b,a) ∈ R

Examles:If A is the set of all males in a family, then the relation "is brother of" is symmetric over A.Because if *a* is brother of *b* then *b* is brother of *a*.

If A is the set of mothers and B is the set of children in a family then a relation R on

AxB is not symmetric because if *a* is mother of *b* then *b* cannot be mother of *a*.

6] **Transitive relation:** A relation *R* is said to be symmetric if (a,b) ∈ R, (b,c) ∈ R => (a,c) ∈R.

Examles: If A is the set of all males in a family, then the relation "is brother of" is transitive over A.Because if *a* is brother of *b* and *b* is brother of *c* then *a* is brother of *c*.

The relation R = {(1,1)(2,2)(3,3)(1,2)(2,3)} is not transitive over the set A = {1,2,3} because thought (1,2), (2,3) ∈R , (1,3) is not in R.

7] **Equivalence relation:** A relation R in a set A is said to be an *equivalence relation* if R is
reflexive, symmetric and transitive.

Exercise1.1

Q1 Let A be the set of all students of a boys school. Show that the relation R in A given by R = {(*a*, *b*) : *a* is sister of *b*} is the empty relation and R/ = {(*a*, *b*) : the difference between heights of *a* and *b* is less than 3 meters} is the universal relation
[NCERT]

Q2 Let T be the set of all triangles in a plane with R a relation in T given by
R = {(T_1, T_2) : T1 is congruent to T_2}. Show that R is an equivalence relation.
[NCERT]

Q3 Let L be the set of all lines in a plane and R be the relation in L defined as R = {(L1, L2) : L1 is perpendicular to L2}. Show that R is symmetric but neither reflexive nor transitive.
[NCERT]

Q4 Show that the relation R in the set {1, 2, 3} given by R = {(1, 1), (2, 2),(3, 3), (1, 2), (2, 3)} is reflexive but neither symmetric nor transitive. [NCERT]

Q5 Show that the relation R in the set **Z** of integers given by R = {(*a, b*) : 2 divides *a* – *b*}is an equivalence relation. [NCERT]

Q6 Let A = {3, 5} and B = {7, 11}. Let R = {(a, b): a ∈ A, b ∈ B, a – b is odd}.Show that R is an empty relation from A into B

Q2 Let A = {3, 5}, B = {7, 11}. Let R = {(a, b): a ∈ A, b ∈B, a – b is even}. Show that R is an universal relation from A into B

Q3 Let A = {1, 2, 3} and R = {(a, b): a, b ∈ A, a divides b and b divides a}. Show that R is an identity relation on A.

Q4 Let R be a relation from N into N defined by R = {(a, b): a, b ∈ N and a = b^2 }. Are the following true?

i) (a, a) ∈ R, for all a ∈ N
ii) (a, b) ∈ R implies (b, a) ∈ R
iii) (a, b) ∈ R, (b, c) ∈ R implies (a, c) ∈ R?
Justify you answer in each case.

Q5 Prove that the relation R defined on the set N of natural numbers by xRy $\Leftrightarrow 2x^2 - 3xy + y^2 = 0$ i.e., by R = {(x, y): x, y ∈ N and

$2x^2-3xy+y^2=0$ is not symmetric but it is reflexive.

Q6 Let N be the set of natural numbers and relation R on N be defined as xRy $\Leftrightarrow$ x divides y i.e., as R = {(x, y): x, y ∈ N and x divides y}. Examine whether R is reflexive, Symmetric, or transitive.

Q7 Let S be any non empty set and P(S) be its power set. We define a relation R on P(S) by ARB to mean A$\subseteq$B; i.e., R= {(A, B): A $\subseteq$B}. Examine whether R is (i) reflexive (ii) Symmetric (iii) transitive.

Q8 Prove that a relation R on a set A is (i) reflexive $\Leftrightarrow$ $I_A \subseteq R, where\ I_A =$ {(x, x): x ∈ A} (ii) symmetric $\Leftrightarrow$ $R^{-1}=R$.

Q9 Give examples of relation which are

- Neither reflexive nor symmetric nor transitive.
- Symmetric and reflexive but not transitive.
- Reflexive and transitive but not symmetric

Past year CBSE Questions

Q1 Let T be the set of all triangles in a plane with R as relation in T given by $R=\{(T_1,T_2):T_1 \cong T_2\}$,Show that R is an equivalence relation.

[CBSE AI-2008]

Q2 Prove that the relation R on the set A={1,2,3,4,5} given by $R=\{(a,b):|a-b|$ is even} is an equivalence relation.

[CBSE Delhi-2009]

Q3 Let Z be the set of all integers and R be the relation on Z defined as R = {(a, b) : a,b∈Z and $a-b$ is divisible by 5.}.Prove that R is an equivalence relation.

[CBSE Delhi-2010]

Q4 Show that the relation S in the set $A=\{x \in Z : 0 \le x \le 12\}$ given by $S=\{(a,b): a,b \in Z, |a-b|$ is divisible by 4} is an equivalence relation.Find the set of all elements related to 1.

[CBSE AI- 2010]

Q5 State the reason for the relation R ,in the set {1,2,3} given by R={(1,2),(2,1)},not a transitive.

[CBSE AI- 2011]

Q6 Show that the relation R defined by : (a,b)R(c,d)⇒a+d=b+c on the set N is an equivalence relation

[HOTS]

Past year AIEEE/IIT Questions

Q1 Let R={(1,3),(4,2),(2,4),(2,3),(3,1)} be a relation on the set A={1,2,3,4}.The relation R is

[AIEEE-2004]

a) not symmetric
b) transitive
c) a function
d) reflexive

Q2 Let R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} be a relation on the set A = {3, 6, 9, 12} be a relation on the set A = {3, 6, 9, 12}. The relation is

[AIEEE-2005]

(a) reflexive and transitive only
(b) reflexive only
(c) an equivalence relation
(d) reflexive and symmetric only

Q3 Let W denote the words in English dictionary.Define the relation R by $R = \{(x, y) \in W \times W$: the words x and y have at least one letter in common}
Then R is

[AIEEE-2006]

a) not reflexive, symmetric and transitive
b) reflexive, symmetric and not transitive
c) reflexive, symmetric and transitive
d) reflexive, not symmetric and transitive

Q4 Let R be the real line .Consider the following subsets of the plane RXR:
$S = \{(x, y) : y = x + 1 \text{ and } 0 < x < 2\}$
$T = \{(x, y) : x - y \text{ is an integer}\}$
Which of the following statement is true?

[AIEEE-2008]

a) T is an equivalence relation on R but S is not.
b) Neither T nor S is an equivalence relation on R
c) Both S and T are is an equivalence relation on R
d) S is is an equivalence relation on R but T is not

Additional Notes:

Previous Year Questions

Mains (SETS,Relations)

2026

Q1: Let R be a relation defined on the set $\{1,2,3,4\}\times\{1,2,3,4\}$ by $R=\{((a,b),(c,d)):2a+3b=3c+4d\}$. Then the number of elements in R is

(a) 6 (b) 15

(c) 12 (d) 18

Q2: Consider two Sets $A=\{x\in Z:|(|x-3|-3)|\le 1\}$ and $B=\left\{x\in\mathbb{R}-\{1,2\}:\dfrac{(x-2)(x-4)}{x-1}\log_e(|x-2|)=0\right\}$. Then the number of onto functions $f:A\to B$ is equal to :

(a) 32 (b) 81

(c) 79 (d) 62

Q3: Let $A=\{0,1,2,.....,9\}$. Let R be a relation on A defined by $(x,y)\in R$ if and only if $|x-y|$ is a multiple of 3. Given below are two statements

Statement I : $n(R)=36$

Statement II : R is an equivalence relation

In the light of the above statements, choose the ***correct*** answer from the options given below :

(a) Statement I is correct but Statement II is incorrect.
(b) Both Statement I and Statement II are correct.
(c) Both Statement I and Statement II are incorrect.
(d) Statement I is incorrect but Statement II is correct.

Q4: Let $A=\{-2,-1,0,1,2,3,4\}$. Let R be a relation on A defined by xRy if and only if $2x+y\le 2$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations respectively. Then $l+m+n$ is equal to :

(a) 34 (b) 32

(c) 33 (d) 35

Q5: The number of elements in the relation $R=\{(x,y):4x^2+y^2<52, x,y\in\mathbb{Z}\}$

(a) 86 (b) 67

(c) 89 (d) 77

Q6: Let the relation R on the set $M=\{1,2,3,......,16\}$ be given by $R=\{(x,y):4y=5x-3, x,y\in M\}$. Then the minimum number of elements required to be added in R, in order to make the relation symmetric , is equal to

(a) 4 (b) 3

(c) 1 (d) 2

Q7: Let $A=\{x:|x^2-10|\le 6\}$ and $B=\{x:|x-2|>1\}$. Then

(a) $A\cup B=(-\infty,1]\cup(2,\infty)$ (b) $B-A=(-\infty,-4)\cup(-2,1)\cup(4,\infty)$

(c) $A-B=[2,3)$ (d) $A\cap B=[-4,-2]\cup[3,4]$

Q8: Let $A=\{2,3,5,7,9\}$. Let R be the relation on A defined by xRy if and only if $2x\le 3y$. Let l be the number of elements in R, and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then $l+m$ is equal to :

(a) 21 (b) 25

(c) 23 (d) 27

Q9: The number of relations, defined on the set {a,b,c,d}, which are both reflexive and symmetric, is equal to :

(a) 16 (b) 64

(c) 256 (d) 1024

2025

Q1: Let **A** = {0, 1, 2, 3, 4, 5}. Let **R** be a relation on **A** defined by (x, y) ∈ **R** if and only if max{x, y} ∈ {3, 4}. Then among the statements
(S_1): The number of elements in **R** is 18, and
(S_2): The relation **R** is symmetric but neither reflexive nor transitive

(a) both are false
(b) only (S_1) is true
(c) only (S_2) is true
(d) both are true

Q2: Let $A=\{(\alpha,\beta)\in\mathbb{R}\times\mathbb{R}:|\alpha-1|\le 4 \text{ and } |\beta-5|\le 6\}$

$B=\{(\alpha,\beta)\in\mathbb{R}\times\mathbb{R}:16(\alpha-2)^2+9(\beta-6)^2\le 144\}$.

Then

(a) $A\subset B$
(b) $B\subset A$
(c) neither $A\subset B$ nor $B\subset A$
(d) $A\cup B=\{(x,y):-4\le x\le 4,-1\le y\le 11\}$

Q3: Let $A=\{-3,-2,-1,0,1,2,3\}$ and R be a relation on A defined by xRy if and only if $2x-y\in\{0,1\}$. Let l be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l+m+n$ is equal to:

(a) 17
(b) 18
(c) 15
(d) 16

Q4: Consider the sets $A=\{(x,y)\in\mathbb{R}\times\mathbb{R}:x^2+y^2=25\}$, $B=\{(x,y)\in\mathbb{R}\times\mathbb{R}:x^2+9y^2=144\}$, $C=\{(x,y)\in\mathbb{Z}\times\mathbb{Z}:x^2+y^2\le 4\}$ and $D=A\cap B$. The total number of one-one functions from the set D to the set C is :

(a) 15120
(b) 18290
(c) 17160
(d) 19320

Q5: Let $A=\{-2,-1,0,1,2,3\}$. Let R be a relation on A defined by xRy if and only if $y=\max\{x,1\}$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l+m+n$ is equal to

(a) 11
(b) 12

(c) 14 (d) 13

Q6: Let $A=\{-3,-2,-1,0,1,2,3\}$. Let R be a relation on A defined by xRy if and only if $0 \le x^2+2y \le 4$. Let l be the number of elements in R and m be the minimum number of elements required to be added in R to make it a reflexive relation. Then $l+m$ is equal to:

(a) 18 (b) 20

(c) 17 (d) 19

Q7: Let $A=\{1,2,3,.....,100\}$ and R be a relation on A such that $R=\{(a,b):a=2b+1\}$. Let (a_1,a_2), $(a_2,a_3),(a_3,a_4),........,(a_k,a_{k+1})$ be a sequence of k elements of R such that the second entry of an ordered pair is equal to the first entry of the next ordered pair. Then the largest integer k, for which such a sequence exists, is equal to

(a) 6 (b) 8

(c) 7 (d) 5

Q8: Let A be the set of all functions $f:\mathbb{Z}\to\mathbb{Z}$ and R be a relation on A such that $R=\{(f,g): f(0)=g(1) \quad and \quad f(1)=g(0)\}$. Then R is :

(a) Symmetric and transitive but not reflexive
(b) Symmetric but neither reflexive nor transitive
(c) Transitive but neither reflexive nor symmetric
(d) Reflexive but neither symmetric nor transitive

Q9: Let $S=\mathbb{N}\cup\{0\}$. Define a relation R from S to $\mathbb{R}$ by:

$R=\left\{(x,y):\log_e y = x\log_e\left(\frac{2}{5}\right), x\in S, y\in\mathbb{R}\right\}$. Then, the sum of all the elements in the range of R is equal to:

(a) $\frac{3}{2}$ (b) $\frac{10}{9}$

(c) $\frac{5}{2}$ (d) $\frac{5}{3}$

Q10: Define a relation R on the interval $\left[0,\frac{\pi}{2}\right)$ by xRy if and only if $\sec^2 x-\tan^2 y=1$. Then R is:

(a) Both reflexive and symmetric but not transitive.
(b) Both reflexive and transitive but not symmetric.

(c) Reflexive but neither symmetric not transitive
(d) An equivalence relation

Q11. The relation $R=\{(x,y): x,y\in\mathbb{Z} \quad and \quad x+y \quad is \quad even\}$ is:

(a) Reflexive and transitive but not symmetric
(b) Reflexive and symmetric but not transitive.
(c) An equivalence relation.
(d) Symmetric and transitive but not reflexive.

Q12: Let $A=\left\{x\in(0,\pi)-\left\{\frac{\pi}{2}\right\}:\log_{(2/\pi)}|\sin x|+\log_{(2/\pi)}|\cos x|=2\right\}$ and $B=\left\{x\geq 0:\sqrt{x}\left(\sqrt{x}-4\right)-3\left|\sqrt{x}-2\right|+6=0\right\}$. Then $n(A\cup B)$ is equal to

(a) 4 (b) 8

(c) 6 (d) 2

13: Let $X=\mathbb{R}\times\mathbb{R}$. Define a relation R on X as: $(a_1,b_1)R(a_2,b_2)\Leftrightarrow b_1=b_2$

Statement I: R is an equivalence relation.

Statement II: For some $(a,b)\in X$, the set $S=\{(x,y)\in X:(x,y)R(a,b)\}$ represents a line parallel to $y=x$.

In the light of above statements,choose the correct answer from the options given below:

(a) Both Statement I and Statement II are true.
(b) Statement I is true but Statement II is false.
(c) Both Statement I and Statement II are false.
(d) Statement I is false but Statement II is true.

14. Let $A=\{(x,y)\in\mathbb{R}\times\mathbb{R}:|x+y|\geq 3\}$ and $B=\{(x,y)\in\mathbb{R}\times\mathbb{R}:|x|+|y|\leq 3\}$. If $C=\{(x,y)\in A\cap B: x=0 \quad or \quad y=0\}$,then $\sum_{(x,y)\in C}|x+y|$ is :

(a) 18 (b) 24

(c) 15 (d) 12

15. Let $R=\{(1,2),(2,3),(3,3)\}$ be a relation defined on the set $\{1,2,3,4\}$. Then the minimum number of elements, needed to be added in R so that R becomes an equivalence relation, is:

(a) 9 (b) 8

(c) 7 (d) 10

16. Let $A=\{1,2,3,....,10\}$ and $B=\left\{\frac{m}{n}: m,n\in A, m<n \quad and \quad \gcd(m,n)=1\right\}$. Then $n(B)$ is equal to:

(a) 29 (b) 31

(c) 37 (d) 36

17. The number of non-empty equivalence relations on the set {1,2,3} is:

(a) 7 (b) 4

(c) 5 (d) 6

2024

Q1: Let $A=\{2,3,6,8,9,11\}$ and $B=\{1,4,5,10,15\}$. Let R be a relation on $A\times B$ defined by $(a,b)R(c,d)$ if and only if $3ad-7bc$ is an even integer. Then the relation R is

(a) Reflexive but not symmetric.
(b) An equivalence relation.
(c) Reflexive and symmetric but not transitive.
(d) Transitive but not symmetric.

Q2: Let $A=\{1,2,3,4,5\}$. Let R be a relation on A defined by xRy if and only if $4x\le 5y$. Let m be the number of elements in R and n be the minimum number of elements from $A\times A$ that are required to be added to R to make it a symmetric relation. Then m+n is equal to:

(a) 23 (b) 26

(c) 25 (d) 24

Q3: Let $A=\{n\in[100,700]\cap\mathbb{N}$: n is neither a multiple of 3 nor a multiple of 4}. Then the number of elements in A is

(a) 300 (b) 310

(c) 290 (d) 280

Q4: Let the relations R_1 and R_2 on the set $X=\{1,2,3,....,20\}$ be given by $R_1=\{(x,y):2x-3y=2\}$ and $R_2=\{(x,y):-5x+4y=0\}$. If M and N be the minimum number of elements required to be added in R_1 and R_2, respectively , in order to make the relations symmetric, then M+N equals

(a) 16 (b) 12

(c) 8 (d) 10

Q5: Let a relation R on $\mathbb{N}\times\mathbb{N}$ be defined as : $(x_1,y_1)R(x_2,y_2)$ if and only if $x_1\le x_2$ or $y_1\le y_2$. Consider the two statements:

(I) R is reflexive but not symmetric.
(II) R is transitive.

Then which one of the following is true?

(a) Only (II) is correct. (b) Both (I) and (II) are correct.

(c) Neither (I) nor (II) is correct. (d) Only (I) is correct.

Q6: Consider the relations R_1 and R_2 defined as $aR_1b \Leftrightarrow a^2+b^2=1$ for all $a,b\in\mathbb{R}$ and $(a,b)R_2(c,d)\Leftrightarrow a+d=b+c$ for all $(a,b),(c,d)\in\mathbb{N}\times\mathbb{N}$. Then

(a) R_1 and R_2 both are equivalence relations.
(b) Only R_1 is an equivalence relation.
(c) Only R_2 is an equivalence relation.
(d) Neither R_1 nor R_2 is an equivalence relation.

Q7: If R is the smallest equivalence relation on the set $\{1,2,3,4\}$ such that $\{(1,2)(1,3)\}\subset R$, then the number of elements in R is

(a) 15 (b) 10

(c) 12 (d) 8

Q8: Let R be a relation on $\mathbb{Z}\times\mathbb{Z}$ defined by $(a,b)R(c,d)$ if and only if $ad-bc$ is divisible by 5. Then R is

(a) Reflexive and transitive but not symmetric.
(b) Reflexive and symmetric but not transitive.
(c) Reflexive but neither symmetric nor transitive.
(d) Reflexive, symmetric and transitive.

Q9: Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point $P(m,n)$ from the point Q(-2,-3) is:

(a) 8 (b) 10

(c) 4 (d) 6

Q10: Let $S=\{1,2,3,....,10\}$. Suppose M is the set of all the subsets of S, then the relation $R=\{(A,B): A\cap B\neq\phi; A,B\in M\}$ is:

(a) Symmetric only. (b) Reflexive only.

(c) Symmetric and Reflexive only. (d) Symmetric and Transitive only.

2023

Q1: Let $A=\{1,3,4,6,9\}$ and $B=\{2,4,5,8,10\}$. Let R be a relation defined on $A\times B$ such that $R=\{((a_1,b_1)(a_2,b_2)): a_1\leq b_2 \quad and \quad b_1\leq a_2\}$. Then the number of elements in the set R is:

(a) 180 (b) 26

(c) 52 (d) 160

Q2: An organization awarded 48 medals in event 'A', 25 in event 'B' and 18 in event 'C'. If these medals went to total 60 men and only five men got medals in all the three events, then, how many received medals in exactly two of three events?

(a) 10 (b) 15

(c) 21 (d) 9

Q3: Let $A=\{2,3,4\}$ and $B=\{8,9,12\}$. Then the number of elements in the relation $R=\{((a_1,b_1),(a_2,b_2))\in(A\times B, A\times B): a_1$ divides b_2 and a_2 divides $b_1\}$ is:

(a) 18 (b) 24

(c) 36 (d) 12

Q4: Let $A=\{1,2,3,4,5,6,7\}$. Then the relation $R=\{(x,y)\in A\times A: x+y=7\}$ is:

(a) Reflexive but neither symmetric nor transitive.
(b) Transitive but neither symmetric nor reflexive.

(c) Symmetric but neither reflexive nor transitive.
(d) An equivalence relation.

Q5: Let $P(S)$ denote the power set of $S=\{1,2,3,....,10\}$. Define the relations R_1 and R_2 on $P(S)$ as AR_1B if $\left(A\cap B^c\right)\cup\left(B\cap A^c\right)=\phi$ and AR_2B if $A\cup B^c=B\cup A^c$, $\forall A,B\in P\left(S\right)$. Then :

(a) only R_2 is an equivalence relation
(b) both R_1 and R_2 are not equivalence relations
(c) both R_1 and R_2 are equivalence relations
(d) only R_1 is an equivalence relation

Q6: Let R be a relation on $\mathbb{R}$, given by $R=\{(a,b):3a-3b+\sqrt{7}$ is an irrational number }. Then R is

(a) an equivalence relation
(b) reflexive and symmetric but not trasitive
(c) reflexive and transitive but not symmetric
(d) reflexive but neither symmetric nor transitive

Q7: Among the relations $S=\left\{(a,b):a,b\in\mathbb{R}-\{0\},2+\frac{a}{b}>0\right\}$

and $T=\left\{(a,b):a,b\in\mathbb{R},a^2-b^2\in\mathbb{Z}\right\}$,

(a) S is transitive but T is not
(b) Both S and T are symmetric
(c) Neither S nor T is transitive
(d) T is symmetric but S is not

Q8: Let R be a relation on $\mathbb{N}\times\mathbb{N}$ defined by $(a,b)R(c,d)$ if and only if $ad(b-c)=bc(a-d)$. Then R is

(a) Symmetric and transitive but not reflexive
(b) Reflexive and symmetric but not transitive
(c) Trasitive but neither reflexive nor symmetric
(d) Symmetric but neither reflexive nor transitive

Q9: The minimum number of elements that must be added to the relation $R=\{(a,b),(b,c)\}$ on the set $\{a,b,c\}$ so that it becomes symmetric and transitive is :

(a) 7 (b) 3

(c) 4 (d) 5

Q10: Let R be a relation defined on $\mathbb{N}$ as aRb if $2a+3b$ is a multiple of $5, a, b \in \mathbb{N}$. Then R is

(a) An equivalence relation (b) non-reflexive

(c) Symmetric but non transitive (d) transitive but not symmetric

Q11: The relation $R=\{(a,b): \gcd(a,b)=1, 2a \neq b, a, b \in \mathbb{Z}\}$ is:

(a) Reflexive but not symmetric (b) Transitive but not reflexive

(c) Symmetric but not transitive (d) neither symmetric nor transitive

2022

Q1: Let R be a relation from the set $\{1,2,3,.....,60\}$ to itself such that $R=\{(a,b): b=pq,$ where $p,q \geq 3$ are prime numbers}. Then, the number of elements in R is :

(a) 600 (b) 660

(c) 540 (d) 720

Q2: For $\alpha \in \mathbb{N}$, consider a relation R on $\mathbb{N}$ given by $R=\{(x,y): 3x+\alpha y$ is a multiple of 7}. The relation R is an equivalence relation if and only if :

(a) $\alpha = 14$ (b) α is a multiple of 4

(c) 4 is the remainder when α is divided by 10

(d) 4 is the remainder when α is divided by 7

Q3: Let R_1 and R_2 be two relations defined on $\mathbb{R}$ by $aR_1b \Leftrightarrow ab \geq 0$ and $aR_2b \Leftrightarrow a \geq b$. Then

(a) R_1 is an equivalence relation but not R_2

(b) R_2 is an equivalence relation but not R_1

(c) Both R_1 and R_2 are equivalence relations

(d) Neither R_1 nor R_2 is an equivalence relation

Q4: Let a set $A=A_1 \cup A_2 \cup .... \cup A_k$, where $A_i \cap A_j = \varphi$ for $i \neq j, 1 \leq i, j \leq k$. Define the relation R from A to A by $R=\{(x,y): y \in A_i$ if and only if $x \in A_i, 1 \leq i \leq k\}$. Then R is :

(a) Reflexive , symmetric but not transitive

(b) Reflexive, transitive but not symmetric
(c) Reflexive but not symmetric and transitive
(d) An equivalence relation

Q5: Let $R_1 = \{(a,b) \in \mathbb{N} \times \mathbb{N} : |a-b| \le 13\}$ and $R_2 = \{(a,b) \in \mathbb{N} \times \mathbb{N} : |a-b| \ne 13\}$. Then on N

(a) Both R_1 and R_2 are equivalence relations
(b) Neither R_1 nor R_2 is an equivalence relation
(c) R_1 is an equivalence relation but R_2 is not
(d) R_2 is an equivalence relation but R_1 is not

2021

Q1: Which of the following is not correct for relation R on the set of real numbers ?

(a) $(x,y) \in R \Leftrightarrow 0 \le |x| - |y| \le 1$ is neither transitive nor symmetric.
(b) $(x,y) \in R \Leftrightarrow 0 < |x-y| \le 1$ is symmetric and transitive.
(c) $(x,y) \in R \Leftrightarrow 0 \le |x| - |y| \le 1$ is reflexive but not symmetric.
(d) $(x,y) \in R \Leftrightarrow |x-y| \le 1$ is reflexive and symmetric.

Q2: Out of all the patients in a hospital 89% are found to be suffering from heart ailment and 98% are suffering from lungs infection. If K% of them are suffering from both ailments, then K can not belong to the set :

(a) $\{80, 83, 86, 89\}$ (b) $\{84, 86, 88, 90\}$

(c) $\{79, 81, 83, 85\}$ (d) $\{84, 87, 90, 93\}$

Q3: Let $\mathbb{N}$ be the set of natural numbers and a relation R on $\mathbb{N}$ be defined by $R = \{(x,y) \in \mathbb{N} \times \mathbb{N} : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is:

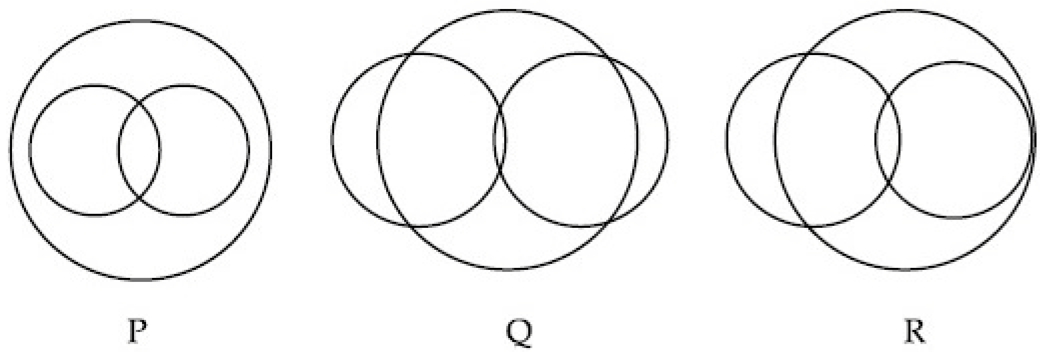
(a) Symmetric but neither reflexive nor transitive
(b) Reflexive but neither symmetric nor transitive
(c) Reflexive and symmetric but not transitive
(d) An equivalence relation

Q4: Define a relation R over a class of $n \times n$ real matrices A and B as "ARB iff there exists a non-singular matrix P such that $PAP^{-1} = B$". Then which of the following is true?

(a) R is reflexive, transitive but not symmetric

(b) R is symmetric, transitive but not reflexive.
(c) R is reflexive, symmetric but not transitive.
(d) R is an equivalence relation.

Q5: In a school, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagrams can justify the above statement?



(a) Q and R
(b) None of these
(c) P and R
(d) P and Q

Q6: Let A = {2, 3, 4, 5, ..., 30} and '$\simeq$' be an equivalence relation on $A \times A$, defined by (a, b) $\simeq$ (c, d), if and only if ad = bc. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair (4, 3) is equal to :

(a) 5
(b) 6
(c) 8
(d) 7

Q7: The number of elements in the set $\{x \in R : (|x|-3)|x+4| = 6\}$ is equal to:

(a) 4
(b) 2
(c) 3
(d) 1

Q8: Let R = {(P, Q) | P and Q are at the same distance from the origin} be a relation, then the equivalence class of (1, -1) is the set :

(a) $S = \{(x, y) \mid x^2 + y^2 = \sqrt{2}\}$
(b) $S = \{(x, y) \mid x^2 + y^2 = 2\}$
(c) $S = \{(x, y) \mid x^2 + y^2 = 1\}$
(d) $S = \{(x, y) \mid x^2 + y^2 = 4\}$

2020

Q1 : Consider the two sets : $A=\{m\in\mathbb{R}:$ both the roots of $x^2-(m+1)x+m+4=0$ are real$\}$ and $B=[-3,5)$. Which of the following is not true?

(a) $A\cup B=(-\infty,-3)\cup(5,\infty)$

(b) $A\cap B=\{-3\}$

(c) $B-A=(-3,5)$

(d) $A\cup B=\mathbb{R}$

Q2 : Let $\bigcup_{i=1}^{50} X_i=\bigcup_{i=1}^{n} Y_i=T$, where each X_i contains 10 elements and each Y_i contains 5 elements. If each element of the set T is an element of exactly 20 of sets X_i's and exactly 6 of sets Y_i's then n is equal to

(a) 15

(b) 30

(c) 50

(d) 45

Q3 : A survey shows that 73% of the persons working in an office like coffee, whereas 65% like tea. If x denotes the percentage of them, who like both coffee and tea, then x cannot be :

(a) 63

(b) 54

(c) 38

(d) 36

Q4 : Let $A=\{x:|x|<2\}$ and $b=\{x:|x-2|\geq 3\}$ then :

(a) $A\cap B=[-2,-1]$

(b) $A\cup B=\mathbb{R}-(2,5)$

(c) $A-B=[-1,2]$

(d) $B-A=\mathbb{R}-(-2,5)$

Q5: A survey shows that 63% of the people in a city read newspaper A whereas 76% read newspaper B. If x% of the people read both the newspapers, then a possible value of x can be:

(a) 37

(b) 65

(c) 29

(d) 55

Q6: Let R_1 and R_2 be two relation defined as follows :
$R_1=\{(a, b)\in R^2 : a^2+b^2\in Q\}$ and
$R_2=\{(a, b)\in R^2 : a^2+b^2\notin Q\}$,
where Q is the set of all rational numbers. Then :

(a) Neither R_1 nor R_2 is transitive.

(b) R_2 is transitive but R_1 is not transitive

(c) R_1 and R_2 are both transitive

(d) R_1 is transitive but R_2 is not transitive

Q7: If R = {(x, y) : x, y $\in$ Z, $x^2 + 3y^2 \leq 8$} is a relation on the set of integers Z, then the domain of R^{-1} is :

(a) {0,1}

(b) {-2,-1,1,2}

(c) {-1,0,1}

(d) {-2,-1,0,1,2}

2019

Q1 : Let A, B and C be sets such that $\varphi \neq A \cap B \subseteq C$. Then which of the following statements are not true?

(a) If $(A - C) \subseteq B$, then $A \subseteq B$

(b) If $(A - B) \subseteq C$,then $A \subseteq C$

(c) $(C \cup A) \cap (C \cup B) = C$

(d) $B \cap C \neq \phi$

Q2: Let $S = \{1, 2, 3, ..., 100\}$. The number of non-empty subsets A of S such that the product of elements in A is even is :

(a) $2^{100} - 1$

(b) $2^{50}(2^{50} - 1)$

(c) $2^{50} - 1$

(d) $2^{50} + 1$

Q3: Let Z be the set of integers.

If $A = \{x \in Z : 2^{(x+2)}(x^2 - 5x + 6) = 1\}$ and

$B = \{x \in Z : -3 < 2x - 1 < 9\}$, then the number of subsets of the set $A \times B$, is :

(a) 2^{15}

(b) 2^{18}

(c) 2^{12}

(d) 2^{10}

Q4: Two newspapers A and B are published in a city. It is known that 25% of the city populations reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisement is :-

(a) 13.5

(b) 13

(c) 12.8 (d) 13.9

Q5: In a class of 140 students numbered 1 to 140, all even numbered students opted Mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is

(a) 42 (b) 102

(c) 1 (d) 38

2018

Q1: Let **N** denote the set of all natural numbers. Define two binary relations on **N** as R = {(x, y) $\in$ **N** $\times$ **N** : 2x + y = 10} and R_2 = {(x, y) $\in$ **N** $\times$ **N** : x + 2y = 10}. Then :

(a) Range of R_1 is {2,4,8}. (b) Range of R_2 is {1,2,3,4}.

(c) Both R_1 and R_2 are symmetric relations

(d) Both R_1 and R_2 are transitive relations

Q2: Two sets A and B are as under:

$$A=\left\{(a,b)\in\mathbb{R}\times\mathbb{R}:|a-5|<1 \quad and \quad |b-5|<1\right\};$$

$$B=\left\{(a,b)\in\mathbb{R}\times\mathbb{R}:4(a-6)^2+9(b-5)^2\le 36\right\};$$

Then

(a) Neither $A\subset B$ nor $B\subset A$ (b) $B\subset A$

(c) $A\subset B$ (d) $A\cap B=\varphi$ (an empty set)

Q3: Consider the following two binary relations on the set A = {a, b, c} :
R_1 = {(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)} and
R_2 = {(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)}.
Then

(a) Both R_1 and R_2 are not symmetric.

(b) R_1 is not symmetric but it is transitive.

(c) R_2 is symmetric but it is not transitive.

(d) Both R_1 and R_2 are transitive.

2016

Q1: Let $P = \left\{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\right\}$ $Q = \left\{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\right\}$ be two seats. Then

(a) $P \subset Q$ and $Q - P \neq \phi$

(b) $Q \not\subset P$

(c) $P \not\subset Q$

(d) $P = Q$

2015

Q1: Let A and B be two sets containing four and two elements respectively. Then the number of subsets of the set $A \times B$, each having at least three elements is:

(1) 256

(2) 275

(3) 510

(4) 219

2014

1. If $X = \{4^n - 3n - 1 : n \in N\}$ and $Y = \left\{9(n-1) : n \in N\right\}$, where N is the set of natural numbers, then $X \cup Y$ is equal to

(1) N

(2) Y – X

(3) X

(4) Y

2013

1. Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of $A \times B$ having 3 or more elements is

(1) 220

(2) 219

(3) 211

(4) 256

2010

1. Let S be a non-empty subset of R. Consider the following statement:

P: There is a rational number $x \in S$ such that x > 0.

Which of the following statements is the negation of the statement P?

(1) There is no rational number $x \in S$ such that $x \leq 0$
(2) Every rational number $x \in S$ satisfies $x \leq 0$
(3) $x \in S$ and $x \leq 0 \Rightarrow x$ is not rational
(4) There is a rational number $x \in S$ such that $x \leq 0$

2. Consider the following relations:

R = {(x,y)|x,y are real numbers and x=wy for some rational number w};

S = $\left\{\left(\frac{m}{n}, \frac{p}{q}\right) \mid m, n, p \, and \, q \, are \, integers \, such \, n, q \neq 0 \, and \, qm = pn\right\}$. Then

(1) neither R nor S is an equivalence relation
(2) S is an equivalence relation but R is not an equivalence relation
(3) R and S both are equivalence relations
(4) R is an equivalence relation but S is not an equivalence relation

2009

1. If A, B nd C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then

(1) A = B
(2) A = C
(3) B = C
(4) $A \cap B = \phi$

2008

1. Let R be the real line. Consider the following subsets of the plane $R \times R$.

S = {(x,y): y = x + 1 and 0 < x <2 }, T = {(x,y):x – y is an integer}. Which one of the following is true?

(1) neither S nor T is an equivalence relation on R
(2) Both S and T are equivalence relations on R
(3) S is an equivalence relation on R but T is not
(4) T is an equivalence relation on R but S is not

2006

1. Let W denote the words in the English dictionary. Define the relation R by :

R = {(x,y)ε$W \times W$ | the words x and y have at least one letter in common}. Then R is

(1) Not reflexive, symmetric and transitive

(2) reflexive, symmetric and not transitive

(3) reflexive, symmetric and transitive

(4) reflexive, not symmetric and transitive

2005

1. Let R = {(3,3), (6,6), (9,9),(12,12),(6,12),(3,9),(3,12),(3,6)} be a relation on the set A = {3,6,9,12} be a relation on the set A = {3,6,9,12}. The relation is

(1) reflexive and transitive only

(2) reflexive only

(3) an equivalence relation

(4) reflexive and symmetric only

2004

1. Let R = {(1,3),(4,2),(2,4),(2,3),(3,1)} be a relation on the set A = {1,2,3,4}. The relation R is

(1) a function

(2) reflexive

(3) not symmetric

(4) transitive

Previous Year Questions

Numeric (Mains)

2026

Q1: Let S be the set of the first 11 natural numbers. Then the number of elements in $A=\{B\subseteq S: n(B)\geq 2$ and the product of all elements of B is even $\}$ is __________.

2025

Q1: The number of relations on the set $A=\{1,2,3\}$, containing at most 6 elements including $(1,2)$, which are reflexive and transitive but not symmetric, is __________.

Q2: For $n\geq 2$, let S_n denote the set of all subsets of $\{1,2,....,n\}$ with no two consecutive numbers. For example $\{1,3,5\}\in S_6$, but $\{1,2,4\}\notin S_6$. Then $n(S_5)$ is equal to __________.

Q3: Let $S=\{p_1,p_2,.....,p_{10}\}$ be the set of first ten prime numbers. Let $A=S\cup P$, where P is the set of all possible products of distinct elements of S. Then the number of all ordered pairs $(x,y), x\in S$, $y\in A,$ such that x divides y, is __________.

Q4: Let $A=\{1,2,3\}$. The number of relations on A, containing $(1,2)$ and $(2,3)$, which are reflexive and transitive but not symmetric, is __________.

2024

Q1: Let $A=\{2,3,6,7\}$ and $B=\{4,5,6,8\}$. Let R be a relation defined on $A\times B$ by $(a_1,b_1)R(a_2,b_2)$ if and only if $a_1+a_2=b_1+b_2$. Then the number of elements in R is __________.

Q2: In a survey of 220 students of a higher secondary school, it was found that at least 125 and at most 130 students studied Mathematics; at least 85 and at most 95 studied Physics; at least 75 and at most

90 studied Chemistry; 30 studied both Physics and Chemistry; 50 studied both Chemistry and Mathematics; 40 studied both Mathematics and Physics and 10 studied none of these subjects. Let m and n respectively be the least and the most number of students who studied all the three subjects. Then m+n is equal to __________.

Q3: Let $A=\{1,2,3,.....,20\}$. Let R_1 and R_2 be two relations on A such that $R_1=\{(a,b): b$ is divisible by a}, $R_2=\{(a,b): a$ is an integral multiple of b}. Then, number of elements in R_1-R_2 is equal to __________.

Q4: Let $A=\{1,2,3,........,100\}$. Let R be a relation on A defined by $(x,y)\in R$ if and only if $2x=3y$. Let R_1 be a symmetric relation on A such that $R\subset R_1$ and the number of elements in R_1 is n. Then the minimum value of n is __________.

Q5: Let $A=\{1,2,3,4\}$ and $R=\{(1,2),(2,3),(1,4)\}$ be a relation on A. Let S be the equivalence relation on A such that $R\subset S$ and the number of elements in S is n. Then the minimum value of n is __________.

Q6: The number of symmetric relations defined on the set $\{1,2,3,4\}$ which are not reflexive is __________.

2023

Q1: The number of elements in the set $\{n\in\mathbb{N}: 10\le n\le 100$ and 3^n-3 is a multible of 7} is __________.

Q2: Let $A=\{1,2,3,4\}$ and R be a relation on the set $A\times A$ defined by $R=\{((a,b),(c,d)): 2a+3b=4c+5d\}$. Then the number of elements in R is __________.

Q3: Let $A=\{-4,-3,-2,0,1,3,4\}$ and $R=\{(a,b)\in A\times A: b=|a|$ or $b^2=a+1\}$ be a relation on A. Then the minimum number of elements, that must be added to the relation R so that it becomes reflexive and symmetric, is __________.

Q4: The number of relations, on the set $\{1,2,3\}$ contaning $(1,2)$ and $(2,3)$, which are reflexive and transitive but not symmetric, is __________.

Q5: The number of elements in the set $\{n\in\mathbb{Z}: |n^2-10n+19|<6\}$ is __________.

Q6: Let $A=\{0,3,4,6,7,8,9,10\}$ and R be a relation defined on A such that $R=\{(x,y)\in A\times A: x-y$ is odd positive integer or $x-y=2\}$. The minimum number of elements that must be added to the relation R, so that it is a symmetric relation, is equal to __________.

Q7: Let $A=\{1,2,3,4,.....,10\}$ and $B=\{0,1,2,3,4\}$. The number of elements in the relaiton $R=\{(a,b)\in A\times A: 2(a-b)^2+3(a-b)\in B\}$ is __________.

Q8: Let S = {1, 2, 3, 5, 7, 10, 11}. The number of non-empty subsets of S that have the sum of all elements a multiple of 3, is __________.

Q9: The minimum number of elements that must be added to the relation R = {(a, b), (b, c), (b, d)} on the set {a, b, c, d} so that it is an equivalence relation, is __________.

2022

Q1: Let $S=\{4,6,9\}$ and $T=\{9,10,11,........,1000\}$. If $A=\{a_1+a_2+...+a_k : k\in\mathbb{N}, a_1,a_2,a_3,...,a_k\in S\}$, then the sum of all the elements in the set T-A is equal to __________.

Q2: Let $A=\{1,2,3,4,5,6,7\}$ and $B=\{3,6,7,9\}$. Then the number of elements in the set $\{C\subseteq A: C\cap B\neq\varphi\}$ is __________.

Q3: Let $A=\{1,2,3,4,5,6,7\}$. Define $B=\{T\subseteq A:$ either $1\notin T$ or $2\in T\}$ and $C=\{T\subseteq A: T$ the sum of all the elements of T is a prime number }. Then the number of elements in the set $B\cup C$ is __________.

Q4: Let R_1 and R_2 be relations on the set {1, 2, ..., 50} such that

R_1 = {(p, p^n) : p is a prime and n $\geq$ 0 is an integer} and

R_2 = {(p, p^n) : p is a prime and n = 0 or 1}.

Then, the number of elements in R_1 - R_2 is __________.

Q5: Let A = {n $\in$ N : H.C.F. (n, 45) = 1} and

Let B = {2k : k $\in$ {1, 2, ..., 100}}. Then the sum of all the elements of A $\cap$ B is __________.

Q6: Let $A=\sum_{i=1}^{10}\sum_{j=1}^{10}\min\{i,j\}$ and $B=\sum_{i=1}^{10}\sum_{j=1}^{10}\max\{i,j\}$. Then $A+B$ is equal to __________.

Q7: The sum of all the elements of the set $\{\alpha \in \{1,2,.....,100\} : HCF(\alpha, 24) = 1\}$ is __________.

2021

Q1: If $A = \{x \in R : |x-2| > 1\}$, $B = \{x \in R : \sqrt{x^2 - 3} > 1\}$, $C = \{x \in R : |x-4| \geq 2\}$ and Z is the set of all integers, then the number of subsets of the set $(A \cap B \cap C)^c \cap Z$ is __________.

Q2: Let $A = \{n \in \mathbb{N} \mid n^2 \leq n + 10,000\}$, $B = \{3k+1 \mid k \in \mathbb{N}\}$ and $C = \{2k \mid k \in \mathbb{N}\}$, then the sum of all the elements of the set $A \cap (B - C)$ is equal to __________.

Q3: Let $A = \{n \in \mathbb{N} : n$ is a 3-digit number$\}$, $B = \{9k+2 \mid k \in \mathbb{N}\}$ and $C = \{9k + l : k \in \mathbb{N}\}$ for some $l\,(0 < l < 9)$. If the sum of all the elements of the set $A \cap (B \cup C)$ is 274×400, then l is equal to __________.

2020

Q1: Set A has m elements and set B has n elements. If the total number of subsets of A is 112 more than the total number of subsets of B, then the value of m.n is __________.

Q2: Let $X = \{n \in \mathbb{N} : 1 \leq n \leq 50\}$. If $A = \{n \in X : n$ is a multiple of 2$\}$ and $B = \{n \in X : n$ is a multiple of 7$\}$, then the number of elements in the smallest subset of X containing both A and B is __________.

CHAPTER 1:FUNCTIONS AND THEIR GRAPHS

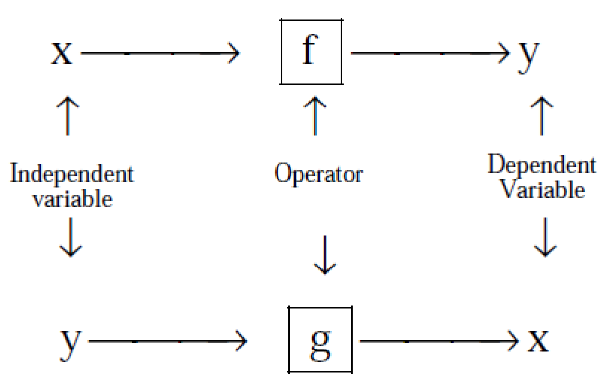
1.1 FUNCTIONS

Definition: If to every value (considered as real unless other-wise stated) of a variable x, which belongs to some collection (Set) A, there corresponds one and only one finite value of the quantity **y**, then **y** is said to be a function (Single valued) of **x** or a dependent variable defined on the set A ; **x** is the argument or independent variable.

If to every value of **x** belonging to some set A there corresponds one or several values of the variable **y**, then **y** is called a multiple valued function of **x** defined on A.Conventionally the word **"FUNCTION"** is used only as the meaning of a single valued function, if not otherwise stated .

See a function as an operator, which takes an input and gives an output. The input is called *the independent variable* and the output, which depends on the input because of the operator, is called the *dependent variable*

Pictorially:



DOMAIN, CO-DOMAIN & RANGE OF A FUNCTION :

Let $f : A \to B$, then the set A is known as the domain of f & the set B is known as co-domain of f . The set of all f images of elements of A is known as the range of f . Thus :

Domain of f = $\{a \mid a \in A, (a, f(a)) \in f\}$

Range of f = $\{f(a) \mid a \in A, f(a) \in B\}$

It should be noted that range is a subset of co-domain . Sometimes if only f (x) is given then domain is set of those values of ' x ' for which f (x) exists or is defined .

Exercise1.1:

Q1: Given a function $f(x)=\dfrac{x+1}{x-1}$,find $f(2x), 2f(x), f(x^2), (f(x))^2$.

Q2: Given a function $f(x)=\dfrac{4^x}{4^x+2}$,prove that $f(x)+f(1-x)=1$,and also find the value of

$f\left(\dfrac{1}{81}\right)+f\left(\dfrac{2}{81}\right)+f\left(\dfrac{3}{81}\right)\ldots\ldots\ldots\ldots+f\left(\dfrac{80}{81}\right)$.

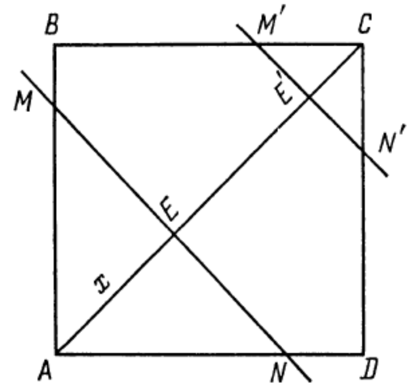
Q3: if $f(x)=\cos(\log x)$, then evaluate $f(x).f(y)-\dfrac{1}{2}\left\{f\left(\dfrac{x}{y}\right)+f(xy)\right\}$

Q4: Given a function $f(x)=\log\dfrac{1-x}{1+x}$.show that at $x_1, x_2 \in (-1,1)$,the following identity holds true.

$f(x_1)+f(x_2)=f\left(\dfrac{x_1+x_2}{1+x_1x_2}\right)$.Also prove that $f\left(\dfrac{2x}{1+x^2}\right)=2f(x)$.

Q5: Given a function $f(x)=\begin{cases} 3^{-x}-1; -1\le x<0, \\ \tan\left(\dfrac{x}{2}\right); 0\le x<\pi, \\ \dfrac{x}{x^2-2}; \pi\le x<6. \end{cases}$ Find $f(-1), f\left(\dfrac{\pi}{2}\right), f\left(\dfrac{2\pi}{3}\right), f(4)$.

Q6: In the square ABCD with side AB=2 a straight line MN is drawn perpendicular to AC .Denoting the distance from the vertex A to the line MN as x ,express through the x the area S of triangle AMN cut off from line MN .Find the area at x=√2/2 and x=2.



Q7: A function *f* satisfies the condition $f\left(x+\frac{1}{x}\right)=x^2+\frac{1}{x^2}, x\neq 0$. Determine it and hence find *f(3)*

Q8: An polynomial expression *f(x)* of degree n satisfies the condition $f(x)+f\left(\frac{1}{x}\right)=f(x).f\left(\frac{1}{x}\right)$, then prove that $f(x)=1\pm x^n$

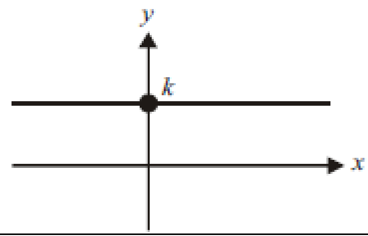
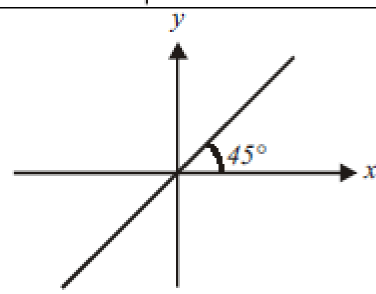
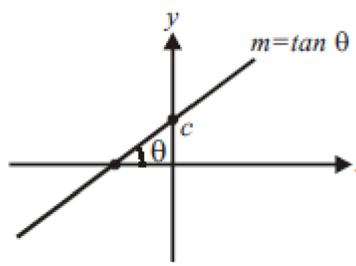
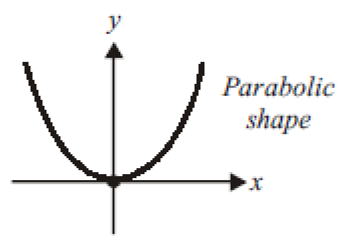
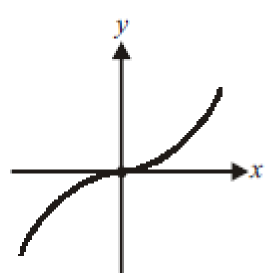
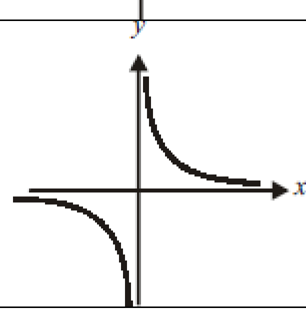
Q9: If for non zero *x*, $a.f(x)+b.f\left(\frac{1}{x}\right)=\frac{1}{x}-5$, where *a≠b*,then *f(x)* is equal to?

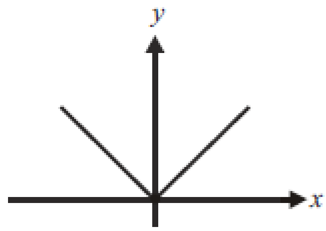
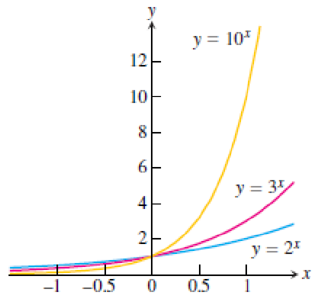
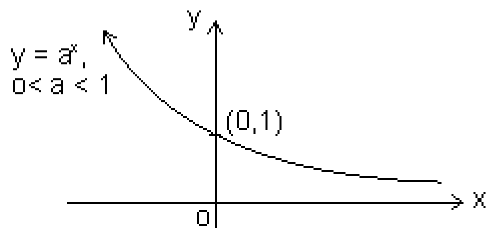
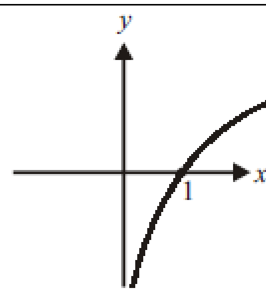
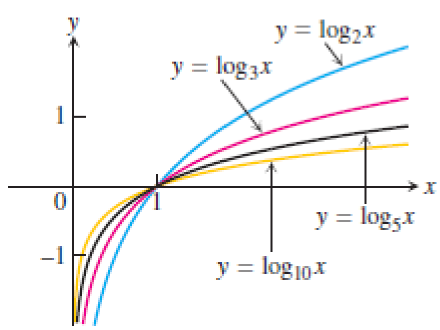
Q10: Let $f_1(x)=\frac{x}{3}+10$, for all x ∈R and $f_n(x)=f_1(f_{n-1}(x))$, for all n≥2.Then find $f_n(x)$.

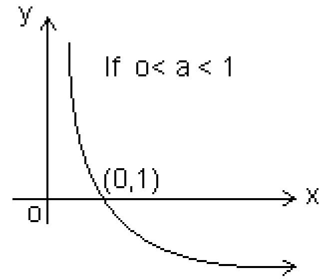
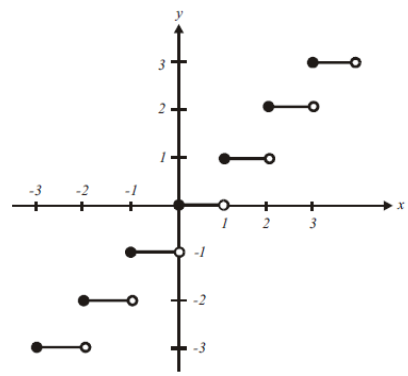
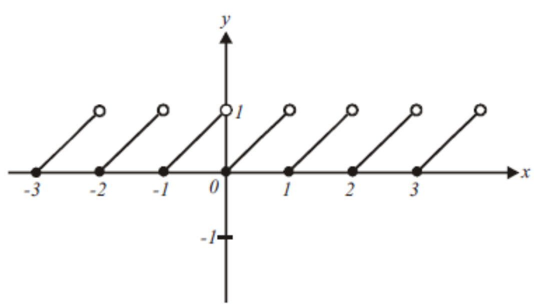
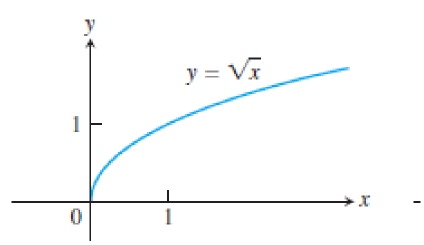
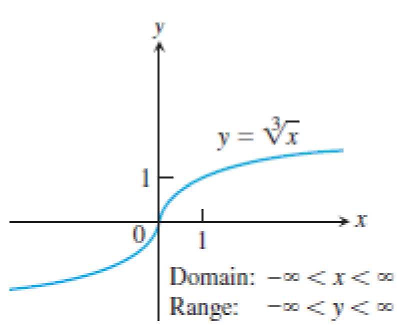
Q11: If $3.f(x)-f\left(\frac{1}{x}\right)=\log_e x^4$,for x>0,Find $f(e^x)$

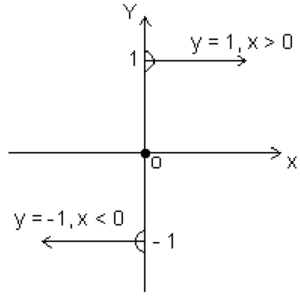
Q12: If $f(x)=\cos\left[\pi^2\right]x+\cos\left[-\pi^2\right]x$ where [x] stands for the greatest integer function, then evaluate f (π / 2), f (π), f (–π) and f (π / 4) .

1.2 GRAPHS OF FUCTIONS

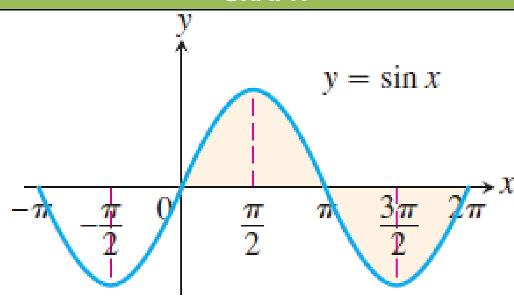
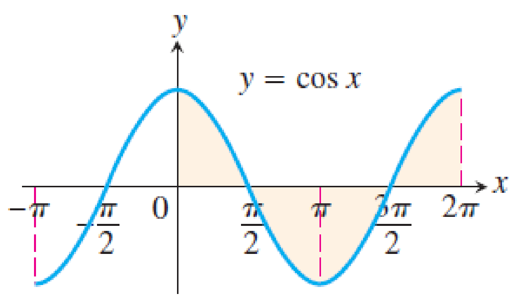
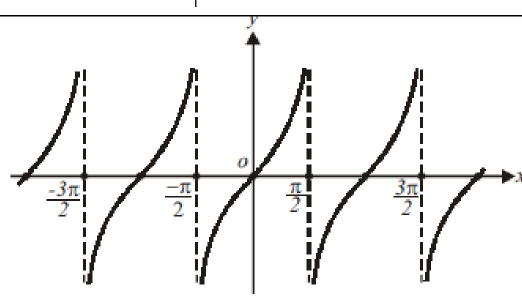
S.NO	FUNCTION		GRAPH	DOMAIN	RANGE
a)	***Constant function***	$f(x)=k$		$\mathbb{R}$	{k}
b)	***Identity function***	$f(x)=x$		$\mathbb{R}$	$\mathbb{R}$
c)	***Linear function***	$f(x)=mx+c$		$\mathbb{R}$	$\mathbb{R}$
d)	***Square function***	$f(x)=x^2$		$\mathbb{R}$	$[0,\infty)$
e)	***Cube function***	$f(x)=x^3$		$\mathbb{R}$	$\mathbb{R}$
f)	***Reciprocal function***	$f(x)=1/x$		$\mathbb{R}-\{0\}$	$\mathbb{R}-\{0\}$

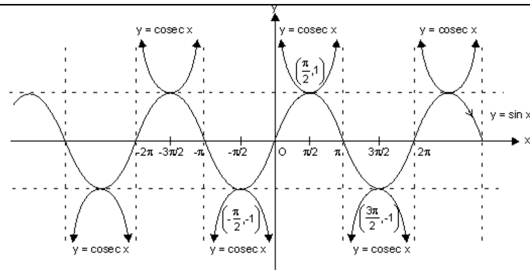
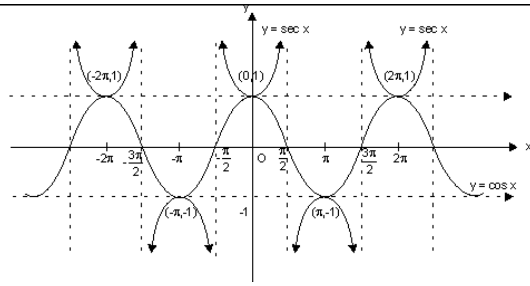
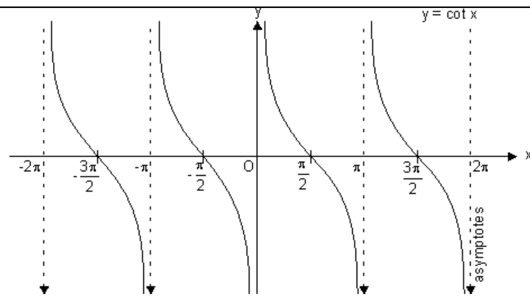
g)	***Modulus Function***	$f(x)=\lvert x\rvert$		$\mathbb{R}$	$[0,\infty)$
h)	***Exponential function***	$f(x)=a^x$ $a>1$		$\mathbb{R}$	$(0,\infty)$
		$f(x)=a^x$ $0<a<1$		$\mathbb{R}$	$(0,\infty)$
i)	***Logarithmic function***	$f(x)=\log_e x$		$(0,\infty)$	$\mathbb{R}$
		$f(x)=\log_a x$ $(a>1)$		$(0,\infty)$	$\mathbb{R}$

		$f(x)=\log_a x$ $(0<a<1)$		$(0,\infty)$	$\mathbb{R}$
j)	***Greatest integer function***	$f(x)=[x]$		$\mathbb{R}$	$\mathbb{Z}$
k)	***Fractional part***	$f(x)=x-[x]$		$\mathbb{R}$	$[0,1)$
l)	***Square root***	$f(x)=\sqrt{x}$		$[0,\infty)$	$[0,\infty)$
m)	***Cube root***	$f(x)=\sqrt[3]{x}$		$\mathbb{R}$	$\mathbb{R}$

n)	***Signum function***	$y = \mathrm{sgn}(x) = \begin{cases} \frac{\lvert x \rvert}{x}, x \neq 0, \\ 0, x = 0 \end{cases}$		ℝ	{- 1, 0,1}

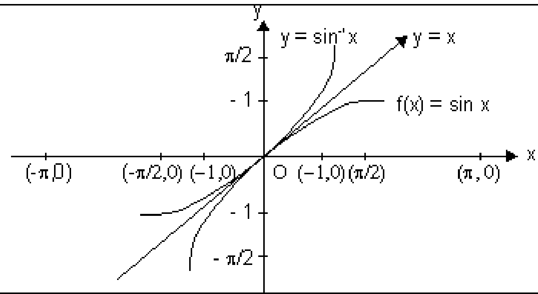
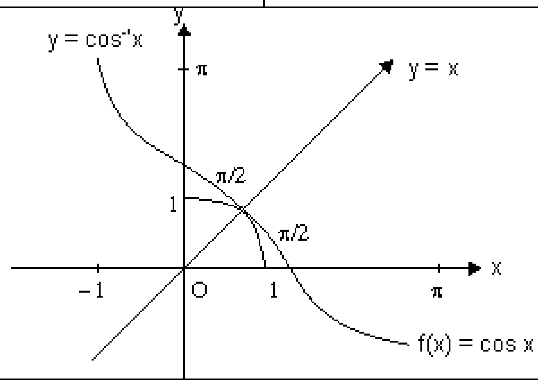
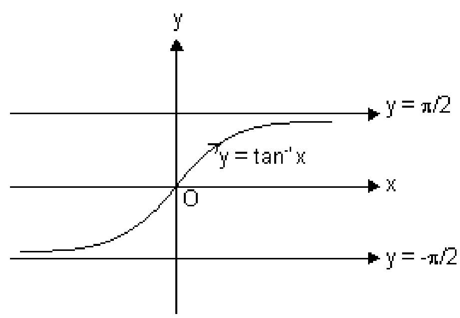
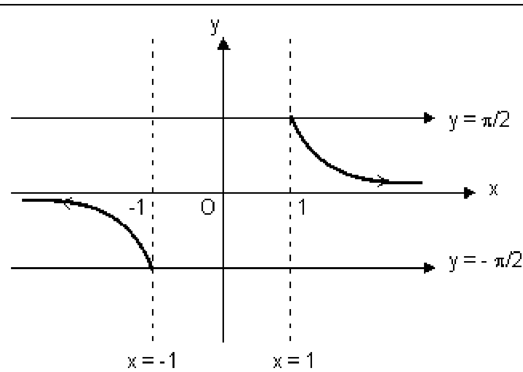
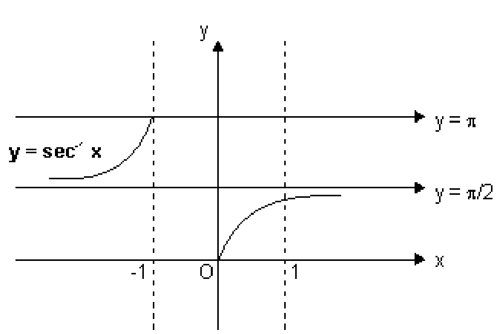
TRIGNOMETRIC FUNCTIONS

	FUNCTION	PERIOD	GRAPH	DOMAIN	RANGE
a)	***sinx***	2π		ℝ	[-1,1]
b)	***cosx***	2π		ℝ	[-1,1]
c)	***tanx***	π		$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}\right\}$	ℝ

d)	***cosecx***	***2π***		$\mathbb{R}-\{n\pi : n\in\mathbb{Z}\}$	$\mathbb{R}-(-1,1)$
e)	***secx***	***2π***		$\mathbb{R}-\left\{(2n+1)\frac{\pi}{2}\right\}$	$\mathbb{R}-(-1,1)$
f)	***cotx***	***π***		$\mathbb{R}-\{n\pi : n\in\mathbb{Z}\}$	$\mathbb{R}$

<u>INVERSE TRIGNOMETRIC FUNCTIONS</u>

	FUNCTION	GRAPH	DOMAIN	RANGE

<table>
<tr><td>a)</td><td>$\sin^{-1} x$</td><td></td><td>[-1, 1]</td><td>$\left[-\dfrac{\pi}{2},\dfrac{\pi}{2}\right]$</td></tr>
<tr><td>b)</td><td>$\cos^{-1} x$</td><td></td><td>[-1, 1]</td><td>$[0,\pi]$</td></tr>
<tr><td>c)</td><td>$\tan^{-1} x$</td><td></td><td>$\mathbb{R}$</td><td>$\left[-\dfrac{\pi}{2},\dfrac{\pi}{2}\right]$</td></tr>
<tr><td>d)</td><td>$\cos ec^{-1} x$</td><td></td><td>$\mathbb{R}-(-1,1)$</td><td>$\left[-\dfrac{\pi}{2},\dfrac{\pi}{2}\right]-\{0\}$</td></tr>
<tr><td>e)</td><td>$\sec^{-1} x$</td><td></td><td>$\mathbb{R}-(-1,1)$</td><td>$[0,\pi]-\left\{\dfrac{\pi}{2}\right\}$</td></tr>
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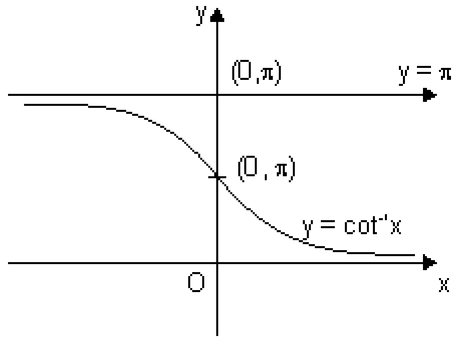
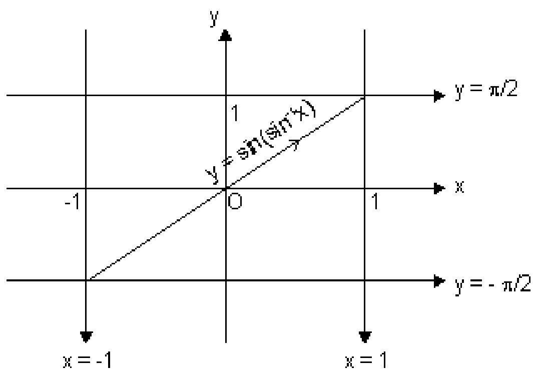
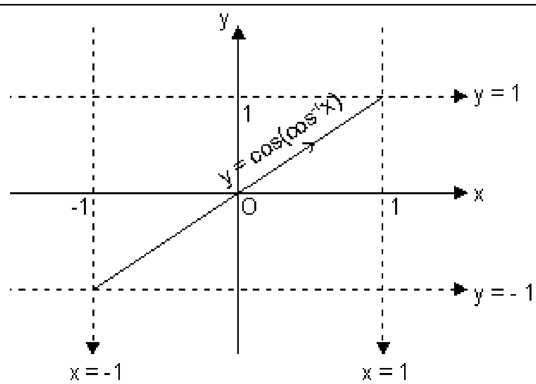
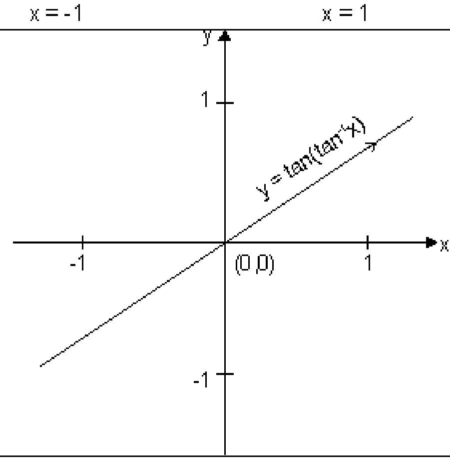
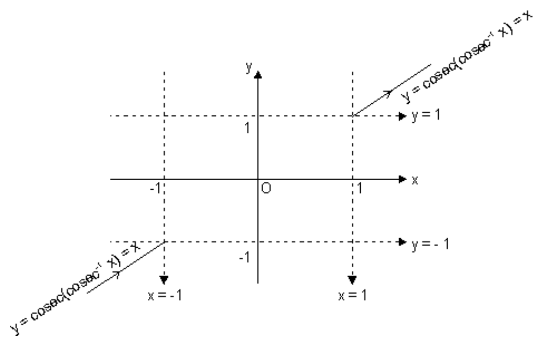
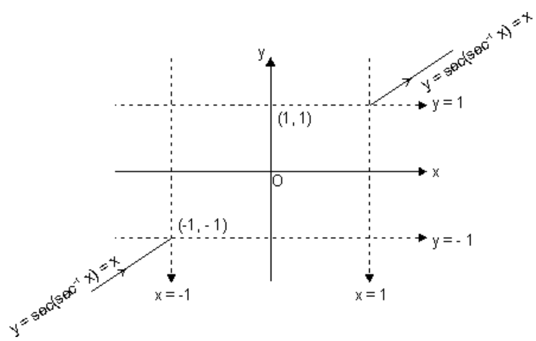
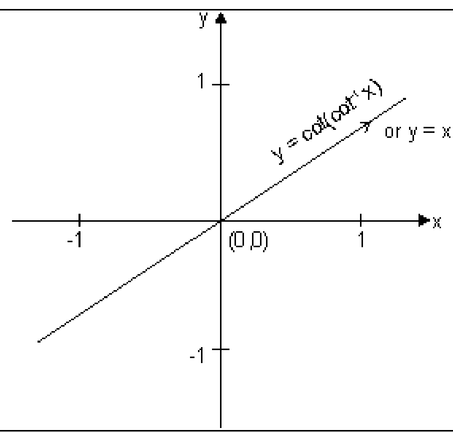
<table>
<tr><td>f)</td><td>$\cot^{-1} x$</td><td>
</td><td>$\mathbb{R}$</td><td>$(0, \pi)$</td></tr>
<tr><td>g)</td><td>sin(sin⁻¹x)</td><td>
</td><td>[-1, 1]</td><td>[-1, 1]</td></tr>
<tr><td>h)</td><td>cos(cos⁻¹x)</td><td>
</td><td>[-1, 1]</td><td>[-1, 1]</td></tr>
<tr><td>i)</td><td>tan(tan⁻¹x)</td><td>
</td><td>$\mathbb{R}$</td><td>$\mathbb{R}$</td></tr>
</table>

j)	**cosec(cosec^{-1}x)**		$\mathbb{R}-(-1,1)$	$\mathbb{R}-(-1,1)$
k)	**sec(sec^{-1}x)**		$\mathbb{R}-(-1,1)$	$\mathbb{R}-(-1,1)$
l)	**cot(cot^{-1}x)**		$\mathbb{R}$	$\mathbb{R}$

1.3

PROPERTIES,DOMAIN AND RANGE

Properties of some standard functions:

1. **Properties of modulus function**:

(i) $|x| \le a \Rightarrow -a \le x \le a \quad (a \ge 0)$

(ii) $|x| \ge a \Rightarrow x \le -a$ or $x \ge a \quad (a \ge\ )$

(iii) $|x + y| \le |x| + |y|$

(iv) $|x+y| \ge \big||x|-|y|\big|$

2. **Properties of greatest integer function**:

(i) $[x] = x$ holds if $x \in I$ (I is Integer).

(ii) $[x + I] = [x] + I$ if I is integer.

(iii) $[x + y] \geq [x] + [y]$

(iv) If $[\phi(x)] \geq I$, then $\phi(x) \geq I$.

(v) If $[\phi(x)] \leq I$, then $\phi(x) < I + 1$

(vi) If $[x] > n \Rightarrow x \geq n + 1$, $n \in I$.

(vii) $[-x] = -[x]$ if $x \in I$
& $[-x] = -[x] - 1$ if $x \notin I$.

(viii) $[x + y] = [x] + [x + y - [x]]$ for all $x, y \in R$.

(ix) $[x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + \ldots\ldots\ldots\ldots + \left[x + \frac{n-1}{n}\right]$
$= [n x]$, $n \in N$ $N \rightarrow$ natural no.

3. **Properties of fraction part of x**

(i) $\{x\} = x$ if $0 \leq x < 1$

(ii) $\{x\} = 0$ if $x \in I$

(iii) $\{-x\} = 1 - \{x\}$ if $x \notin I$

4. **Properties of logarithmic function:**

(i) $\log_a a = 1$ $(a>0, a\neq 1)$

(ii) $\log_e(ab) = \log_e a + \log_e b$ $(a,b>0)$

(iii) $\log_e(a/b) = \log_e a - \log_e b$ $(a,b>0)$

(iv) $\log_e a^m = m \log_e a$ $(a>0, m\in R)$

(v) $\log_b a = 1/(\log_a b)$ $(a,b>0$ and $a,b\neq 1)$

(vi) $\log_b a = \log_m a / \log_m a$ $(a,b,m>0 \neq 1)$

(vii) $a^{\log_a m} = m$ $(a>0, a\neq 1; m>0)$

(viii) $\log_m x > \log_m y \Rightarrow \begin{Bmatrix} x > y; m > 1 \\ x < y; 0 < m < 1 \end{Bmatrix}$

(ix) $\log_m a > b \Rightarrow \begin{Bmatrix} a > m^b; m > 1 \\ a < m^b; 0 < m < 1 \end{Bmatrix}$

(x) $\log_m a < b \Rightarrow \begin{Bmatrix} a < m^b; m > 1 \\ a > m^b; 0 < m < 1 \end{Bmatrix}$

(xi) $\log_m a = b \Rightarrow a = m^b$ (m,a>0;m≠1b∈R)

Exercise1.2:

Q1: Solve the following equations:

I. $x^2 + 3x + 2 > 0$

II. $x^2 + 3x + 5 < 0$

III. $x^2 + 2x + 1 > 0$

IV. $\dfrac{(x-1)(x-2)}{(x-3)(x+1)} > 0$

V. $1 \le |x-1| \le 3$

VI. $\dfrac{2x}{2x^2 + 5x + 2} > \dfrac{1}{x+1}$

VII. $\left| \dfrac{x^2 - 5x + 4}{x^2 - 4} \right| \le 1$

VIII. $\log_{10}(x^2 - 3x + 3) \ge 0$

IX. $\dfrac{(2x-1)(x-1)^2(x-2)^2}{(x-4)^4} > 0$

X. $\dfrac{|x|-1}{|x|-2} \ge 0, x \in \mathbb{R}, x \ne \pm 2$

XI. $\dfrac{|x+3| + x}{x+2} > 1$

XII. $|x+1| + \sqrt{x-1} = 0$

XIII. $\dfrac{x^2 - 2x + 2^{|a|}}{x^2 - a^2} > 0$

XIV. $\log_3(5 + 4\log_3(x-1)) = 2$

XV. $\log_2(3-x) + \log_2(1-x) = 3$

XVI. $\log_{\frac{1}{5}} \dfrac{4x+6}{x} \ge 0$

XVII. $\log_{\frac{x+4}{2}} \log_2 \dfrac{2x-1}{3+x} < 0$

XVIII. $4\{x\} = x + [x]$

Exercise1.3:

Q1: Find the domains of the following functions, assuming the functions to be real for given $f(x) =$

I. $\sqrt{1-x^2}$

II. $\dfrac{1}{\sqrt{1-x^2}}$

III. $\dfrac{1}{1+\sin x}$

IV. $\sqrt{2-x}+\sqrt{3+x}$

V. $\log_e \left|x^2-3x+2\right|$

VI. $\sqrt{\sin x - 1}$

VII. $\dfrac{1}{[x]}$

VIII. $\log[x]$

IX. $\sin\sqrt{x}$

X. $\dfrac{1}{\sqrt{[x]^2-3[x]+2}}$

XI. $\log_2 \log_3 \log_4 \log_5 x$

XII. $\sqrt{x^2-1}+\dfrac{1}{\sqrt{x^2-5x+4}}$

XIII. $\sin\left(\ln\left(\dfrac{\sqrt{4-x^2}}{1-x}\right)\right)$

XIV. $\dfrac{1}{\sqrt{[x]-x}}$

XV. $\dfrac{\sqrt{1}}{\sqrt{|x|-x}}$

XVI. $\log\left(\log_{|\sin x|}(x^2-8x+23)-\dfrac{3}{\log_2|\sin x|}\right)$

XVII. $\sqrt{\log_{.4}\left(\dfrac{x-1}{x+5}\right)}$

XVIII. $\cos^{-1}\left(\dfrac{1-2|x|}{3}\right)+\log_{|x-1|} x$

XIX. $[\sin x]\cos\left(\dfrac{\pi}{[x-1]}\right)$

XX. $\dfrac{1}{\sqrt{\log_{1/2}(x^2-7x+13)}}$

Q2: Find the domain of single valued function y = f(x) given by the equation $10^x+10^y=10$

Exercise1.4:

Q1: Find the range of the following functions, assuming real functions given by $f(x) =$

I. x

II. x^2+1

III. $\sqrt{\sin x}$

IV. $[\sin x]$

V. $\dfrac{1}{[x]}$

VI. $\dfrac{1}{x+1}$

VII. $\left[\dfrac{1}{x}\right], x>0$

VIII. $[|\cos x|]$

IX. $|[\cos x]|$

X. x^2+3x+2

XI. $\dfrac{x^2-3x+2}{x^2+x-6}$

XII. $\dfrac{x^2+x-1}{x^2-x+2}$

XIII. $\dfrac{1}{2+\sin 3x+\cos 3x}$

XIV. $[x^2]-[x]^2$

XV. x^3+3x^2+4x+5

XVI. $\dfrac{x-[x]}{1-[x]+x}$

XVII. $\log_e(3x^2-4x+5)$

XVIII. $\log_3(\log_{1/2}(x^2+4x+4))$

1.4

ANALYSIS OF A FUNCTION

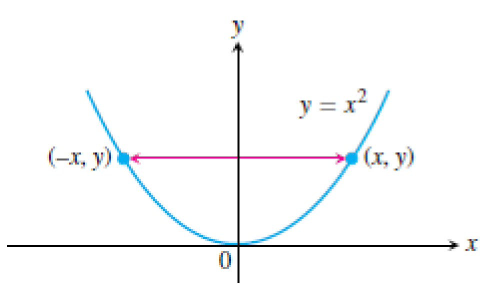
Even and odd functions

Definition: A function y=f(x) is an

Even function of x if $f(-x)=f(x)$

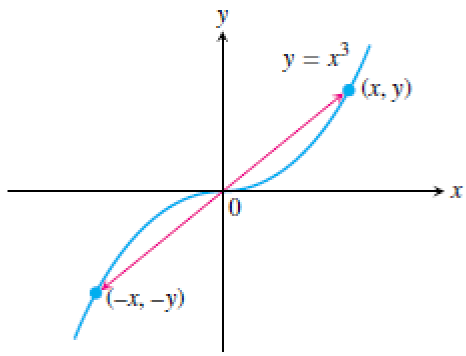
Odd function of x if $f(-x)=-f(x)$

Graphs of even function are symmetric about y axis ,example $y=x^2$



(a)

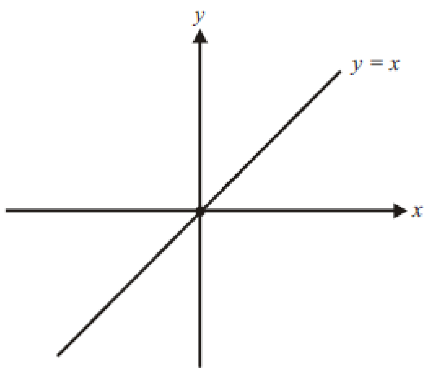
Graphs of odd function are symmetric about origin ,example $y = x^3$



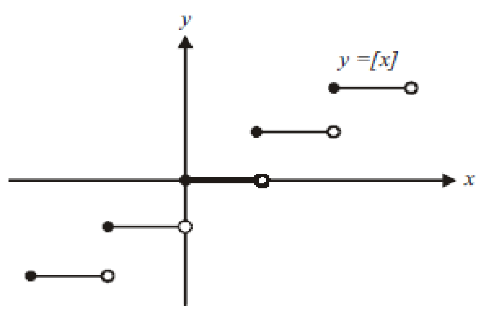
Increasing and decreasing function:

If the graph of a function *climbs* or *rises* as you move from left to right, we say that the function is *increasing*. If the graph *descends* or *falls* as you move from left to right, the function is *decreasing*

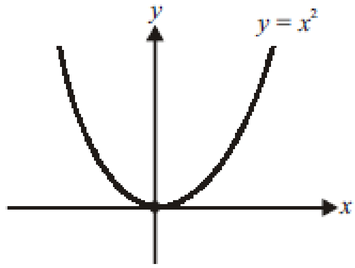
If	$x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$	function is strictly increasing
If	$x_1 < x_2 \Rightarrow f(x_1) \le f(x_2)$	function is increasing
If	$x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$	function is strictly deccreasing
If	$x_1 > x_2 \Rightarrow f(x_1) \ge f(x_2)$	function is decreasing



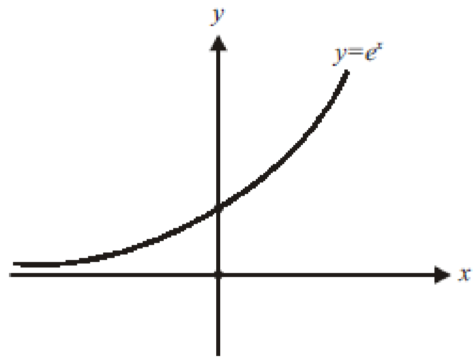
strictly increasing



increasing



strictly decreasing on $(-\infty,0]$
strictly increasing on $[0, \infty)$



strictly increasing on $[-\infty, 0)$

Periodic function:

A function $f(x)$ is **periodic** if there is a positive number T such that $f(x+T) = f(x)$ for every value of x. The smallest such value of T is the **period** of f.
For example, sin x has periods 2π , 4π ,6π , -4π , $2n\pi$ etc. but the fundamental period is 2π.

Properties of periodic function:

(i) If f(x) is periodic with period T, then, c.f(x), f(x + c) & f(x) ± c is periodic with period T.
eg. if sin x has period 2π . Then f(x) = 5 sinx + 4 is also periodic with period 2π .

(ii) If f(x) is periodic with period T, then, kf(cx + d) has period $\frac{T}{|C|}$

f(x) = 7 sin2x - 12 has period $\frac{2\pi}{|2|} = \pi$ as sinx is periodic with period .

(iii) If $f_1(x)$, $f_2(x)$ are periodic functions with periods T_1, T_2 respectively then; h(x) = $af_1(x) \pm bf_2(x)$ has period as

$$= \begin{cases} \text{LCm of } \{T_1, T_2\}; \text{ If } h(x) \text{ is not an even function} \\ \qquad\qquad\text{or} \\ \frac{1}{2} \text{ LCM of } \{T_1, T_2\}, f_1(x) \ \& f_2(x) \text{ are complementar pair wise comparable functions} \end{cases}$$

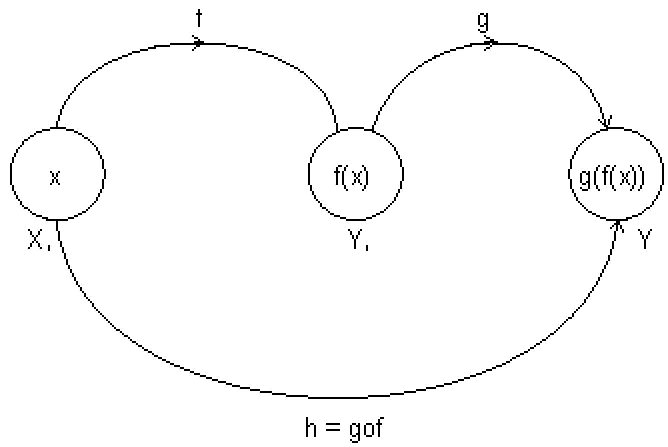
LCM of $\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{\text{LCM of } (a, b, c)}{\text{HCF of } (b, d, f)}$

Note: LCM of rational and irrational is not possible. eg. 2π . is irrational & 1 is rational.

Composite Function:

Let two function,

f : x $\rightarrow$ Y_1 and g : $Y_1 \rightarrow$ Y



Let another function h : X $\rightarrow$ Y

Such that

h(x) = g(f(x)) = (gof)(x)

domain (gof) = {x : x $\in$ domain (f), f(x) $\in$ domain (a)}

domain fog(x) = {x : x $\in$ domain (g), g(x) $\in$ domain f}

gof exist iff range of f $\subseteq$domain of g

fog exist iff range of g $\subseteq$domain of f.

<u>Properties of composite function</u>:

(i) f is even, g is even $\Rightarrow$ fog is even function.

(ii) f is odd, g is odd $\Rightarrow$ fog is odd function.

(iii) f is even, g is odd $\Rightarrow$ fog is even function.

(iv) f is odd, g is even $\Rightarrow$ fog is even function.

<u>Exercise1.5:</u>

Q1: Check weather the following functions $f(x)$ are even or odd

I. $\sin x - \cos x$

II. $\frac{x}{e^x - 1} + \frac{x}{2} + 1$

III. $\sqrt[3]{(1-x)^2} + \sqrt[3]{(1+x)^2}$

IV. $\dfrac{x(\sin x+\tan x)}{\left[\dfrac{x+\pi}{\pi}\right]-\dfrac{1}{2}}$

Q2: If the function f satisfies $f(x+y)+f(x-y)=2f(x).f(y)\ \forall\ x, y\in R$, and $f(0)\neq 0$, prove that $f(x)$ is even.

Q3: If f is non-periodic and g is periodic, what can you say about $f($

$g(x))$ and $g\,(f(x))$?

Q4: What are the periods of the following functions $f(x)$=

I. $|\sin 2x|+|\sin 3x|$

II. $\sin^4 x+\cos^4 x$

III. $2\cos\left(\dfrac{x-\pi}{3}\right)$

IV. $\sin^2 \pi x$

Q5: Prove that if f is increasing, f^{-1} is increasing and if f is decreasing, f^{-1} is decreasing

Q6: If $f(x)=\sin x+\cos x$ and $g(x)=x^2-1$ then find the interval in which g(f(x) is invertible?

Q7: If $f(x+y)=f(x).f(y)\ \forall$ real x,y and $f(0)\neq 0$ then prove that the function $f(x)=\dfrac{f(x)}{1+(f(x))^2}$ is an even function.

Q8: If $f:[1,\infty)\to[2,\infty)$ is given by $f(x)=x+\dfrac{1}{x}$,then find $f^{-1}(x)$ (assuming bijective)

Q9: If $f:[1/2,\infty)\to[3/4,\infty)$ is given by $f(x)=x^2-x+1$,then find $f^{-1}(x)$ (assuming bijective)

Q10: Find the inverse of the following functions (assuming bijective)

I. $f(x)=\sin^{-1}(x/3), x\in[-3,3]$

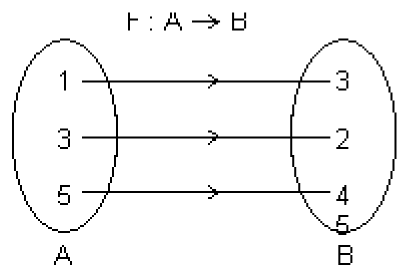
II. $f(x)=\ln(x^2+3x+1), x\in[1,3]$

Q11: If $f(x)=\begin{cases} x-1; -1\le x\le 0 \\ x^2; 0\le x\le 1\end{cases}$ and $g(x)=\sin x$. Then find $h(x)=f\,|g(x)|+|f(g(x))|$

Mapping of a Function:

Mapping of function: A function exists only if," to every element in domain there exists unique image in co domain.
$f : A \rightarrow B$

One - One Mapping (Injective): A function f: $A \rightarrow B$ is said to bwe one-one mapping if different elements of A have different images in B. $\Rightarrow$ no two elements of set A can have same 'f' image
eg.



f(x) is one-one

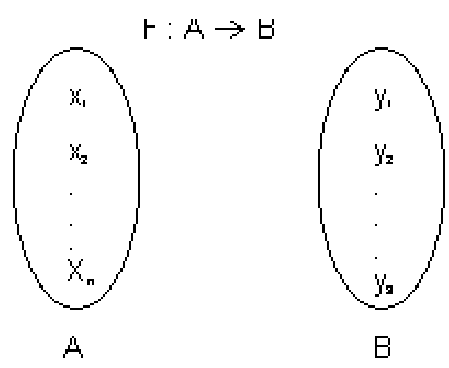
Methods to find one-one mapping:

(1) **Graphically**: A fun. is one-one if no line 11 to x-axis meets graph of function at more than one point.

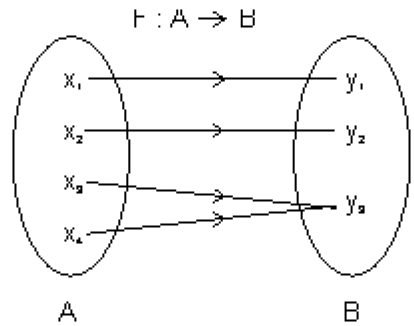
By Calculus: If f(x) is function and if $f'(x) \geq 0 \; \forall x \in$ domain i.e. increasing or $f'(x) \leq 0 \; \forall x \in$ domain i.e. decreasing, then one-one.

No. of one-one Mapping: If A & B are finite sets having m & h elements, then, no. of one-one function from A to B

$$= \begin{cases} {}^{n}P_m & \text{if } n \geq m \\ 0 & \text{if } n < m \end{cases}$$



Many one Mapping: A mapping $f : A \rightarrow B$ is many one if two or more elements of set A have the same image in B.

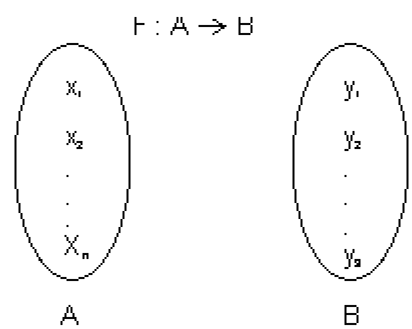


i.e. $f(x^3) = f(x_4) = y_3$

<u>Onto Function (Surjective)</u>: If function $f : A \rightarrow B$ is such that each element of B is the 'f' image of at least one element in A.
i.e. range of f = co domain $\Rightarrow f(A) = B$

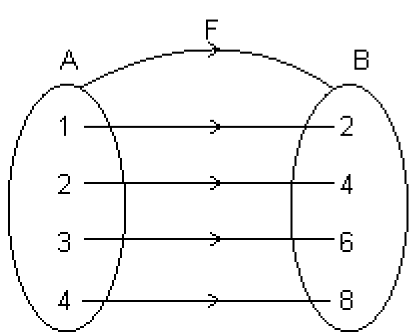
<u>Into Function</u>: A function of $f : A \rightarrow B$ is into if there exists an element in B having no pre image in A.

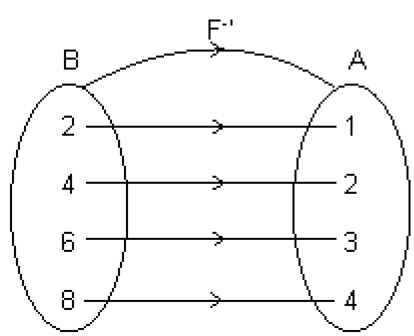
<u>No. of one-one onto mapping (bijection)</u>: If A & B are finite sets & $f : A \rightarrow B$ is a bijection.
Then A & B have same no. of elements.



Total no. of bijection $A \rightarrow B = n\,!$

<u>Inverse function</u>: Let $f : A \rightarrow B$ be one-one & onto function then there exists a unique function.





$g : B \rightarrow A$ such that $f(x) = y \Leftrightarrow g(y) = x, \forall x \in A$ & $y \in B$
Then g is c/d inverse of F.
So, $g = f^{-1} : B \rightarrow A = \{(f(x), x) \mid (x, f(x) \in f\}$

Domain of f = {1, 2, 3, 4} = range of f^{-1}
Range of f = {2, 4, 6, 8} = domain of f^{-1}.

Note: (1) Graph of y = f(x) and y = f^{-1}(x) are mirror images to each other.

(2) If f : A $\rightarrow$ B & g : B $\rightarrow$ C are two bijections, then gof : A $\rightarrow$ C is bijection & $(gof)^{-1} = (f^{-1}og^{-1})$.

(3) If fog = gof then either f^{-1} = g or g^{-1} = f and (fog)(x) = (gof)(x) = (x)

Exercise1.6:

Q1: Is $f(x) = \dfrac{x+1}{x+2}$ one one or many one?

Q2: Is $f(x) = \dfrac{x^2-8x+18}{x^2+4x+30}$ one one or many one?

Q3: Are the following functions into or onto?

I. $f:\mathbb{R}\rightarrow\mathbb{R}$ $\quad f(x)=\sin x$

II. $f:[-1,1]\rightarrow[0,1]$ $\quad f(x)=x^2$

III. $f:(0,\infty)\rightarrow\mathbb{R}$ $\quad f(x)=\ln x$

IV. $f:(1,\infty)\rightarrow\mathbb{R}$ $\quad f(x)=\ln x$

Exercise level 1:

1. The period of the function, $f(x)=[\sin 3x]+|\cos 6x|$ is : ([.] denotes the greatest integer less than or equal to x)

(a) π (b) $\frac{2\pi}{3}$ (c) 2π (d) $\frac{\pi}{2}$

---

2. The function $f(x)=\frac{\sec^{-1} x}{\sqrt{x-[x]}}$, where $[x]$ denotes the greatest integer less than or equal to x is defined for all x belonging to :

(a) R (b) R - {(-1, 1) $\cup$ {n : n $\in$ I)} (c) R' - (0, 1) (d) R' - {n : n $\in$ N}

---

3. The function $f(x)=\sqrt{\log_{x^2}(x)}$ is defined for x belonging to :

(a) $(-\infty, 0)$ (b) $(1, \infty)$ (c) $(0, \infty)$ (d) none of these

---

4. If $f(x)=\sin^2 x+\sin^2\left(x+\frac{\pi}{3}\right)+\cos x.\cos\left(x+\frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right)=1$, then (gof)(x) =

(a) 1 (b) -1 (c) x (d) none of these

---

5. Let [x] denote the greatest integer $\le x$. The domain of definition of function $f(x)=\sqrt{\frac{4-x^2}{[x]+2}}$ is

(a) $(-\infty, -2)\cup[-1, 2]$ (b) [0, 2] (c) [-1, 2] (d) (0, 2)

---

6. The domain of definition of the function $f(x)=\sqrt{\log_{10}\left(\frac{5x-x^2}{4}\right)}$ is

(a) [1, 4] (b) (1, 4) (c) (0, 5) (d) [0, 5]

---

7. The range of the function $f(x)=\frac{1}{2-\cos 3x}$ is

(a) $\left[-\frac{1}{3},0\right]$ (b) R (c) $\left[\frac{1}{3},1\right]$ (d) none of these

---

8. The domain of definition of the function $f(x)=\frac{1}{\sqrt{|x|-x}}$ is

(a) R (b) $(0, \infty)$ (c) $(-\infty, 0)$ (d) none of these

---

9. The function $f(x)=\log(x+\sqrt{x^2+1})$ is

(a) an even function (b) an odd function (c) periodic function (d) none of these

10. The function $f(x)=\cos\left(\log(x+\sqrt{x^2+1})\right)$ is

(a) even (b) odd (c) constant (d) none of these

11. Thr period of the function $f(x) = \sin^4 x + \cos^4 x$ is

(a) π (b) $\pi/2$ (c) 2π (d) none of these

12. Which of the following functions is an odd function

(a) $f(x) = \text{constant}$ (b) $f(x) = \sin x + \cos x$ (c) $f(x)=\sin\left(\log(x+\sqrt{x^2+1})\right)$ (d) $f(x) = 1 + x + 2x^3$

13. The domain of definition of the function $f(x) = {}^{7-x}P_{x-3}$ is

(a) $[3, 7]$ (b) $\{3,4,5,6,7\}$ (c) $\{3,4,5\}$ (d) none of these

14. The function $f(x)=\dfrac{\sin^4 x+\cos^4 x}{x+x^2\tan x}$ is

(a) even (b) odd (c) periodic with period π (d) periodic with period 2π

15. Range of $f(x) = |x| + |x+1|$ is

(a) $[0, \infty)$ (b) $(0, \infty)$ (c) $(1, \infty)$ (d) $[1, \infty)$

16. Range of $\tan^{-1}x - \cot^{-1}x$ is

(a) $(0, \pi)$ (b) $\left[\dfrac{-3\pi}{2},\dfrac{\pi}{2}\right]$ (c) $\left(\dfrac{2}{3},1\right)$ (d) $\left[\dfrac{2}{3},1\right]$

17. The range of $\sqrt{|x|-x}$ is

(a) $(0, \infty)$ (b) $[0, \infty)$ (c) $(-\infty, 0)$ (d) $(-\infty, 0]$

18. If $f(x)=x^2\left(\dfrac{10^{2x}-1}{10^{2x}+1}\right)$ then 'f' is

(a) an even function (b) an odd function (c) neither even nor odd (d) cannot be determined

19. The domain of the function $f(x)=\dfrac{1}{\log_{10}(1-x)}+\sqrt{(x+2)}$ is

(a) $[-3, -2]$ excluding (-2.5) (b) $[0, 1]$ excluding 0.5

(c) $[-2, 1]$, excluding 0 (d) None of these

20. The period of $e^{\cos^4 \pi x+x-[x]+\cos^2 \pi x}$ is ([.] denotes the greatest integer function)

(a) 2 (b) 1 (c) 0 (d) -1

21. The domain of the function $f(x)=\sqrt{\sin^{-1}(\log_2 x)}+\sqrt{\cos(\sin x)}+\sin^{-1}\left(\dfrac{1+x^2}{2x}\right)$

(a) $\{x : 1 \le x \le 2\}$ (b) $\{1\}$ (c) Not defined for any value x (d) $\{-1, 1\}$

22. If [.] denotes the greatest integer function then the domain of the real valued function $\log_{[x+1/2]}|x^2-x-2|$ is

(a) $\left[\frac{3}{2},\infty\right)$ (b) $\left[\frac{3}{2},2\right)\cup(2,\infty)$ (c) $\left(\frac{1}{2},2\right)\cup(2,\infty)$ (d) None of these

23. The domain of the function $f(x)=\sqrt[6]{4^x+8^{\frac{2}{3}(x-2)}-52-2^{2(x-1)}}$ is

(a) (0, 1) (b) $[3, \infty)$ (c) (1, 0) (d) none

24. A function whose graph is symmetrical about the y-axis is given by

(a) $f(x)=\log_e(x+\sqrt{x^2+1})$ (b) $f(x+y)=f(x)+f(y)$ for all $x, y \in R$

(c) $f(x)=\cos x+\sin x$ (d) none of these

25. If $f(x)=x^2-x+1,\ x\in\left[\frac{1}{2},\infty\right)$ then value of 'x' satisfying $f(x)=f^{-1}(x)$ is

(a) 1 (b) 2 (c) $\frac{1}{2}$ (d) none of these

26. If $f(x)=\sqrt{\log_{0.4}\left(\frac{x-1}{x-5}\right)}$ and $g(x)=x^2-36$, then $D_{f/g}$ is

(a) $(-\infty, 0)\sim\{-6\}$ (b) $(0,\infty)\sim\{1, 6\}$

(c) $(1,\infty)\sim\{6\}$ (d) $[1,\infty)\sim\{6\}$

27. The domain of the real-valued function $f(x)=\log_e|\log_e x|$ is

(a) $(0,1)\cup(1,\infty)$ (b) $(0,\infty)$ (c) (e,∞) (d) $(1,\infty)$

28. If f(x) and g(x) are two functions of 'x' such that $f(x)+g(x)=e^x$ and $f(x)-g(x)=e^{-x}$, then

(a) f(x) is odd, g(x) is odd (b) f(x) is even, g(x) is even

(c) f(x) is even, g(x) is odd (d) f(x) is odd, g(x) is even

29. The inverse of the function $y=\log_a(x+\sqrt{x^2+1})\ (a>0,\ a\neq 1)$ is

(a) $\frac{1}{2}(a^x-a^{-x})$ (b) not defined for all x

(c) defined for only positive x (d) none of these

30. Let $f(x)=\cos\sqrt{p}\,x$, where p = [a] = the greatest integer less than or equal to 'a'. If the period of f(x) is π, then

(a) $a\in[4,5]$ (b) $a=[4,5]$ (c) $a\in[4,5)$ (d) none of these

31. If $f(x)=e^{x-[x]+|\cos\pi x|+|\cos 2\pi x|+........+|\cos n\pi x|}$, then period of f(x) is

(a) 1 (b) $\frac{1}{n}$ (c) $\frac{n^2-1}{n^2+1}$ (d) $\frac{1}{1.2.3......n}$

32. Let $f(x)=\sin^{-1}\left(\frac{2-|x|}{4}\right)+\cos^{-1}\left(\frac{2-|x|}{4}\right)+\tan^{-1}\left(\frac{2-|x|}{4}\right)$, then D_f is

(a) [-6, 6] (b) $[6, \infty)$ (c) [-6, 3] (d) [-3, 6]

33. Identify the statement(s) which is/are incorrect ?

(a) The function $f(x) = \cos(\cos^{-1} x)$ is neither odd nor even

(b) The fundamental period of $f(x)=\cos(\sin x)+\cos(\cos x)$ is π

(c) The range of the function $f(x) = \cos(3\sin x)$ is [-1, 1]

(d) None of these

34. Let $f(x)=4\cos\sqrt{x^2-\frac{\pi^2}{9}}$. Then

(a) $D_f=\left[\frac{\pi}{3}, \infty,\right), R_f=[-1, 1]$ (b) $D_f=\left[\frac{\pi}{3}, \infty,\right), R_f=[-2, 2]$

(c) $D_f=\left(-\infty, -\frac{\pi}{3}\right]\cup\left[\frac{\pi}{3}, \infty\right)$ and $R_f=[-4, 4]$ (d) $D_f=\left(-\infty, -\frac{\pi}{3}\right), R_f=(0, 4]$

35. The range of the function $f(x)=\sin(xe^{[x]}+x^2-x),\ x\in(-1, \infty)$ where [x] denotes the greatest integer function is:

(a) ϕ (b) [0, 1] (c) [-1, 1] (d) R

36. The domain of $f(x) = \log_2 \log_3 \log_{4/\pi} (\tan^{-1} x)^{-1} =$

(a) R (b) $(4/\pi, \infty)$ (c) (0, 1) (d) None of these

37. If f(x) is a periodic function of the period 'λ' then $f(\lambda x + a)$, where a is a constant, is a periodic function of the period

(a) 1 (b) λ (c) $\frac{\lambda}{a}$ (d) none of these

38. If $f(x)=\cos^{-1}(x-x^2)+\sqrt{\left(1-\frac{1}{|x|}\right)}+\frac{1}{[x^2-1]}$ then domain of f(x) is (where [.] is the greatest integer)

(a) $\left[\sqrt{2}, \frac{1+\sqrt{5}}{2}\right]$ (b) $\left(\sqrt{2}, \frac{1+\sqrt{5}}{2}\right]$ (c) $\left(-\sqrt{2}, \frac{1-\sqrt{2}}{2}\right)$ (d) none of these

39. Let $f(x) = \sin x$, $g(x) = \ln|x|$. If the ranges of the composite functions fog and gof are R_1 and R_2 respectively, then

(a) $R_1=(-1,1), R_2=(-\infty,0)$ (b) $R_1=(-\infty,0], R_2=[-1,1]$

(c) $R_1=(-1,1), R_2=(-\infty,0]$ (d) $R_1=[-1,1], R_2=(-\infty,0]$

40. If $f(x)=\log\frac{1+x}{1-x}$, then

(a) f(x) is even (b) $f(x_1).f(x_2)=f(x_1+x_2)$ (c) $\frac{f(x_1)}{f(x_2)}=f(x_1-x_2)$ (d) f(x) is odd

Exercise level 2:

1. The domain of the function $f(x) = \log(1-x) + \sqrt{x^2 - 1}$ is

   (a) [-1, 1] (b) $(1, \infty)$ (c) (0, 1) (d) $(-\infty, -1]$

2. The range of the function $f(x) = \dfrac{1+x^2}{x^2}$ is equal to

   (a) [0, 1] (b) (0, 1) (c) $(1, \infty)$ (d) $[1, \infty)$

3. The curves $y = |x|^3 + 3|x|^2 + 2$ and $y = x^3 + 3x^2 + 2$ have the same graph for

   (a) $x > 0$ (b) $x \geq 0$ (c) all x except 0 (d) all x

4. Domain of the function $f(x) = \dfrac{1}{x^2} + 2^{\sin^{-1} x} + \dfrac{1}{\sqrt{x-3}} + 7$ is

   (a) ϕ (b) R - {0} (c) R (d) None of these

5. The domain of definition of the function $y = 3e^{\sqrt{x^2-1}} \log(x-1)$ is

   (a) $(1, \infty)$ (b) $[1, \infty)$ (c) $R \sim \{1\}$ (d) $(-\infty, -1) \cup (1, \infty)$

6. The range of the function f(x) = cos [x], where $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$, is

   (a) {-1, 1, 0} (b) {cos 1, 1, cos 2} (c) $\{\cos 1, -\cos 1, 1\}$ (d) none of these

7. If $b^2 - 4ac = 0$ and $a > 0$, then domain of the function $y = \log(ax^3 + (a+b)x^2 + (b+c)x + c)$ is

   (a) $R \sim \left\{-\dfrac{b^2}{2a}\right\}$ (b) $R \sim \left(\left\{-\dfrac{b}{2a}\right\} \cup \{x \mid x \geq -1\}\right)$

   (c) $R \sim \left(\left\{-\dfrac{b}{2a}\right\} \cup (-\infty, -1]\right)$ (d) none of these

8. Which of the following functions is an even function?

   (a) $f(x) = \dfrac{a^x + a^{-x}}{a^x - a^{-x}}$ (b) $f(x) = \dfrac{a^x + 1}{a^x - 1}$ (c) $f(x) = x\dfrac{a^x - 1}{a^x + 1}$ (d) $f(x) = \log_2\left(x + \sqrt{x^2 + 1}\right)$

9. If $(\log_3 x)(\log_x 2x)(\log_{2x} y) = \log_x x^2$, then y is equal to

   (a) 9 (b) 18 (c) 27 (d) 81

10. The range of the function $f(x) = \dfrac{x^2 - x + 1}{x^2 + x + 1}$ is

(a) R (b) $[3, \infty)$ (c) $\left[\dfrac{1}{3}, 3\right]$ (d) none of these

11. The domain of the function $f(x) = \sqrt{2 - 2x - x^2}$ is

(a) $-3 \le x \le \sqrt{3}$ (b) $-1-\sqrt{3} \le x \le -1+\sqrt{3}$ (c) $-2 \le x \le 2$ (d) none of these

12. If the domain of the function $f(x) = x^2 - 6x + 7$ is $(-\infty, \infty)$, then range of the function is

(a) $(-\infty, \infty)$ (b) $[-2, \infty)$ (c) (-2, 3) (d) $(-\infty, -2)$

13. The domain of the function $f(x) = \sin^{-1}\left(\log_2 \dfrac{x^2}{2}\right)$ is

(a) $-1 \le x \le 1$ (b) $0 \le x \le 1$ (c) $1 \le x \le 2$ (d) none of these

14. If f(x) = cos (log x) then $f(x)f(y) - \dfrac{1}{2}\left(f\left(\dfrac{x}{y}\right) + f(xy)\right)$ has the value

(a) 1 (b) $\dfrac{1}{2}$ (c) -2 (d) 0

15. The inverse of the function $f(x) = \dfrac{e^x - e^{-x}}{e^x + e^{-x}} + 2$ is given by

(a) $\log_e\left(\dfrac{x-2}{x-1}\right)^{1/2}$ (b) $\log_e\left(\dfrac{x-1}{3-x}\right)^{1/3}$ (c) $\log_e\left(\dfrac{x}{2-x}\right)^{1/2}$ (d) $\log_e\left(\dfrac{x-1}{x+1}\right)^{-2}$

16. The range of the function for real x of $y = \dfrac{1}{2 - \sin 3x}$ is

(a) $\dfrac{1}{3} \le y \le 1$ (b) $-\dfrac{1}{3} \le y < 1$ (c) $-\dfrac{1}{3} > y > 1$ (d) $\dfrac{1}{3} > y > 1$

17. The domain of the function $\dfrac{1}{[x]} + \sqrt{2x - x^2}$, where [.] denotes greatest integer function.

(a) [1, 2] (b) [0, 2] (c) [0, 1) (d) [1, 2)

18. Domain of $\sin^{-1}(\log_3(x/3))$ is

(a) [1, 9] (b) [-1, 9] (c) [-9, 1] (d) [-9, -1]

19. The range of $f(x) = {}^{7-x}P_{x-3}$ is

(a) {1, 2, 3} (b) {1, 2, 3, 4, 5, 6} (c) {1, 2, 3, 4} (d) {1, 2, 3, 4, 5}

20. The value of $n \in Z$ for which the function $f(x) = \dfrac{\sin nx}{\sin\left(\dfrac{x}{n}\right)}$ has 4π as its period is

(a) 2 (b) 3 (c) 5 (d) 4

21. The domain of the function $f(x) = \log_2(\log_3(\log_4 x))$ is

(a) $x < 4$ (b) $x > 4$ (c) $0 < x < 2$ (d) $2 < x < 4$

22. If f(x) is an odd periodic with period 2, then f(4) is

(a) 0 (b) 2 (c) 4 (d) -4

23. If $f(x) = 1 + \alpha x$, ($\alpha \neq 0$) is the inverse of itself then the value of α is

(a) -2 (b) -1 (c) 0 (d) 2

24. If g(x) be a function defined on [-1, 1] and if the area of the equilateral triangle with two of its vertices at (0, 0) and (x, g(x))) is $\dfrac{\sqrt{3}}{4}$, then the function is

(a) $g(x) = \pm\sqrt{1-x^2}$ (b) $g(x) = -\sqrt{1-x^2}$ (c) $g(x) = \sqrt{1-x^2}$ (d) $g(x) = \sqrt{1+x^2}$

25. Which of the following functions is periodic

(a) $f(x) = x - [x]$ (b) $f(x) = \begin{cases} x\sin\dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ (c) $f(x) = x \cos x$ (d) none of these

26. $f(x) = \sqrt{\dfrac{(x+1)(x-3)}{x-2}}$ is real valued in the domain

(a) $(-\infty, -1] \cup [3, \infty)$ (b) $(-\infty, -1] \cup (2, 3]$ (c) $[-1, 2) \cup [3, \infty)$ (d) none of these

27. The domain of definition of the function y(x) given by the equation $2^x + 2^y = 2$ is

(a) $0 < x \leq 1$ (b) $0 \leq x \leq 1$ (c) $-\infty < x \leq 0$ (d) $-\infty < x < 1$

28. If the function $f : [1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$ then $f^{-1}(x)$ is

(a) $\left(\dfrac{1}{2}\right)^{x(x-1)}$ (b) $\dfrac{1}{2}\left(1+\sqrt{1+4\log_2 x}\right)$ (c) $\dfrac{1}{2}\left(1-\sqrt{1+4\log_2 x}\right)$ (d) not defined

29. If $\log_{0.3}(x - 1) < \log_{0.09}(x - 1)$, then x lies in the interval

(a) $(2, \infty)$ (b) (1, 2) (c) (-2, -1) (d) none of these

30. If $g(f(x)) = |\sin x|$ and $f(g(x)) = (\sin\sqrt{x})^2$, then

(a) $f(x) = \sin^2 x,\ g(x) = \sqrt{x}$ (b) $f(x) = \sin x,\ g(x) = |x|$

(c) $f(x) = x^2,\ g(x) = \sin\sqrt{x}$ (d) f and g cannot be determined

31. Let $g(x)=1+x-[x]$ and $f(x)=\begin{cases}-1, & x<0\\ 0, & x=0\\ 1, & x>0\end{cases}$. Then for all x, f(g(x)) is equal to

(a) x (b) 1 (c) f(x) (d) g(x).

---

32. The domain of definition of $f(x)=\dfrac{\log_2(x+3)}{x^2+3x+2}$ is

(a) R ~ {-1, -2} (b) $(-2, \infty)$ (c) R ~ {-1, -2, -3} (d) $(-3,\infty)\sim\{-1,-2\}$

---

33. If $f(x)=\dfrac{\alpha x}{x+1}$, $x\neq-1$, then for what value of α is f(f(x)) = x ?

(a) $\sqrt{2}$ (b) $-\sqrt{2}$ (c) 1 (d) -1

---

34. The set of all real numbers x for which $x^2 - |x+2| + x > 0$, is

(a) $(-\infty, -2)\cup(2, \infty)$ (b) $(-\infty, -\sqrt{2})\cup(\sqrt{2}, \infty)$

(c) $(-\infty, -1)\cup(1, \infty)$ (d) $(\sqrt{2}, \infty)$

---

35. Range of the function $f(x)=\dfrac{x^2+x+2}{x^2+x+1}$, $x\in R$ is

(a) $(1, \infty)$ (b) $\left(1, \dfrac{11}{7}\right)$ (c) $\left(1, \dfrac{7}{3}\right]$ (d) $\left[1, \dfrac{7}{5}\right]$

Exercise level 3 :

1. Find the domain of the following functions, where [.]denotes the greatest integer function.

(i) $f(X) = \dfrac{3}{\left[\dfrac{x}{2}\right]} - 5^{\cos^{-1}x^2} + \dfrac{(2x+1)!}{\sqrt{x+1}}$

(ii) $f(x) = \dfrac{1}{[|x-1|]+[|12-x|]-11}$

2. (a) Find the domain and range of $f(x) = log\dfrac{1}{\sqrt{[\cos x]-[\sin x]}}$, [.]denotes the greatest integer function.

(b). Solve the equation $||x^2-5x+4|-|2x-3|| = |x^2-3x+1|$

(c). Find domain and range of function $f(x) = \sqrt{\log_{1/2}\log_2[x^2+4x+5]}$where [.]denotes the greatest integer function.

3. If a function is defined as $f(x) = \sqrt{\log_{\phi(x)} g(x)}$, where $g(x) = |\sin x| + \sin x, \phi(x) = \sin x + \cos x, 0 \le x \le \pi$, Then find the domain of f (x).

4(a). Find the range of the function $f(x) = \sqrt{\sin\left(\ln\left(\dfrac{x^2+e}{x^2+1}\right)\right)} + \sqrt{\cos\left(\ln\left(\dfrac{x^2+e}{x^2+1}\right)\right)}$

(b). Find the range of function $f : R \to R$ defined by $f(x) = \left(\tan^{-1}x\right)^2 + \dfrac{2}{\sqrt{1+x^2}}$

5(a). Find period of the functions

(i) $f(x) = e^{x-[x]+|\sin \pi x|+|\sin 2\pi x|+....+|\sin n\pi x|}$, [.]denotes the greatest integer.

(ii) $f(x) = \dfrac{1}{|\sin x| + |\cos x|}$

(b). Prove that $f(x) = \dfrac{2x(\sin x + \tan x)}{2\left[\dfrac{x+2\pi}{\pi}\right]-3}$is an odd functin. [.]dentoes greatest integer function.

6. Let $f(x,y)$be a periodic function, satisfying the condition $f(x,y) = f(2x+2y, 2y-2x)\,\forall x, y \in R$and let g (x) be a function defined as $g(x) = f(2^x, 0)$, Prove that g (x) is periodic function and find its period,

7(a). Solve the equation $x^3 - [x] = 5$, where [.]denotes the integral part of the number x.

(b). Find all the real solutions of the equation $3x^2 - 8[x] + 1 = 0$, where [.]denotes the greatest integer function.

8. Prove that $[x] = \left[\dfrac{x}{2}\right] + \left[\dfrac{x+1}{2}\right]$where [.]denote greatest integer function and n be any positive integer then show that $\left[\dfrac{n+1}{2}\right] + \left[\dfrac{n+2}{4}\right] + \left[\dfrac{n+4}{8}\right] + \left[\dfrac{n+8}{16}\right] + ... = n$

9(a). A function f (x) is defined for all $x \in R$ and satisfies, $f(x+y) = f(x) + 2y^2 + kxy\,\forall x, y \in R$, where k is a given constant. If $f(1) = 2$and $f(2) = 8$,find $f(x)$and show that $f(x+y).f\left(\dfrac{1}{x+y}\right) = k, x+y \ne 0$.

(b). Find all $f(x)$satisfying the relation $\{f(x)\}^2 + 4f(x).f'(x) + \{f'(x)\}^2 = 0$.

10. Find all possible values of real parameter 'a' so that the range of $f(x) = \dfrac{x-1}{a-x^2+1}$does not contain any value belonging to the interval $\left[-1, -\dfrac{1}{3}\right]$.

Exercise level 1 :

ANSWER KEY LEVEL 1

1.	b	21.	b
2.	b	22.	b
3.	b	23.	b
4.	a	24.	d
5.	a	25.	a
6.	a	26.	c
7.	c	27.	a
8.	c	28.	c
9.	b	29.	a
10.	a	30.	c
11.	b	31.	a
12.	c	32.	a
13.	c	33.	a
14.	b	34.	c
15.	d	35.	c
16.	b	36.	c
17.	b	37.	a
18.	b	38.	a
19.	d	39.	d
20.	b	40.	d

Exercise level 2:

1.	d	19.	a
2.	c	20.	a
3.	b	21.	b
4.	a	22.	a
5.	a	23.	b
6.	b	24.	b
7.	c	25.	a
8.	c	26.	c
9.	a	27.	d
10.	c	28.	b
11.	b	29.	a
12.	b	30.	a
13.	d	31.	b
14.	d	32.	d
15.	b	33.	d
16.	a	34.	b
17.	a	35.	c
18.	a		

Previous Year Questions

Advanced (Functions)

Numerical

2025

Q1: Let $\mathbb{R}$ denote the set of all real numbers. Let $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to(0,4)$ be functions defined by $f(x)=\log_e\left(x^2+2x+4\right)$, and $g(x)=\dfrac{4}{1+e^{-2x}}$

Define the composite function fog^{-1} by $\left(fog^{-1}\right)(x)=f\left(g^{-1}(x)\right)$, where g^{-1} is the inverse of function g. Then the value of derivative of the composite function fog^{-1} at $x=2$ is __________.

Q2: Let $\mathbb{R}$ denote the set of all real numbers. Let $f:\mathbb{R}\to\mathbb{R}$ be a function such that $f(x)>0$ for all $x\in\mathbb{R}$, and $f(x+y)=f(x)f(y)$ for all $x,y\in\mathbb{R}$. Let the real numbers $a_1,a_2,\ldots\ldots,a_{50}$ be in an arithmetic progression. If $f(a_{31})=64f(a_{25})$, and $\sum_{i=1}^{50}f(a_i)=3\left(2^{25}+1\right)$, then the value of $\sum_{i=6}^{30}f(a_i)$ is __________.

2024

Q1: Let $f:\mathbb{R}\to\mathbb{R}$ be a function such that $f(x+y)=f(x)+f(y)$ for all $x,y\in\mathbb{R}$ and $g:\mathbb{R}\to(0,\infty)$ be a function such that $g(x+y)=g(x)g(y)$ for all $x,y\in\mathbb{R}$. If $f\left(\dfrac{-3}{5}\right)=12$ and $g\left(\dfrac{-1}{3}\right)=2$, then the value of $\left(f\left(\dfrac{1}{4}\right)+g(-2)-8\right)g(0)$ is __________.

Q2: Let the function $f:\mathbb{R}\to\mathbb{R}$ be defined by

$$f(x)=\frac{\sin x}{e^{\pi x}}\frac{\left(x^{2023}+2024x+2025\right)}{\left(x^2-x+3\right)}+\frac{2}{e^{\pi x}}\frac{\left(x^{2023}+2024x+2025\right)}{\left(x^2-x+3\right)}$$. Then the number of solutions of $f(x)=0$ in $\mathbb{R}$ is __________.

2020

Q1: Let the function $f:[0,1]\to\mathbb{R}$ be defined by $f(x)=\frac{4^x}{4^x+2}$. Then the value of $f\left(\frac{1}{40}\right)+f\left(\frac{2}{40}\right)+f\left(\frac{3}{40}\right)+.....+f\left(\frac{39}{40}\right)-f\left(\frac{1}{2}\right)$ is __________.

Q2: Let the function $f:(0,\pi)\to\mathbb{R}$ be defined by $f(\theta)=(\sin\theta+\cos\theta)^2+(\sin\theta-\cos\theta)^4$. Suppose the function f has a local minimum at θ precisely when $\theta\in(\lambda_1\pi,.......,\lambda_r\pi)$, where $0<\lambda_1<...\lambda_r<1$. Then the value of $\lambda_1+...+\lambda_r$ is __________.

Q3: Let $f:[0,2]\to\mathbb{R}$ be a function defined by $f(x)=(3-\sin(2\pi x))\sin\left(\pi x-\frac{\pi}{4}\right)-\sin\left(3\pi x+\frac{\pi}{4}\right)$. If $\alpha,\beta\in[0,2]$ are such that $\{x\in[0,2]:f(x)\ge 0\}=[\alpha,\beta]$, then the value of $\beta-\alpha$ is __________.

Q4: For a polynomial g(x) with real coefficients, let m_g denote the number of distinct real roots of g(x). Suppose S is the set of polynomials with real coefficients defined by

$S=\left\{\left(x^2-1\right)^2\left(a_0+a_1x+a_2x^2+a_3x^3\right):a_0,a_1,a_2,a_3\in\mathbb{R}\right\}$; For a polynomial f, let f' and f'' denote its first and second order derivatives, respectively. Then the minimum possible value of ($m_{f'}+m_{f''}$), where $f\in S$, is __________.

2018

Q1: Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X, then the value of $\frac{1}{5!}(\beta-\alpha)$ is __________.

2009

Q1: If the function $f(x)=x^3+e^{x/2}$ and $g(x)=f^{-1}(x)$, then the value of $g'(1)$ is __________.

MCQ (More than One Correct Answer)

2025

Q1: Let ℕ denote the set of all natural numbers, and ℤ denote the set of all integers. Consider the functions $f: \mathbb{N} \to \mathbb{Z}$ and $g: \mathbb{Z} \to \mathbb{N}$ defined by

$$f(n) = \begin{cases} \dfrac{(n+1)}{2} & if \ldots n \ldots is \ldots odd \\ \dfrac{(4-n)}{2} & if \ldots n \ldots is \ldots even, \end{cases}$$

and

$$g(n) = \begin{cases} 3+2n & if \ldots n \geq 0, \\ -2n & if \ldots n < 0. \end{cases}$$

Define $(gof)(n) = g(f(n))$ for all $n \in \mathbb{N}$ and $(fog)(n) = f(g(n))$ for all $n \in \mathbb{Z}$.

Then which of the statements is (are) true?

(a) gof is NOT one-one and gof is NOT onto.
(b) fog is NOT one-one but fog is onto.
(c) g is one-one and g is onto.
(d) f is NOT one-one but f is onto.

2023

Q1: Let $S = (0,1) \cup (1,2) \cup (3,4)$ and $T = \{0,1,2,3\}$. Then which of the following statements is(are) true?

(a) There are infinitely many functions from S to T.
(b) There are infinitely many strictly increasing functions from S to T.
(c) The number of continuous functions from S to T is is at most 120.
(d) Every continuous function from S to T is differentiable.

Q2: Let $f:[0,1] \to [0,1]$ be the function defined by $f(x) = \dfrac{x^3}{3} - x^2 + \dfrac{5}{9}x + \dfrac{17}{36}$. Consider the square region $S = [0,1] \times [0,1]$. Let $G = \{(x,y) \in S : y > f(x)\}$ be called the green region and

$R=\{(x,y)\in S: y<f(x)\}$ be called the red region. Let $L_h=\{(x,h)\in S: x\in[0,1]\}$ be the horizontal line drawn at a height $h\in[0,1]$. Then which of the statements is(are) true?

(a) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the green region below the line L_h.

(b) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the red region below the line L_h.

(c) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the red region below the line L_h.

(d) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the green region below the line L_h.

2022

Q1: Let $|M|$ denote the determinant of a square Matrix M. Let $g:\left[0,\frac{\pi}{2}\right]\to\mathbb{R}$ be the function defined by $g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$ where

$$f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi\end{vmatrix}$$

. Let $p(x)$ be a quadratic polynomial whose roots are the maximum and minimum values of the function $g(\theta)$, and $p(2)=2-\sqrt{2}$. Then which of the following is (are) true?

(a) $p\left(\frac{3+\sqrt{2}}{4}\right)<0$ (b) $p\left(\frac{1+3\sqrt{2}}{4}\right)>0$

(c) $p\left(\frac{5\sqrt{2}-1}{4}\right) > 0$

(d) $p\left(\frac{5-\sqrt{2}}{4}\right) < 0$

2015

Q1: Let $f(x) = \sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)\right)$ for all $x \in \mathbb{R}$ and $g(x) = \frac{\pi}{2}\sin x$ for all $x \in \mathbb{R}$. Let $(fog)(x)$ denote $f(g(x))$ and $(gof)(x)$ denote $g(f(x))$. The which of the following is/are true?

(a) Range of f is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(b) Range of fog is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(c) $\lim\limits_{x \to 0} \frac{f(x)}{g(x)} = \frac{\pi}{6}$

(d) There is an $x \in \mathbb{R}$ such that $(gof)(x) = 1$.

2014

Q1: For every pair of continuous function $f, g : [0,1] \to \mathbb{R}$ such that $\max\{f(x) : x \in [0,1]\} = \max\{g(x) : x \in [0,1]\}$. The correct statements is(are)

(a) $[f(c)]^2 + 3f(c) = [g(c)]^2 + 3g(c)$ for some $c \in [0,1]$

(b) $[f(c)]^2 + f(c) = [g(c)]^2 + 3g(c)$ for some $c \in [0,1]$

(c) $[f(c)]^2 + 3f(c) = [g(c)]^2 + g(c)$ for some $c \in [0,1]$

(d) $[f(c)]^2 = [g(c)]^2$ for some $c \in [0,1]$

Q2: Let $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$ be given by $f(x) = [\log(\sec x + \tan x)]^3$ .Then,

(a) $f(x)$ is an odd function

(b) $f(x)$ is a one-one function

(c) $f(x)$ is an onto function

(d) $f(x)$ is an even function

2012

Q1: Let $f:(-1,1)\to\mathbb{R}$ be such that $f(\cos 4\theta)=\dfrac{2}{2-\sec^2\theta}$ for $\theta\in\left(0,\dfrac{\pi}{4}\right)\cup\left(\dfrac{\pi}{4},\dfrac{\pi}{2}\right)$. Then the value(s) of $f\left(\dfrac{1}{3}\right)$ is(are):

(a) $1-\sqrt{\dfrac{3}{2}}$ (b) $1+\sqrt{\dfrac{3}{2}}$

(c) $1-\sqrt{\dfrac{2}{3}}$ (d) $1+\sqrt{\dfrac{2}{3}}$

2011

Q1: Let $f:(0,1)\to\mathbb{R}$ be defined by $f(x)=\dfrac{b-x}{1-bx}$, where b is a constant such that $0<b<1$. Then

(a) f is not invertible on $(0,1)$

(b) $f\neq f^{-1}$ on $(0,1)$ and $f'(b)=\dfrac{1}{f'(0)}$.

(c) $f=f^{-1}$ on $(0,1)$ and $f'(b)=\dfrac{1}{f'(0)}$.

(d) f^{-1} is differentiable on $(0,1)$.

MCQ (Single Correct Answer)

2020

Q1: If the function $f:\mathbb{R}\to\mathbb{R}$ is defined by $f(x)=|x|(x-\sin x)$, then which of the following statements is TRUE?

(a) f is one-one, but NOT onto.
(b) f is onto, but NOT one-one.
(c) f is BOTH one-one and onto.
(d) f is NEITHER one-one NOR onto.

2018

Q1: Let $E_1 = \left\{ x \in \mathbb{R} : x \neq 1 \ldots and \ldots \frac{x}{x-1} > 0 \right\}$ and $E_2 = \left\{ \begin{array}{c} x \in E_1 : \sin^{-1}\left(\log_e \left(\frac{x}{x-1} \right) \right) \\ is \ldots a \ldots real \ldots number \end{array} \right\}$.

(Here the inverse trigonometric function $\sin^{-1} x$ assumes values in $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$.).Let $f : E_1 \to \mathbb{R}$ be the function defined by $f(x) = \log_e \left(\frac{x}{x-1} \right)$ and $g : E_2 \to \mathbb{R}$ be the function defined by

$g(x) = \sin^{-1}\left(\log_e \left(\frac{x}{x-1} \right) \right)$.

List-I	List-II
P. The range of f is	1. $\left(-\infty, \frac{1}{1-e} \right] \cup \left[\frac{e}{e-1}, \infty \right)$
Q. The range of g contains	2. $(0,1)$
R. The domain of f contains	3. $\left[-\frac{1}{2}, \frac{1}{2} \right]$
S. The domain of g is	4. $(-\infty, 0) \cup (0, \infty)$
	5. $\left(-\infty, \frac{e}{e-1} \right)$
	6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{e-1} \right]$

The correct option is:

(a) $P \to 4; Q \to 2; R \to 1; S \to 1$
(b) $P \to 3; Q \to 3; R \to 6; S \to 5$
(c) $P \to 4; Q \to 2; R \to 1; S \to 6$
(d) $P \to 4; Q \to 3; R \to 6; S \to 5$

2017

Q1: Let $S = \{1, 2, 3, ..., 9\}$. For $k = 1, 2, ..., 5$,let N_k be the number of subsets of S, each containing five elements out of which exactly k are odd. Then $N_1 + N_2 + N_3 + N_4 + N_5 =$

(a) 210 (b) 252

(c) 126 (d) 125

2014

Q1: Let $f_1 \to \mathbb{R} \to \mathbb{R}$, $f_2 : [0,\infty) \to \mathbb{R}$, $f_3 \to \mathbb{R} \to \mathbb{R}$ and $f_4 \to \mathbb{R} \to [0,\infty)$ be defined by

$$f_1(x) = \begin{cases} |x| & if \ldots x < 0, \\ e^x & if \ldots x \geq 0; \end{cases}$$

$$f_2(x) = x^2;$$

$$f_3(x) = \begin{cases} \sin x & if \ldots x < 0, \\ x & if \ldots x \geq 0; \end{cases}$$

and

$$f_4(x) = \begin{cases} f_2(f_1(x)) & if \ldots x < 0, \\ f_2(f_1(x)) - 1 & if \ldots x \geq 0; \end{cases}$$

List-I	List-II
P. f_4 is	(i) onto but not one-one
Q. f_3 is	(ii) neither continuous nor one-one
R. $f_2 o f_1$ is	(iii) differentiable but not one-one
S. f_2 is	(iv) continuous and one-one

(a) $P \to 3, Q \to 1, R \to 4, S \to 2$
(b) $P \to 1, Q \to 3, R \to 4, S \to 2$
(c) $P \to 3, Q \to 1, R \to 2, S \to 4$
(d) $P \to 1, Q \to 3, R \to 2, S \to 4$

2012

Q1: The function $f : [0,3] \to [1,29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is

(a) one-one and onto.
(b) onto but not one-one.
(c) one-one but not onto.
(d) neither one-one nor onto.

2011

Q1: Let $f(x)=x^2$ and $g(x)=\sin x$ for all $x\in\mathbb{R}$. Then the set of all x satisfying $(fogogof)(x)=(gogof)(x)$, where $(fog)(x)=f(g(x))$, is

(a) $\pm\sqrt{n\pi}, n\in\{0,1,2,.....\}$ (b) $\pm\sqrt{n\pi}, n\in\{1,2,.....\}$

(c) $\frac{\pi}{2}+2n\pi, n\in\{....,-2,-1,0,1,2,....\}$ (d) $2n\pi, n\in\{.....,-2,-1,0,1,2,....\}$

Q2: Match the statements given in Column I with the intervals/union of intervals given in Column II:

Column I	Column II
(A) The set $\left\{\mathrm{Re}\left(\frac{2iz}{1-z^2}\right): z \text{ is a complex number}, \lvert z\rvert=1,\ z\neq\pm1\right\}$ is	(p) $(-\infty,-1)\cup(1,\infty)$
(B) The domain of the function $f(x)=\sin^{-1}\left(\frac{8(3)^{x-2}}{1-3^{2(x-1)}}\right)$ is	(q) $(-\infty,0)\cup(0,\infty)$
(C) If $f(\theta)=\begin{vmatrix}1 & \tan\theta & 1\\ -\tan\theta & 1 & \tan\theta\\ -1 & -\tan\theta & 1\end{vmatrix}$, then the set $\left\{f(\theta): 0\le\theta<\frac{\pi}{2}\right\}$ is	(r) $[2,\infty)$
(D) If $f(x)=x^{3/2}(3x-10)$, $x\ge0$, then $f(x)$ is increasing in	(s) $(-\infty,-1]\cup[1,\infty)$
	(t) $(-\infty,0]\cup[2,\infty)$

(a) $A\to S, B\to T, C\to P, D\to Q$

(b) $A\to S, B\to T, C\to R, D\to P$

(c) $A\to S, B\to T, C\to R, D\to R$

(d) $A\to P, B\to Q, C\to R, D\to R$

2010

Q1: Let f, g and h be real valued functions defined on the interval $[0,1]$ by

$f(x)=e^{x^2}+e^{-x^2}$

$g(x) = xe^{x^2} + e^{-x^2}$

and $h(x) = x^2 e^{x^2} + e^{-x^2}$.

If a, b and c denote, respectively, the absolute maximum of f, g and h on $[0,1]$, then

(a) $a = b$ and $c \neq b$ (b) $a = c$ and $a \neq b$

(c) $a \neq b$ and $c \neq b$ (d) $a = b = c$

Q2: Let $S = \{1,2,3,4\}$. The total number of unordered pairs of disjoint subsets of S is equal to :

(a) 25 (b) 34

(c) 42 (d) 41

2006

Q1: If $f''(x) = -f(x)$ and $g(x) = f'(x)$ and $g(x) = f'(x)$ and $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ and given that $F(5) = 5$, then $F(10)$ is equal to:

(a) 5 (b) 10

(c) 0 (d) 15

2005

Q1: Find the range of value of t for which $2\sin t = \dfrac{1-2x+5x^2}{3x^2-2x-1}, t \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$

(a) $\left[-\dfrac{\pi}{3}, -\dfrac{\pi}{10}\right] \cup \left[\dfrac{\pi}{10}, \dfrac{\pi}{2}\right]$ (b) $\left[-\dfrac{\pi}{2}, -\dfrac{\pi}{10}\right] \cup \left[\dfrac{3\pi}{10}, \dfrac{\pi}{2}\right]$

(c) $\left[-\dfrac{\pi}{2}, -\dfrac{\pi}{6}\right] \cup \left[\dfrac{3\pi}{10}, \dfrac{\pi}{3}\right]$ (d) $\left[\dfrac{\pi}{2}, -\dfrac{\pi}{10}\right] \cup \left[\dfrac{\pi}{10}, \dfrac{\pi}{2}\right]$

Previous Year Questions

Mains (Functions)

MCQ (Single Correct Answer)

2026

Q1: Given below are two statements:

Statement I : The function $f:\mathbb{R}\to\mathbb{R}$ defined by $f(x)=\dfrac{x}{1+|x|}$ is one-one.

Statement II: The function $f:\mathbb{R}\to\mathbb{R}$ defined by $f(x)=\dfrac{x^2+4x-30}{x^2-8x+18}$ is many-one.

In the light of the above statements, choose the **correct answer** from the options given below :

(a) Statement I is true but Statement II is false.
(b) Both Statement I and Statement II are false.
(c) Both Statement I and Statement II are true.
(d) Statement I is false but Statement II is true.

Q2: The sum of all the elements in the range of

$f(x)=Sgn(\sin x)+Sgn(\cos x)+Sgn(\tan x)+Sgn(\cot x)$, $x\neq\dfrac{n\pi}{2}, n\in\mathbb{Z}$, where

$Sgn(t)=\begin{cases}1, & if \quad t>0\\ -1, & if \quad t<0\end{cases}$ is:

(a) 4   (b) 0

(c) 2   (d) -2

Q3: If $g(x)=3x^2+2x-3,\ f(0)=-3$ and $4g(f(x))=3x^2-32x+72,$ then $f(g(2))$ is equal to:

(a) $\frac{7}{2}$

(b) $-\frac{25}{6}$

(c) $\frac{25}{6}$

(d) $-\frac{7}{2}$

Q4: Let f be a function such that $3f(x)+2f\left(\frac{m}{19x}\right)=5x, x\neq 0,$ where $m=\sum_{i=1}^{9}(i)^2$. Then $f(5)-f(2)$ is equal to

(a) 36

(b) 9

(c) -9

(d) 18

Q5: Let $f(x)=[x]^2-[x+3]-3, x\in\mathbb{R}$, where $[.]$ is the greatest integer function. Then

(a) $f(x)=0$ for finitely many values of x.

(b) $f(x)<0$ only for $x\in[-1,3)$.

(c) $\int_0^2 f(x)dx=-6$

(d) $f(x)>0$ only for $x\in[4,\infty)$

Q6: Let the domain of the function $f(x)=\log_3\log_5\left(7-\log_2\left(x^2-10x+85\right)\right)$ + $\sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right)$ be $(\alpha,\beta]$. Then $\alpha+\beta$ is equal to:

(a) 12

(b) 8

(c) 10

(d) 9

Q7: Let f and g be functions satisfying $f(x+y)=f(x)f(y)$, $f(1)=7$ and $g(x+y)=g(xy)$, $g(1)=1$, for all $x,y\in\mathbb{N}$. If $\sum_{x=1}^{n}\left(\frac{f(x)}{g(x)}\right)=19607,$ then n is equal to:

(a) 6

(b) 7

(c) 4

(d) 5

Q8: If the domain of the function $f(x)=\sin^{-1}\left(\frac{5-x}{3+2x}\right)+\frac{1}{\log_e(10-x)}$ is $(-\infty,\alpha]\cup[\beta,\gamma)-\{\delta\}$, then $6(\alpha+\beta+\gamma+\delta)$ is equal to

(a) 66 (b) 68

(c) 70 (d) 67

2025

Q1: If the range of the function $f(x)=\frac{5-x}{x^2-3x+2}, x\neq 1,2,$ is $(-\infty,\alpha]\cup[\beta,\infty)$. Then $\alpha^2+\beta^2$ is equal to

(a) 188 (b) 192

(c) 190 (d) 194

Q2: Let the domains of the functions $f(x)=\log_4\log_3\log_7\left(8-\log_2\left(x^2+4x+5\right)\right)$ and $g(x)=\sin^{-1}\left(\frac{7x+10}{x-2}\right)$ be (α,β) and $[\gamma,\delta]$ respectively. Then $\alpha^2+\beta^2+\gamma^2+\delta^2$ is equal to:

(a) 15 (b) 13

(c) 16 (d) 14

Q3: Let $f,g:(1,\infty)\to\mathbb{R}$ be defined as $f(x)=\frac{2x+3}{5x+2}$ and $g(x)=\frac{2-3x}{1-x}$. If the range of the function $fog:[2,4]\to\mathbb{R}$ is $[\alpha,\beta]$, then $\frac{1}{\beta-\alpha}$ is equal to

(a) 56 (b) 2

(c) 29 (d) 68

Q4: Let f be a function such that $f(x)+3f\left(\frac{24}{x}\right)=4x, x\neq 0.$ The $f(3)+f(8)$ is

(a) 13 (b) 11

(c) 10 (d) 12

Q5: If the domain of the function $f(x)=\log_7\left(1-\log_4\left(x^2-9x+18\right)\right)$ is $(\alpha,\beta)\cup(\gamma,o)$, then $\alpha+\beta+\gamma+\hat{o}$ is equal to

(a) 17 (b) 15

(c) 16 (d) 18

Q6: If the domain of the function $f(x)=\log_e\left(\frac{2x-3}{5+4x}\right)+\sin^{-1}\left(\frac{4+3x}{2-x}\right)$ is $[\alpha,\beta)$, then $\alpha^2+4\beta$ is equal to

(a) 4 (b) 3

(c) 7 (d) 5

Q7: If the domain of the function $f(x)=\frac{1}{\sqrt{10+3x-x^2}}+\frac{1}{\sqrt{x+|x|}}$ is (a,b), then $(1+a)^2+b^2$ is equal to:

(a) 29 (b) 30

(c) 25 (d) 26

Q8: If the domain of the function $\log_5\left(18x-x^2-77\right)$ is (α,β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ,δ), then $\alpha^2+\beta^2+\gamma^2$ is equal to

(a) 186 (b) 179

(c) 195 (d) 174

Q9: Let $f:[0,3]\to A$ be defined by $f(x)=2x^3-15x^2+36x+7$ and $g:[0,\infty)\to B$ be defined by $g(x)=\frac{x^{2025}}{x^{2025}+1}$, If both the functions are onto and $S=\{x\in\mathbb{Z}; x\in A \text{ or } x\in B\}$ then $n(S)$ is equal to:

(a) 29 (b) 31

(c) 30 (d) 36

Q10: If $f(x)=\frac{2^x}{2^x+\sqrt{2}}, x\in\mathbb{R}$, then $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)$ is equal to

(a) 82

(b) $81\sqrt{2}$

(c) 41

(d) $\frac{81}{2}$

Q11: Let $f:\mathbb{R}\times\mathbb{R}$ be a function defined by

$f(x)=(2+3a)x^2+\left(\frac{a+2}{a-1}\right)x+b, a\neq 1$. If $f(x+y)=f(x)+f(y)+1-\frac{2}{7}xy$, then the value of $28\sum_{i=1}^{5}|f(i)|$ is

(a) 735

(b) 675

(c) 715

(d) 545

Q12: The function $f:(-\infty,\infty)\rightarrow(-\infty,1)$, defined by $f(x)=\frac{2^x-2^{-x}}{2^x+2^{-x}}$ is :

(a) One-one but not onto.
(b) Onto but not one-one.
(c) Both one-one and onto.
(d) Neither one-one nor onto.

Q13: Let $f(x)=\frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}$. Then the value of $8\left(f\left(\frac{1}{15}\right)+f\left(\frac{2}{15}\right)+......+f\left(\frac{59}{15}\right)\right)$ is equal to

(a) 108

(b) 92

(c) 118

(b) 102

Q14: Let $f(x)=\log_e x$ and $g(x)=\frac{x^4-2x^3+3x^2-2x+2}{2x^2-2x+1}$. Then the domain of $f og$ is

(a) $(0,\infty)$

(b) $[1,\infty)$

(c) $\mathbb{R}$

(d) $[0,\infty)$

Q15: Let $A=\{1,2,3,4\}$ and $B=\{1,4,9,16\}$. Then the number of many-one functions $f:A\rightarrow B$ such that $1\in f(A)$ is equal to:

(a) 151

(b) 139

(c) 163

(d) 127

2024

Q1: Let the range of the function $f(x)=\dfrac{1}{2+\sin 3x+\cos 3x}, x\in\mathbb{R}$ be $[a,b]$. If α and β are respectively the A.M. and the G.M. of a and b, then $\dfrac{\alpha}{\beta}$ is equal to

(a) π

(b) $\sqrt{\pi}$

(c) $\sqrt{2}$

(d) 2

Q2: If the domain of the function $f(x)=\sin^{-1}\left(\dfrac{x-1}{2x+3}\right)$ is $\mathbb{R}-(\alpha,\beta)$, the $12\alpha\beta$ is equal to

(a) 40

(b) 36

(c) 24

(d) 32

Q3: Let $f(x)=\begin{cases} -a & if \quad -a\le x\le 0 \\ x+a & if \quad 0<x\le a \end{cases}$ where $a>0$ and $g(x)=\left(f(|x|)-|f(x)|\right)/2$. Then the function $g:[-a,a]\to[-a,a]$ is

(a) Neither one-one nor onto.
(b) Both one-one and onto.
(c) One-one.
(d) Onto.

Q4: If the function $f(x)=\left(\dfrac{1}{x}\right)^{2x}; x>0$ attains the maximum value at $x=\dfrac{1}{e}$ then :

(a) $e^{\pi}<\pi^{e}$

(b) $e^{2\pi}<(2\pi)^{e}$

(c) $(2e)^{\pi}>\pi^{2e}$

(d) $e^{\pi}>\pi^{e}$

Q5: Let $f(x)=\dfrac{1}{7-\sin 5x}$ be a function defined on $\mathbb{R}$. Then the range of the function $f(x)$ is equal to

(a) $\left[\dfrac{1}{8},\dfrac{1}{5}\right]$

(b) $\left[\dfrac{1}{7},\dfrac{1}{6}\right]$

(c) $\left[\frac{1}{7},\frac{1}{5}\right]$ (d) $\left[\frac{1}{8},\frac{1}{6}\right]$

Q6: The function $f(x)=\frac{x^2+2x-15}{x^2-4x+9}, x\in\mathbb{R}$ is

(a) Both one-one and onto.
(b) Onto but not one-one.
(c) Neither one-one nor onto.
(d) One-one but not onto.

Q7: Let $f,g:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=|x-1|$ and $g(x)=\begin{cases} e^x, & x\ge 0 \\ x+1, & x\le 0\end{cases}$. Then the function $f(g(x))$ is

(a) neither one-one nor onto.
(b) One-one but not onto.
(c) Both one-one and onto.
(d) Onto but not one-one.

Q8: Let $A=\{1,3,7,9,11\}$ and $B=\{2,4,5,7,8,10,12\}$. Then the total number of one-one maps $f:A\to B$, such that $f(1)+f(3)=14$, is :

(a) 120 (b) 180

(c) 240 (d) 480

Q9: If the domain of the function $f(x)=\frac{\sqrt{x^2-25}}{(4-x^2)}+\log_{10}(x^2+2x-15)$ is $(-\infty,\alpha)\cup[\beta,\infty)$, then $\alpha^2+\beta^3$ is equal to :

(a) 140 (b) 175

(c) 125 (d) 150

Q10: Let $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=\begin{cases}\log_e x, & x>0 \\ e^{-x}, & x\le 0\end{cases}$ and $g(x)=\begin{cases} x, & x\ge 0 \\ e^x, & x<0\end{cases}$.
Then $gof:\mathbb{R}\to\mathbb{R}$ is:

(a) One-one but not onto. (b) neither one-one nor onto.

(c) Onto but not one-one. (d) both one-one and onto.

Q11: If $f(x)=\frac{4x+3}{6x-4}$, $x\neq\frac{2}{3}$ and $(fof)(x)=g(x)$, where $g:\mathbb{R}-\left\{\frac{2}{3}\right\}\to\mathbb{R}-\left\{\frac{2}{3}\right\}$, then $(gogog)(4)$ is equal to

(a) -4

(b) $\frac{19}{20}$

(c) $-\frac{19}{20}$

(d) 4

Q12: If the domain of the function $f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ is $(\alpha,\beta]$, then the value of $5\beta-4\alpha$ is equal to

(a) 9

(b) 12

(c) 11

(d) 10

Q13: If the domain of the function $f(x)=\cos^{-1}\left(\frac{2-|x|}{4}\right)+\{\log_e(3-x)\}^{-1}$ is $[-\alpha,\beta)-\{\gamma\}$, then $\alpha+\beta+\gamma$ is equal to

(a) 11

(b) 12

(c) 9

(d) 8

Q14: If $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}$, $g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, then range of $(fog)(x)$ is

(a) $[0,1)$

(b) $[0,3)$

(c) $(0,1]$

(d) $[0,1]$

Q15: Let $f:\mathbb{R}-\left\{\frac{-1}{2}\right\}\to\mathbb{R}$ and $g:\mathbb{R}-\left\{\frac{-5}{2}\right\}\to\mathbb{R}$ be defined as $f(x)=\frac{2x+3}{2x+1}$ and $g(x)=\frac{|x|+1}{2x+5}$. Then the domain of the function fog is

(a) $\mathbb{R}-\left\{-\frac{7}{4}\right\}$

(b) $\mathbb{R}$

(c) $\mathbb{R}-\left\{-\frac{5}{2},-\frac{7}{2}\right\}$

(d) $\mathbb{R}-\left\{-\frac{5}{2}\right\}$

Q16: The functions $f:\mathbb{N}-\{1\}\to\mathbb{N}$; defined by $f(n)=$ the highest prime factor of n, is :

(a) One-one only

(b) neither one-one nor onto.

(c) onto only

(d) both one-one and onto

2023

Q1: The range of $f(x)=4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is

(a) $[0,2\pi]$

(b) $[0,2\pi)$

(c) $[0,\pi)$

(d) $[0,\pi]$

Q2: For $x\in\mathbb{R}$, two real valued functions $f(x)$ and $g(x)$ are such that, $g(x)=\sqrt{x}+1$ and $fog(x)=x+3-\sqrt{x}$. Then $f(0)$ is equal to

(a) 5

(b) 0

(c) -3

(d) 1

Q3: Let D be the domain of the function $f(x)=\sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$. If the range of the function $g:D\to\mathbb{R}$ defined by $g(x)=x-[x]$, ([x] is the greatest integer function), is (α,β), then $\alpha^2+\frac{5}{\beta}$ is equal to

(a) 45

(b) 136

(c) 46

(d) nearly 135

Q4: The domain of the function $f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$ is : (where [x] denotes the greatest integer less than or equal to x)

(a) $(-\infty,-2)\cup[6,\infty)$

(b) $(-\infty,-3]\cup[6,\infty)$

(c) $(-\infty,-2)\cup(5,\infty)$ (d) $(-\infty,-3]\cup(5,\infty)$

Q5: If $f(x)=\dfrac{(\tan 1^o)x+\log_e(123)}{x\log_e(1234)-\tan 1^o}, x>0,$ then the least value of $f(f(x))+f\left(f\left(\dfrac{4}{x}\right)\right)$ is :

(a) 2 (b) 4

(c) 0 (d) 8

Q6: Let the sets A and B denote the domain and range respectively of the function $f(x)=\dfrac{1}{\sqrt{\lceil x\rceil-x}}$,

where $\lceil x\rceil$ denotes the smallest integer greater than or equal to x

Then among the statements

(S1): $A\cap B=(1,\infty)-\mathbb{N}$ and

(S2): $A\cup B=(1,\infty)$

(a) Only (S2) is true (b) Only (S1) is true

(c) Neither (S1) nor (S2) is true (d) Both (S1) and (S2) are true

Q7: Let $f:\mathbb{R}-\{0,1\}\to\mathbb{R}$ be a function such that $f(x)+f\left(\dfrac{1}{1-x}\right)=1+x.$ Then $f(2)$ is equal to

(a) $\dfrac{9}{4}$ (b) $\dfrac{7}{4}$

(c) $\dfrac{7}{3}$ (d) $\dfrac{9}{2}$

Q8: Let $f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & \sin 2x\\ \sin^2 x & 1+\cos^2 x & \sin 2x\\ \sin^2 x & \cos^2 x & 1+\sin 2x\end{vmatrix}, x\in\left[\dfrac{\pi}{6},\dfrac{\pi}{3}\right].$ If α and β respectively are the maximum and minimum values of f, then

(a) $\alpha^2-\beta^2=4\sqrt{3}$ (b) $\beta^2-2\sqrt{\alpha}=\dfrac{19}{4}$

(c) $\beta^2+2\sqrt{\alpha}=\dfrac{19}{4}$ (d) $\alpha^2+\beta^2=\dfrac{9}{2}$

Q9: Let $f:\mathbb{R}-\{2,6\}\to\mathbb{R}$ be real valued function defined as $f(x)=\dfrac{x^2+2x+1}{x^2-8x+12}$. Then range of f is

(a) $\left(-\infty,-\dfrac{21}{4}\right]\cup[1,\infty)$ (b) $\left(-\infty,-\dfrac{21}{4}\right)\cup(0,\infty)$

(c) $\left(-\infty,-\dfrac{21}{4}\right]\cup[0,\infty)$ (d) $\left(-\infty,-\dfrac{21}{4}\right]\cup\left[\dfrac{21}{4},\infty\right)$

Q10: The absolute minimum value, of the function $f(x)=\left|x^2-x+1\right|+\left[x^2-x+1\right]$, where $[t]$ denotes the greatest integer function, in the interval $[-1,2]$, is :

(a) $\dfrac{3}{4}$ (b) $\dfrac{3}{2}$

(c) $\dfrac{1}{4}$ (d) $\dfrac{5}{4}$

Q11: If the domain of the function $f(x)=\dfrac{[x]}{1+x^2}$, where [x] is the greatest integer $\le x$, is $[2,6)$, then it's range is

(a) $\left(\dfrac{5}{37},\dfrac{2}{5}\right]-\left\{\dfrac{9}{29},\dfrac{27}{109},\dfrac{18}{89},\dfrac{9}{53}\right\}$ (b) $\left(\dfrac{5}{37},\dfrac{2}{5}\right]$

(c) $\left(\dfrac{5}{26},\dfrac{2}{5}\right]$ (d) $\left(\dfrac{5}{26},\dfrac{2}{5}\right]-\left\{\dfrac{9}{29},\dfrac{27}{109},\dfrac{18}{89},\dfrac{9}{53}\right\}$

Q12: The range of the function $f(x)=\sqrt{3-x}+\sqrt{2+x}$ is :

(a) $\left[2\sqrt{2},\sqrt{11}\right]$ (b) $\left(\sqrt{5},\sqrt{13}\right)$

(c) $\left(\sqrt{2},\sqrt{7}\right)$ (d) $\left(\sqrt{5},\sqrt{10}\right)$

Q13: Consider the function $f:\mathbb{N}\to\mathbb{R}$, satisfying

$f(1)+2f(2)+3f(3)+....+xf(x)=x(x+1)f(x);\ x\ge 2$ with $f(1)=1$. Then

$\dfrac{1}{f(2022)}+\dfrac{1}{f(2028)}$ is equal to

(a) 8000

(b) 8400

(c) 8100

(d) 8200

Q14: The domain of $f(x)=\dfrac{\log_{(x+1)}(x-2)}{e^{2\log_e x}-(2x+3)}, x\in\mathbb{R}$ is

(a) $(-1,\infty)-\{3\}$

(b) $\mathbb{R}-\{-1,3\}$

(c) $(2,\infty)-\{3\}$

(d) $\mathbb{R}-\{3\}$

Q15: Let $f:\mathbb{R}\to\mathbb{R}$ be a function such that $f(x)=\dfrac{x^2+2x+1}{x^2+1}$. Then

(a) $f(x)$ is many-one in $(-\infty,-1)$

(b) $f(x)$ is one-one in $(-\infty,\infty)$

(c) $f(x)$ is one-one in $[1,\infty)$ but not in $(-\infty,\infty)$

(d) $f(x)$ is many-one in $(1,\infty)$

Q16: The number of functions $f:\{1,2,3,4\}\to\{a\in\mathbb{Z};|a|\le 8\}$ satisfying $f(n)+\dfrac{1}{n}f(n+1)=1$, $\forall n\in\{1,2,3\}$ is

(a) 2

(b) 3

(c) 1

(d) 4

Q17: Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=\log_{\sqrt{m}}\left\{\sqrt{2}(\sin x-\cos x)+m-2\right\}$, for some m, such that the range of f is $[0,2]$. Then the value of m is

(a) 4

(b) 3

(c) 5

(d) 2

Q18: Let $f(x)=2x^n+\lambda, \lambda\in\mathbb{R}, n\in\mathbb{N}$, and $f(4)=133$, $f(5)=255$. Then the sum of all the positive integer divisors of $(f(3)-f(2))$ is

(a) 60

(b) 58

(c) 61

(d) 59

Q19: Let $f(x)$ be a function such that $f(x+y)=f(x).f(y)$ for all $x,y\in\mathbb{N}$. If $f(1)=3$ and $\sum_{k=1}^{n} f(k)=3279,$ then the value of n is

(a) 9
(b) 7
(c) 6
(d) 8

Q20: If $f(x)=\dfrac{2^{2x}}{2^{2x}+3}, x\in\mathbb{R}$, then $f\left(\dfrac{1}{2023}\right)+f\left(\dfrac{2}{2023}\right)+.......+f\left(\dfrac{2022}{2023}\right)$ is equal to

(a) 2011
(b) 2010
(c) 1010
(d) 1011

2022

Q1: Let $f(x)=ax^2+bx+c$ be such that $f(1)$, $f(-2)=\lambda$ and $f(3)=4.$ If $f(0)+f(1)+f(-2)+f(3)=14$, then λ is equal to

(a) -4
(b) $\dfrac{13}{2}$
(c) $\dfrac{23}{2}$
(d) 4

Q2: Let α,β and γ be three positive real numbers. Let $f(x)=\alpha x^5+\beta x^3+\gamma x, x\in\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ be such that $g(f(x))=x$ for all $x\in\mathbb{R}$. If $a_1,a_2,a_3,.......,a_n$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\dfrac{1}{n}\sum_{i=1}^{n} f(a_i)\right)\right)$ is equal to

(a) 0
(b) 3
(c) 9
(d) 27

Q3: Let $f,g:\mathbb{N}-\{1\}\to\mathbb{N}$ be functions defined by $f(a)=\alpha$, where α is the maximum of the powers of those primes p such that p^{α} divides a, and $g(a)=a+1$, for all $a\in\mathbb{N}-\{1\}$. Then , the function $f+g$

(a) One-one but not onto
(b) Onto but not one-one

(c) both one-one and onto (d) neither one-one nor onto

Q4: The number of bijective functions $f:\{1,3,5,7,....,99\}\to\{2,4,6,8,....,100\}$ such that $f(3)\geq f(9)\geq f(15)\geq f(21)\geq .....f(99)$, is

(a) ${}^{50}P_{17}$ (b) ${}^{50}P_{33}$

(c) $33!\times 17!$ (d) $\frac{50!}{2}$

Q5: The total number of functions $f:\{1,2,3,4\}\to\{1,2,3,4,5,6\}$ such that $f(1)+f(2)=f(3)$, is equal to:

(a) 60 (b) 90

(c) 108 (d) 126

Q6: Let the function $f:\mathbb{N}\to\mathbb{N}$ be defined by $f(n)=\begin{cases} 2n, & n=2,4,6,8,..... \\ n-1, & n=3,7,11,15,... \\ \frac{n+1}{2}, & n=1,5,9,13,..... \end{cases}$, then f is

(a) One-one but not onto (b) onto but not one-one

(c) neither one-one nor onto (d) one-one and onto

Q7: Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=x-1$ and $g:\mathbb{R}-\{-1,1\}\to\mathbb{R}$ be defined as $g(x)=\frac{x^2}{x^2-1}$. Then the function fog is :

(a) One-one but not onto (b) onto but not one-one.

(c) Both one-one and onto (d) neither one-one nor onto

Q8: Let $f(x)=\frac{x-1}{x+1},\ x\in\mathbb{R}-\{0,-1,1\}$. If $f^{n+1}(x)=f\left(f^{n}(x)\right)$ for all $n\in\mathbb{N}$, then $f^{6}(6)+f^{7}(7)$ is equal to:

(a) $\frac{7}{6}$ (b) $-\frac{3}{2}$

(c) $\frac{7}{12}$

(b) $-\frac{11}{12}$

Q9: Let $f:\mathbb{N}\to\mathbb{R}$ be a function such that $f(x+y)=2f(x)f(y)$ for natural numbers x and y. If $f(1)=2,$ then the value of α for which $\sum_{k=1}^{10} f(\alpha+k)=\frac{512}{3}\left(2^{20}-1\right)$ holds, is

(a) 2

(b) 3

(c) 4

(d) 6

Q10: Let $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ be two functions defined by $f(x)=\log_e\left(x^2+1\right)-e^{-x}+1$ and $g(x)=\frac{1-2e^{2x}}{e^x}$. Then, for which of the following range of α, the inequality $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right)>f\left(g\left(\alpha-\frac{5}{3}\right)\right)$ holds?

(a) $(2,3)$

(b) $(-2,-1)$

(c) $(1,2)$

(d) $(-1,1)$

2021

Q1: The range of the function,

$f(x)=\log_{\sqrt{5}}\left(3+\cos\left(\frac{3\pi}{4}+x\right)+\cos\left(\frac{\pi}{4}+x\right)+\cos\left(\frac{\pi}{4}-x\right)-\cos\left(\frac{3\pi}{4}-x\right)\right)$ is

(a) $\left(0,\sqrt{5}\right)$

(b) $[-2,2]$

(c) $\left[\frac{1}{\sqrt{5}},\sqrt{5}\right]$

(d) $[0,2]$

Q2: Let $f:\mathbb{N}\to\mathbb{N}$ be a function such that $f(m+n)=f(m)+f(n)$ for every $m,n\in\mathbb{N}$. If $f(6)=18$. Then $f(2).f(3)$ is equal to

(a) 6

(b) 54

(c) 18

(d) 36

Q3: Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x+y)+f(x-y)=2f(x)f(y),\ f\left(\frac{1}{2}\right)=-1$, Then, the value of $\sum_{k=1}^{20}\frac{1}{\sin(k)\sin(k+f(k))}$ is equal to:

(a) $\cos ec^2(21)\cos(20)\cos(2)$
(b) $\sec^2(1)\sec(21)\cos(20)$
(c) $\cos ec^2(1)\cos ec(21)\sin(20)$
(d) $\sec^2(21)\sin(20)\sin(2)$

Q4: Consider the function $f:A\to B$ and $g:B\to C$ $(A,B,C\subseteq R)$ such that $(gof)^{-1}$ exists, then

(a) f and g both are one-one
(b) f and g both are onto
(c) f is one-one and g is onto
(d) f is onto and g is one-one

Q5: Let g : N → N be defined as

g(3n + 1) = 3n + 2,

g(3n + 2) = 3n + 3,

g(3n + 3) = 3n + 1, for all n ≥ 0.

Then which of the following statements is true?

(a) There exists an onto function $f:\mathbb{N}\to\mathbb{N}$ such that $fog = f$
(b) There exists a one-one function $f:\mathbb{N}\to\mathbb{N}$ such that $fog = f$
(c) gogog=g
(d) There exists a function $f:\mathbb{N}\to\mathbb{N}$ such that $gof = f$

Q6: Let $f:\mathbb{R}-\left\{\frac{\alpha}{6}\right\}\to\mathbb{R}$ be defined by $f(x)=\frac{5x+3}{6x-\alpha}$. Then the value of α for which $(fof)(x)=x$, for all $x\in\mathbb{R}-\left\{\frac{\alpha}{6}\right\}$, is

(a) No such α exists
(b) 5
(c) 8
(d) 6

Q7: Let $[x]$ denote the greatest integer ≤x, where $x\in\mathbb{R}$. If the domain of the real valued function $f(x)=\sqrt{\frac{|[x]|-2}{|[x]|-3}}$ is $(-\infty,a]\cup[b,c)\cup[4,\infty)$, $a<b<c$, then the value of $a+b+c$ is

(a) 8
(b) 1
(c) -2
(d) -3

Q8: Let $f:\mathbb{R}-\{3\}\to\mathbb{R}-\{1\}$ be defined by $f(x)=\frac{x-2}{x-3}$. Let $g:\mathbb{R}\to\mathbb{R}$ be given as $g(x)=2x-3$, Then, the sum of all the values of x for which $f^{-1}(x)+g^{-1}(x)=\frac{13}{2}$ is equal to

(a) 3 (b) 5

(c) 2 (d) 7

Q9: The real valued function $f(x)=\frac{\cos ec^{-1}x}{\sqrt{x-[x]}}$, where $[x]$ denotes the greatest integer less than or equal to x, is defined for all x belonging to:

(a) all real except integers (b) All non-integers except the interval $[-1,1]$

(c) all integers except 0, -1, 1 (d) all real except the interval $[-1,1]$

Q10: If the functions are defined as $f(x)=\sqrt{x}$ and $g(x)=\sqrt{1-x}$, then what is the common domain of the following functions: $f+g,\ f-g,\ f/g,\ g/f,\ g-f$ where $(f\pm g)(x)=f(x)\pm g(x)$, $(f/g)x=\frac{f(x)}{g(x)}$

(a) $0\le x\le 1$ (b) $0\le x<1$

(c) $0<x<1$ (d) $0<x\le 1$

Q11: The inverse of $y=5^{\log x}$ is

(a) $x=5^{\log y}$ (b) $x=y^{\frac{1}{\log 5}}$

(c) $x=5^{\frac{1}{\log y}}$ (d) $x=y^{\log 5}$

Q12: The range of $a\in\mathbb{R}$ for which the function

$f(x)=(4a-3)(x+\log_e 5)+2(a-7)\cot\left(\frac{x}{2}\right)\sin^2\left(\frac{x}{2}\right), x\ne 2n\pi, n\in\mathbb{N}$ has critical points, is:

(a) $[1,\infty)$ (b) $(-3,1)$

(c) $\left[-\frac{4}{3},2\right]$ (d) $(-\infty,-1]$

Q13: Let $A=\{1,2,3,....,10\}$ and $f:A\to A$ be defined as $f(k)=\begin{cases} k+1 & if \quad k \quad is \quad odd \\ k & if \quad k \quad is \quad even \end{cases}$. Then the number of possible functions $g:A\to A$ such that $gof = f$ is:

(a) 5^5

(b) 10^5

(c) 5!

(d) ${}^{10}C_5$

Q14: A function $f(x)$ is given by $f(x)=\dfrac{5^x}{5^x+5}$, then the sum of the series

$f\left(\dfrac{1}{20}\right)+f\left(\dfrac{2}{20}\right)+f\left(\dfrac{3}{20}\right)+.....+f\left(\dfrac{39}{20}\right)$ is equal to:

(a) $\dfrac{39}{2}$

(b) $\dfrac{19}{2}$

(c) $\dfrac{49}{2}$

(d) $\dfrac{29}{2}$

Q15: Let x denote the total number of one-one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one-one functions form the set A to the set A$\times$B. Then :

(a) $2y=273x$

(b) $y=91x$

(c) $2y=91x$

(d) $y=273x$

Q16: Let $f,g:\mathbb{N}\to\mathbb{N}$ such that $f(n+1)=f(n)+f(1)\ \forall\ n\in\mathbb{N}$ and g be any arbitrary function. Which of the following statements is NOT true?

(a) If g is onto , then fog is one-one

(b) f is one-one

(c) If f is onto, then $f(n)=n\ \forall\ n\in\mathbb{N}$

(d) If fog is one-one, then g is one-one

Q17: Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=2x-1$ and $g:\mathbb{R}-\{1\}\to\mathbb{R}$ be defined as $g(x)=\dfrac{x-\frac{1}{2}}{x-1}$.

Then the composite function $f(g(x))$ is :

(a) One-one but not onto.

(b) onto but not one-one

(c) both one-one and onto

(d) neither one-one not onto

2020

Q1 : Find the number of solutions of $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$, $\|x \in [0, 2\pi]\|$

(a) 2 (b) 4

(c) 6 (d) 8

Q2: Solution set of $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$ contains :

(a) Singleton set (b) two elements

(c) at least four elements (d) infinite elements

Q3: Let $f(x) = \dfrac{x[x]}{x^2+1}$: $(1,3) \to \mathbb{R}$ then range of $f(x)$ is (where $[f(x)]$ denotes the greatest integer function)

(a) $\left(0, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{7}{5}\right]$ (b) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$

(c) $\left(\frac{2}{5}, 1\right) \cup \left(1, \frac{4}{5}\right]$ (d) $\left(0, \frac{1}{3}\right) \cup \left(\frac{2}{5}, \frac{4}{5}\right]$

2017

1. The function f : $R \to \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as $f(x) = \dfrac{x}{1+x^2}$, is :

(1) invertible. (2) injective but not surjective.

(3) surjective but not injective. (4) neither injective nor surjective. 2.

Let a , b , c ε R. If f(x) = $ax^2 + bx + c$ is such that a + b + c = 3 and f(x+y) = f(x) +f(y) +xy $\forall$ x , y ε R, then $\sum_{n=1}^{10} f(n)$ is equal to :

(1) 330 (2) 165

(3) 190 (4) 255

2016

1. If $f(x)+2f\left(\frac{1}{x}\right)=3x, x\neq 0$ and $S=\{x\in R: f(x)=f(-x)\}$; then S:

(1) contains exactly one element.

(2) contains exactly two elements.

(3) contains more than two elements

(4) is an empty set

2014

1. If $a\in R$ and the equation $-3(x-[x])^2+2(x-[x])+a^2=0$ (where [x] denotes the greatest integer ≤ x) has no integral solution, then all possible values of a lie in the interval

(1) $(-1,0)\cup(0,1)$

(2) (1,2)

(3) $(-2,-1)$

(3) $\dfrac{1}{1+\{g(x)\}^5}$

(4) $1+\{g(x)\}^5$

2009

1. For real x, let $f(x)=x^3+5x+1$, then

(1) f is one-one but not onto R

(2) f is onto R but not one-one

(3) f is one-one and onto R

(4) f is neither one-one nor onto R

2008

1. Let f : $N\to Y$ be a function defined as f(x) = 4x +3, where Y = {y ε N : y = 4x +3 for some xεN }. Show that f is invertible and its inverse is

(1) g(y) = $\dfrac{3y+4}{4}$

(2) g(y) = $4+\dfrac{y+3}{4}$

(3) g(y) = $\dfrac{y+3}{4}$

(4) g(y) = $\log\left|\sin\left(x-\frac{\pi}{4}\right)\right|+c$

2007

1. The largest interval lying in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which the function

$[f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)]$ is defined , is

(1) $[0, \pi]$ (2) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(3) $\left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$ (4) $\left[0, \frac{\pi}{2}\right)$

2006

1. A real valued function f(x) satisfies the functional equation f(x-y) = f(x) f(y) – f(a – x) f(a + y) where a is a given constant and f(0) = 1, f(2a – x) is equal to

(1) -f(x) (2) f(x)

(3) f(a) + f(a-x) (4) f(-x)

2005

1. Let f : $(-1,1) \to B$, be a function defined by $f(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$, then f is both one-one and onto when B is the interval

(1) $\left(0, \frac{\pi}{2}\right)$ (2) $\left[0, \frac{\pi}{2}\right)$

(3) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (4) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

2004

1. The range of the function f(x) = ${}^{7-x}P_{x-3}$ is

(1) { 1, 2, 3} (2) {1,2,3,4,5}

(3) {1,2,3,4} (4) {1,2,3,4,5,6}

2. If f : $R \to S$, defined by $f(x) = \sin x - \sqrt{3}\cos x + 1$, is onto, then the interval of S is

(1) [0,3]

(2) [-1,1]

(3) [0,1]

(4) [-1,-3]

3. The graph of the function $y = f(x)$ is symmetrical about the line x = 2 , then

(1) f(x + 2) = f(x -2)

(2) f(2+x) = f(2-x)

(3) f(x) = f(-x)

(4) f(x) = -f(-x)

4. The domain of the function $f(x) = \dfrac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is

(1) [2,3]

(2) [2,3)

(3) [1,2]

(4) [1,2)

2003

1. A function f from the set of natural numbers to integers defined by

f(n) = $\begin{cases} \dfrac{n-1}{2}, & when\, n\, is\, odd \\ -\dfrac{n}{2}, & when\, n\, is\, even \end{cases}$ is

(1) one-one but not onto

(2) onto but not one-one

(3) one-one and onto both

(4) neither one-one nor onto

2. If $f : R \to R$ satisfies f(x+y) = f(x) + f(y), for all x,y ε R and f(1) = 7, then $\sum_{r=1}^{n} f(r)$ is

(1) $\dfrac{7n}{2}$

(2) $\dfrac{7(n+1)}{2}$

(3) 7n(n+1)

(4) $\dfrac{7n(n+1)}{2}$

3. Domain of definition of the function f(x) = $\dfrac{3}{4-x^2} + \log_{10}\left(x^3 - x\right)$, is

(1) (1,2)

(2) $(-1,0) \cup (1,2)$

(3) $(1,2) \cup (2,\infty)$ (4) $(-1,0) \cup (1,2) \cup (2,\infty)$

4. The function f(x) = $\log\left(x + \sqrt{x^2+1}\right)$, is

(1) an even function (2) an odd function

(3) a periodic function (4) neither an even nor an odd function

2002

1. The period of $\sin^2 \theta$ is

(1) π^2 (2) π

(3) π^3 (4) $\pi / 2$

2. Which one is not periodic

(1) $|\sin 3x| + \sin^2 x$ (2) $\cos \sqrt{x} + \cos^2 x$

(3) cos4x + $\tan^2 x$ (4) $\cos 2x + \sin x$

3. If f(x+y) = f(x).f(y) $\forall$ x,y and f(5) = 2, f'(0) = 3, then f'(5) is

(1) 0 (2) 1

(3) 6 (4) 2

4. The domain of $\sin^{-1}\left[\log_3\left(\frac{x}{3}\right)\right]$ is

(1) [1,9] (2) [-1,9]

(3) [-9,1] (4) [-9,-1]

Assertion – Reason Type

1. Let f be a function defined by $f(x)=(x-1)^2+1, (x\geq 1)$

   Statement – I : The set { x : f(x) = $f^{-1}(x)$ } = {1,2}.

   Statement – II : f is bijection and $f^{-1}(x) = 1+\sqrt{x-1}$, $x\geq 1$.

2. Let f(x) = $(x+1)^2-1$, $x\geq -1$

   Statement – I : The set { x : f(x) = $f^{-1}(x)$ } = {0,-1}

   Statement – II : f is a bijection.